
Probability

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10.1 Revision

EMBJP

Terminology

EMBJQ

Outcome: a single observation of an uncertain or random process (called an *experiment*). For example, when you accidentally drop a book, it might fall on its cover, on its back or on its side. Each of these options is a possible outcome.

Sample space of an experiment: the set of all possible outcomes of the experiment. For example, the sample space when you roll a single 6-sided die is the set $\{1; 2; 3; 4; 5; 6\}$. For a given experiment, there is exactly one sample space. The sample space is denoted by the letter S .

Event: a set of outcomes of an experiment. For example, during radioactive decay of 1 gramme of uranium-234, one possible event is that the number of alpha-particles emitted during 1 microsecond is between 225 and 235.

Probability of an event: a real number between 0 and 1 that describes how likely it is that the event will occur. A probability of 0 means the outcome of the experiment will never be in the event set. A probability of 1 means the outcome of the experiment will always be in the event set. When all possible outcomes of an experiment have equal chance of occurring, the probability of an event is the number of outcomes in the event set as a fraction of the number of outcomes in the sample space.

Relative frequency of an event: the number of times that the event occurs during experimental trials, divided by the total number of trials conducted. For example, if we flip a coin 10 times and it landed on heads 3 times, then the relative frequency of the heads event is $\frac{3}{10} = 0,3$.

Union of events: the set of all outcomes that occur in at least one of the events. For 2 events called A and B , we write the union as “ A or B ”. Another way of writing the union is using set notation: $A \cup B$.

Intersection of events: the set of all outcomes that occur in all of the events. For 2 events called A and B , we write the intersection as “ A and B ”. Another way of writing the intersection is using set notation: $A \cap B$.

Mutually exclusive events: events with no outcomes in common, that is $(A \text{ and } B) = \emptyset$. Mutually exclusive events can never occur simultaneously. For example the event that a number is even and the event that the same number is odd are mutually exclusive, since a number can never be both even and odd.

Complementary events: two mutually exclusive events that together contain all the outcomes in the sample space. For an event called A , we write the complement as “not A ”. Another way of writing the complement is as A' .

📺 See video: 239K at www.everythingmaths.co.za

The **addition rule** (also called the sum rule) for any 2 events, A and B is

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

This rule relates the probabilities of 2 events with the probabilities of their union and intersection.

The **addition rule for 2 mutually exclusive events** is

$$P(A \text{ or } B) = P(A) + P(B)$$

This rule is a special case of the previous rule. Because the events are mutually exclusive, $P(A \text{ and } B) = 0$.

The **complementary rule** is

$$P(\text{not } A) = 1 - P(A)$$

This rule is a special case of the previous rule. Since A and $(\text{not } A)$ are mutually exclusive, $P(A \text{ or } (\text{not } A)) = 1$.

🔊 See video: 239M at www.everythingmaths.co.za

Worked example 1: Events

QUESTION

You take all the hearts from a deck of cards. You then select a random card from the set of hearts. What is the sample space? What is the probability of each of the following events?

1. The card is the ace of hearts.
2. The card has a prime number on it.
3. The card has a letter of the alphabet on it.

SOLUTION

Step 1: Write down the sample space

Since we are considering only one suit from the deck of cards (the hearts), we need to write down only the letters and numbers on the cards. Therefore the sample space is

$$S = \{A; 2; 3; 4; 5; 6; 7; 8; 9; 10; J; Q; K\}$$

Step 2: Write down the event sets

- ace of hearts: $\{A\}$
- prime number: $\{2; 3; 5; 7\}$
- letter of alphabet: $\{A; J; Q; K\}$

Step 3: Compute the probabilities

The probability of an event is defined as the number of elements in the event set divided by the number of elements in the sample space. There are 13 elements in the sample space. So the probability of each event is

- ace of hearts: $\frac{1}{13}$
- prime number: $\frac{4}{13}$
- letter of alphabet: $\frac{4}{13}$

Worked example 2: Events

QUESTION

You roll two 6-sided dice. Let E be the event that the total number of dots on the dice is 10. Let F be the event that at least one die is a 3.

1. Write down the event sets for E and F .
2. Determine the probabilities for E and F .
3. Are E and F mutually exclusive? Why or why not?

SOLUTION

Step 1: Write down the sample space

The sample space of a single 6-sided die is just $\{1; 2; 3; 4; 5; 6\}$. To get the sample space of two 6-sided dice, we have to take every possible pair of numbers from 1 to 6.

$$S = \left\{ \begin{array}{cccccc} (1; 1) & (1; 2) & (1; 3) & (1; 4) & (1; 5) & (1; 6) \\ (2; 1) & (2; 2) & (2; 3) & (2; 4) & (2; 5) & (2; 6) \\ (3; 1) & (3; 2) & (3; 3) & (3; 4) & (3; 5) & (3; 6) \\ (4; 1) & (4; 2) & (4; 3) & (4; 4) & (4; 5) & (4; 6) \\ (5; 1) & (5; 2) & (5; 3) & (5; 4) & (5; 5) & (5; 6) \\ (6; 1) & (6; 2) & (6; 3) & (6; 4) & (6; 5) & (6; 6) \end{array} \right\}$$

Step 2: Write down the events

For E the dice have to add to 10.

$$E = \{(4; 6); (5; 5); (6; 4)\}$$

For F at least one die has to be 3.

$$F = \{(1; 3); (3; 1); (2; 3); (3; 2); (3; 3); (4; 3); (3; 4); (5; 3); (3; 5); (6; 3); (3; 6)\}$$

Step 3: Compute the probabilities

The probability of an event is defined as the number of elements in the event set divided by the number of elements in the sample space. There are

- $6 \times 6 = 36$ outcomes in the sample space, S ;
- 3 outcomes in event E ; and
- 11 outcomes in event F .

Therefore

$$P(E) = \frac{3}{36} = \frac{1}{12}$$

and

$$P(F) = \frac{11}{36}$$

Step 4: Are they mutually exclusive

To test whether two events are mutually exclusive, we have to test whether their intersection is empty. Since E has no outcomes that contain a 3 on one of the dice, the intersection of E and F is empty: $(E \text{ and } F) = \emptyset$. This means that the events are mutually exclusive.

► See video: 239N at www.everythingmaths.co.za

Exercise 10 – 1: Revision

1. A bag contains r red balls, b blue balls and y yellow balls. What is the probability that a ball drawn from the bag at random is yellow?
2. A packet has yellow and pink sweets. The probability of taking out a pink sweet is $\frac{7}{12}$. What is the probability of taking out a yellow sweet?
3. You flip a coin 4 times. What is the probability that you get 2 heads and 2 tails? Write down the sample space and the event set to determine the probability of this event.
4. In a class of 37 children, 15 children walk to school, 20 children have pets at home and 12 children who have a pet at home also walk to school. How many children walk to school and do not have a pet at home?
5. You roll two 6-sided dice and are interested in the following two events:
 - A : the sum of the dice equals 8
 - B : at least one of the dice shows a 1

Show that these events are mutually exclusive.

6. You ask a friend to think of a number from 1 to 100. You then ask her the following questions:

- Is the number even?
- Is the number divisible by 7?

How many possible numbers are less than 80 if she answered “yes” to both questions?

7. In a group of 42 pupils, all but 3 had a packet of chips or a Fanta or both. If 23 had a packet of chips and 7 of these also had a Fanta, what is the probability that one pupil chosen at random has:

- a) both chips and Fanta
- b) only Fanta

8. Tamara has 18 loose socks in a drawer. Eight of these are orange and two are pink. Calculate the probability that the first sock taken out at random is:

- a) orange
- b) not orange
- c) pink
- d) not pink
- e) orange or pink
- f) neither orange nor pink

9. A box contains coloured blocks. The number of blocks of each colour is given in the following table.

Colour	Purple	Orange	White	Pink
Number of blocks	24	32	41	19

A block is selected randomly. What is the probability that the block will be:

- a) purple
- b) purple or white
- c) pink and orange
- d) not orange?

10. The surface of a soccer ball is made up of 32 faces. 12 faces are regular pentagons, each with a surface area of about 37 cm^2 . The other 20 faces are regular hexagons, each with a surface area of about 56 cm^2 .

You roll the soccer ball. What is the probability that it stops with a pentagon touching the ground?

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1. 239P 2. 239Q 3. 239R 4. 239S 5. 239T 6. 239V
7. 239W 8. 239X 9. 239Y 10. 239Z

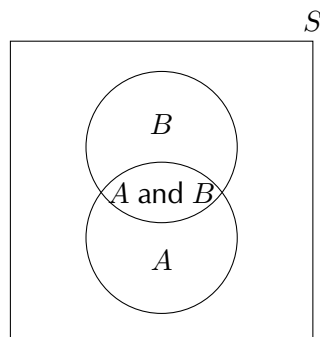


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A Venn diagram is used to show how events are related to one another. A Venn diagram can be very helpful when doing calculations with probabilities. In a Venn diagram each event is represented by a shape, often a circle or a rectangle. The region inside the shape represents the outcomes included in the event and the region outside the shape represents the outcomes that are not in the event.



A Venn diagram representing a sample space, S , as a square; and two events, A and B , as circles. The intersection of the two circles contains outcomes that are in both A and B .

Venn diagrams can be used in slightly different ways and it is important to notice the differences between them. The following 3 examples show how a Venn diagram is used to represent

- the outcomes included in each event;
- the number of outcomes in each event; and
- the probability of each event.

Worked example 3: Venn diagrams with outcomes

QUESTION

Choose a number between 1 and 20. Draw a Venn diagram to answer the following questions.

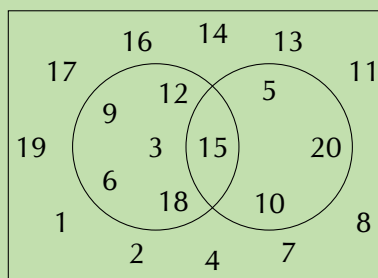
1. What is the probability that the number is a multiple of 3?
2. What is the probability that the number is a multiple of 5?
3. What is the probability that the number is a multiple of 3 or 5?
4. What is the probability that the number is a multiple of 3 and 5?

SOLUTION

Step 1: Draw a Venn diagram

The Venn diagram should show the sample space of all numbers from 1 to 20. It should also show an event set that contains all the multiples of 3, let $A = \{3; 6; 9; 12; 15; 18\}$,

and another event set that contains all the multiples of 5, let $B = \{5; 10; 15; 20\}$. Note that there is one shared outcome between these two events, namely 15.



Step 2: Compute probabilities

The probability of an event is the number of outcomes in the event set divided by the number of outcomes in the sample space. There are 20 outcomes in the sample space.

1. Since there are 6 outcomes in the multiples of 3 event set, the probability of a multiple of 3 is $P(A) = \frac{6}{20} = \frac{3}{10}$.
2. Since there are 4 outcomes in the multiples of 5 event set, the probability of a multiple of 5 is $P(B) = \frac{4}{20} = \frac{1}{5}$.
3. The event that the number is a multiple of 3 or 5 is the union of the above two event sets. There are 9 elements in the union of the event sets, so the probability is $\frac{9}{20}$.
4. The event that the number is a multiple of 3 and 5 is the intersection of the two event sets. There is 1 element in the intersection of the event sets, so the probability is $\frac{1}{20}$.

Worked example 4: Venn diagrams with counts

QUESTION

In a group of 50 learners, 35 take Mathematics and 30 take History, while 12 take neither of the two subjects. Draw a Venn diagram representing this information. If a learner is chosen at random from this group, what is the probability that he takes both Mathematics and History?

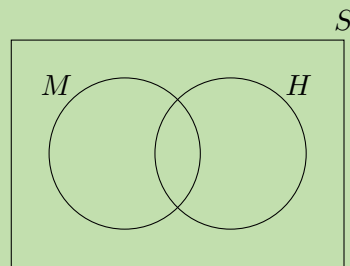
SOLUTION

Step 1: Draw outline of Venn diagram

There are 2 events in this question, namely

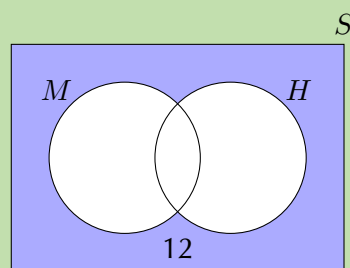
- M : that a learner takes Mathematics; and
- H : that a learner takes History.

We need to do some calculations before drawing the full Venn diagram, but with the information above we can already draw the outline.



Step 2: Write down sizes of the event sets, their union and intersection

We are told that 12 learners take neither of the two subjects. Graphically we can represent this as:



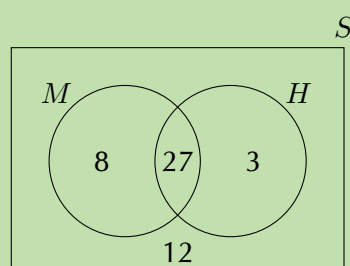
Since there are 50 elements in the sample space, we can see from this figure that there are $50 - 12 = 38$ elements in $(M \text{ or } H)$. So far we know

- $n(M) = 35$
- $n(H) = 30$
- $n(M \text{ or } H) = 38$

From the addition rule,

$$\begin{aligned} n(M \text{ or } H) &= n(M) + n(H) - n(M \text{ and } H) \\ \therefore n(M \text{ and } H) &= 35 + 30 - 38 \\ &= 27 \end{aligned}$$

Step 3: Draw the final Venn diagram



Worked example 5: Venn diagrams with probabilities

QUESTION

Draw a Venn diagram to represent the same information as in the previous example, except showing the probabilities of the different events, rather than the counts.

If a learner is chosen at random from this group, what is the probability that she takes both Mathematics and History?

SOLUTION

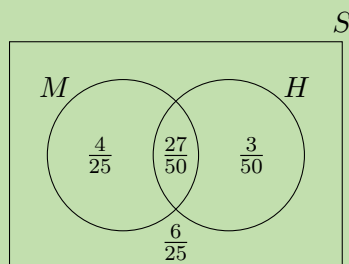
Step 1: Use counts to compute probabilities

Since there are 50 elements (learners) in the sample space, we can compute the probability of any event by dividing the size of the event set by 50. This gives the following probabilities:

- $P(M) = \frac{35}{50} = \frac{7}{10}$
- $P(H) = \frac{30}{50} = \frac{3}{5}$
- $P(M \text{ or } H) = \frac{38}{50} = \frac{19}{25}$
- $P(M \text{ and } H) = \frac{27}{50}$

Step 2: Draw the Venn diagram

Next we replace each count from the Venn diagram in the previous example with a probability.



Step 3: Find the answer

The probability that a random learner will take both Mathematics and History is $P(M \text{ and } H) = \frac{27}{50}$.

► See video: 23B2 at www.everythingmaths.co.za

Exercise 10 – 2: Venn diagram revision

1. Given the following information:

- $P(A) = 0,3$
- $P(B \text{ and } A) = 0,2$
- $P(B) = 0,7$

First draw a Venn diagram to represent this information. Then compute the value of $P(B \text{ and (not } A))$.

2. You are given the following information:

- $P(A) = 0,5$
- $P(A \text{ and } B) = 0,2$
- $P(\text{not } B) = 0,6$

Draw a Venn diagram to represent this information and determine $P(A \text{ or } B)$.

3. A study was undertaken to see how many people in Port Elizabeth owned either a Volkswagen or a Toyota. 3% owned both, 25% owned a Toyota and 60% owned a Volkswagen. What percentage of people owned neither car?
4. Let S denote the set of whole numbers from 1 to 15, X denote the set of even numbers from 1 to 15 and Y denote the set of prime numbers from 1 to 15. Draw a Venn diagram depicting S , X and Y .

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1. 23B3 2. 23B4 3. 23B5 4. 23B6



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10.2 Dependent and independent events

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Sometimes the presence or absence of one event tells us something about other events. We call events dependent if knowing whether one of them happened tells us something about whether the others happened. Independent events give us no information about one another; the probability of one event occurring does not affect the probability of the other events occurring.

DEFINITION: Independent events

Two events, A and B are independent if and only if

$$P(A \text{ and } B) = P(A) \times P(B)$$

At first it might not be clear why we should call events that satisfy the equation above independent. We will explore this further using a number of examples.

Investigation: Independence

Roll a single 6-sided die and consider the following two events:

- E : you get an even number
- T : you get a number that is divisible by three

Now answer the following questions:

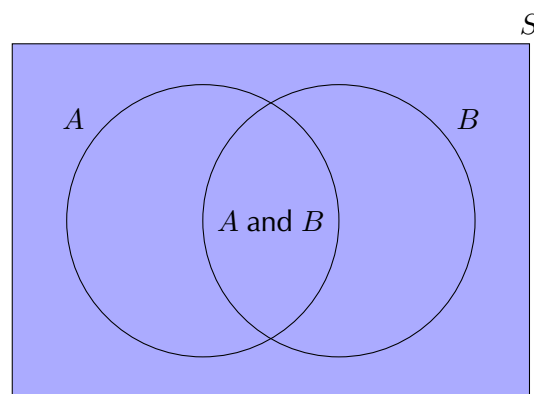
- What is the probability of E ?
- What is the probability of getting an even number if you are told that the number was also divisible by three?
- Does knowing that the number was divisible by three change the probability that the number was even?

Are the events E and T dependent or independent according to the definition (hint: compute the probabilities in the definition of independence)?

▶ See video: 23B7 at www.everythingmaths.co.za

So, why do we call it **independence** when $P(A \text{ and } B) = P(A) \times P(B)$? For two events, A and B , independence means that knowing the outcome of B does not affect the probability of A .

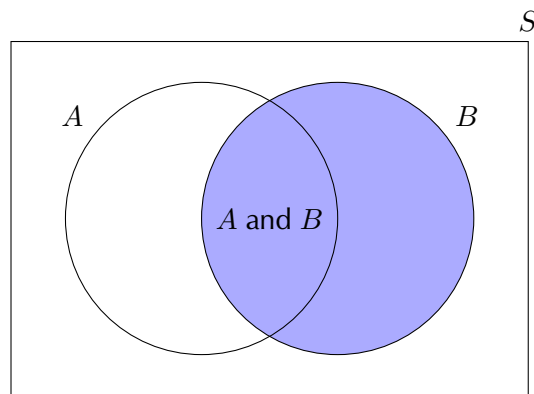
Consider the following Venn diagram.



The probability of A is the ratio between the number of outcomes in A and the number of outcomes in the sample space, S .

$$P(A) = \frac{n(A)}{n(S)}$$

Now, let's say that we **know** that event B happened. How does this affect the probability of A ? Here is how the Venn diagram changes:



A lot of the possible outcomes (all of the outcomes outside B) are now out of the picture, because we know that they did not happen. Now the probability of A happening, given that we know that B happened, is the ratio between the size of the region where A is present (A and B) and the size of all possible events (B).

$$P(A \text{ if we know } B) = \frac{n(A \text{ and } B)}{n(B)}$$

If $P(A) = P(A \text{ if we know } B)$ we call them independent, because knowing B does not change the probability of A .

With some algebra, we can prove that this statement of independence is the same as the definition of independence that we saw at the beginning of this section. For independent events

$$P(A \text{ and } B) = P(A) \times P(B)$$

This is equivalent to

$$\begin{aligned} P(A) &= P(A \text{ and } B) \div P(B) \\ &= \frac{n(A \text{ and } B)}{n(S)} \div \frac{n(B)}{n(S)} \\ &= \frac{n(A \text{ and } B)}{n(B)} \\ &= P(A \text{ if we know } B) \end{aligned}$$

That is why we call events independent!

(For enrichment only):

The ratio

$$\frac{P(A \text{ and } B)}{P(B)}$$

is called a **conditional probability** and written using the notation $P(A \mid B)$. This notation is read as "the probability of A given B ."

If (and only if) A and B are independent: $P(A \mid B) = P(A)$ and $P(B \mid A) = P(B)$. Try to prove this using the definition of independence.

QUESTION

A bag contains 5 red and 5 blue balls. We remove a random ball from the bag, record its colour and put it back into the bag. We then remove another random ball from the bag and record its colour.

1. What is the probability that the first ball is red?
2. What is the probability that the second ball is blue?
3. What is the probability that the first ball is red and the second ball is blue?
4. Are the first ball being red and the second ball being blue independent events?

SOLUTION**Step 1: Probability of a red ball first**

Since there are a total of 10 balls, of which 5 are red, the probability of getting a red ball is

$$P(\text{first ball red}) = \frac{5}{10} = \frac{1}{2}$$

Step 2: Probability of a blue ball second

The problem states that the first ball is placed back into the bag before we take the second ball. This means that when we draw the second ball, there are again a total of 10 balls in the bag, of which 5 are blue. Therefore the probability of drawing a blue ball is

$$P(\text{second ball blue}) = \frac{5}{10} = \frac{1}{2}$$

Step 3: Probability of red first and blue second

When drawing two balls from the bag, there are 4 possibilities. We can get

- a red ball and then another red ball;
- a red ball and then a blue ball;
- a blue ball and then a red ball;
- a blue ball and then another blue ball.

We want to know the probability of the second outcome, where we have to get a red ball first. Since there are 5 red balls and 10 balls in total, there are $\frac{5}{10}$ ways to get a red ball first. Now we put the first ball back, so there are again 5 red balls and 5 blue balls in the bag. Therefore there are $\frac{5}{10}$ ways to get a blue ball second if the first ball was red. This means that there are

$$\frac{5}{10} \times \frac{5}{10} = \frac{25}{100}$$

ways to get a red ball first and a blue ball second. So, the probability of getting a red ball first and a blue ball second is $\frac{1}{4}$.

Step 4: Dependent or independent?

According to the definition, events are independent if and only if

$$P(A \text{ and } B) = P(A) \times P(B)$$

In this problem:

- $P(\text{first ball red}) = \frac{1}{2}$
- $P(\text{second ball blue}) = \frac{1}{2}$
- $P(\text{first ball red and second ball blue}) = \frac{1}{4}$

Since $\frac{1}{4} = \frac{1}{2} \times \frac{1}{2}$, the events are independent.

► See video: 23B8 at www.everythingmaths.co.za

Worked example 7: Independent and dependent events

QUESTION

In the previous example, we picked a random ball and put it back into the bag before continuing. This is called **sampling with replacement**. In this example, we will follow the same process, except that we will not put the first ball back into the bag. This is called **sampling without replacement**.

So, from a bag with 5 red and 5 blue balls, we remove a random ball and record its colour. Then, without putting back the first ball, we remove another random ball from the bag and record its colour.

1. What is the probability that the first ball is red?
2. What is the probability that the second ball is blue?
3. What is the probability that the first ball is red and the second ball is blue?
4. Are the first ball being red and the second ball being blue independent events?

SOLUTION

Step 1: Count the number of outcomes

We will look directly at the number of possible ways in which we can get the 4 possible outcomes when removing 2 balls. In the previous example, we saw that the 4 possible outcomes are

- a red ball and then another red ball;
- a red ball and then a blue ball;
- a blue ball and then a red ball;
- a blue ball and then another blue ball.

For the first outcome, we have to get a red ball first. Since there are 5 red balls and 10 balls in total, there are $\frac{5}{10}$ ways to get a red ball first. After we have taken out a red ball, there are now 4 red balls and 5 blue balls left. Therefore there are $\frac{4}{9}$ ways to get a red ball second if the first ball was also red. This means that there are

$$\frac{5}{10} \times \frac{4}{9} = \frac{20}{90}$$

ways to get a red ball first and a red ball second. The probability of the first outcome is $\frac{2}{9}$.

For the second outcome, we have to get a red ball first. As in the first outcome, there are $\frac{5}{10}$ ways to get a red ball first; and there are now 4 red balls and 5 blue balls left. Therefore there are $\frac{5}{9}$ ways to get a blue ball second if the first ball was red. This means that there are

$$\frac{5}{10} \times \frac{5}{9} = \frac{25}{90}$$

ways to get a red ball first and a blue ball second. The probability of the second outcome is $\frac{5}{18}$.

We can compute the probabilities of the third and fourth outcomes in the same way as the first two, but there is an easier way. Notice that there are only 2 types of ball and that there are exactly equal numbers of them at the start. This means that the problem is completely symmetric in red and blue. We can use this symmetry to compute the probabilities of the other two outcomes.

In the third outcome, the first ball is blue and the second ball is red. Because of symmetry this outcome must have the same probability as the second outcome (when the first ball is red and the second ball is blue). Therefore the probability of the third outcome is $\frac{5}{18}$.

In the fourth outcome, the first and second balls are both blue. From symmetry, this outcome must have the same probability as the first outcome (when both balls are red). Therefore the probability of the fourth outcome is $\frac{2}{9}$.

To summarise, these are the possible outcomes and their probabilities:

- first ball red and second ball red: $\frac{2}{9}$;
- first ball red and second ball blue: $\frac{5}{18}$;
- first ball blue and second ball red: $\frac{5}{18}$;
- first ball blue and second ball blue: $\frac{2}{9}$.

Step 2: Probability of a red ball first

To determine the probability of getting a red ball on the first draw, we look at all of the outcomes that contain a red ball first. These are

- a red ball and then another red ball;
- a red ball and then a blue ball.

The probability of the first outcome is $\frac{2}{9}$ and the probability of the second outcome is $\frac{5}{18}$. By adding these two probabilities, we see that the probability of getting a red ball first is

$$P(\text{first ball red}) = \frac{2}{9} + \frac{5}{18} = \frac{1}{2}$$

This is the same as in the previous exercise, which should not be too surprising since the probability of the first ball being red is not affected by whether or not we put it back into the bag before drawing the second ball.

Step 3: Probability of a blue ball second

To determine the probability of getting a blue ball on the second draw, we look at all of the outcomes that contain a blue ball second. These are

- a red ball and then a blue ball;
- a blue ball and then another blue ball.

The probability of the first outcome is $\frac{5}{18}$ and the probability of the second outcome is $\frac{2}{9}$. By adding these two probabilities, we see that the probability of getting a blue ball second is

$$P(\text{second ball blue}) = \frac{5}{18} + \frac{2}{9} = \frac{1}{2}$$

This is also the same as in the previous exercise! You might find it surprising that the probability of the second ball is not affected by whether or not we replace the first ball. The reason why this probability is still $\frac{1}{2}$ is that we are computing the probability that the second ball is blue without knowing the colour of the first ball. Because there are only two equal possibilities for the second ball (red and blue) and because we don't know whether the first ball is red or blue, there is an equal chance that the second ball will be one colour or the other.

Step 4: Probability of red first and blue second

We have already calculated the probability that the first ball is red and the second ball is blue. It is $\frac{5}{18}$.

Step 5: Dependent or independent?

According to the definition, events are independent if and only if

$$P(A \text{ and } B) = P(A) \times P(B)$$

In this problem:

- $P(\text{first ball red}) = \frac{1}{2}$
- $P(\text{second ball blue}) = \frac{1}{2}$
- $P(\text{first ball red and second ball blue}) = \frac{5}{18}$

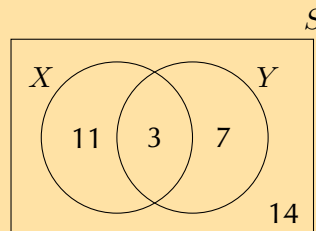
Since $\frac{5}{18} \neq \frac{1}{2} \times \frac{1}{2}$, the events are dependent.

WARNING!

Just because two events are mutually exclusive does not necessarily mean that they are independent. To test whether events are mutually exclusive, always check that $P(A \text{ and } B) = 0$. To test whether events are independent, always check that $P(A \text{ and } B) = P(A) \times P(B)$. See the exercises below for examples of events that are mutually exclusive and independent in different combinations.

Exercise 10 – 3: Dependent and independent events

1. Use the following Venn diagram to determine whether events X and Y are
 - a) mutually exclusive or not mutually exclusive;
 - b) dependent or independent.



2. Of the 30 learners in a class 17 have black hair, 11 have brown hair and 2 have red hair. A learner is selected from the class at random.
 - a) What is the probability that the learner has black hair?
 - b) What is the probability that the learner has brown hair?
 - c) Are these two events mutually exclusive?
 - d) Are these two events independent?
3. $P(M) = 0,45$; $P(N) = 0,3$ and $P(M \text{ or } N) = 0,615$. Are the events M and N mutually exclusive, independent or neither mutually exclusive nor independent?
4. (For enrichment)
Prove that if event A and event B are mutually exclusive with $P(A) \neq 0$ and $P(B) \neq 0$, then A and B are always dependent.

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1. 23B9 2. 23BB 3. 23BC 4. 23BD



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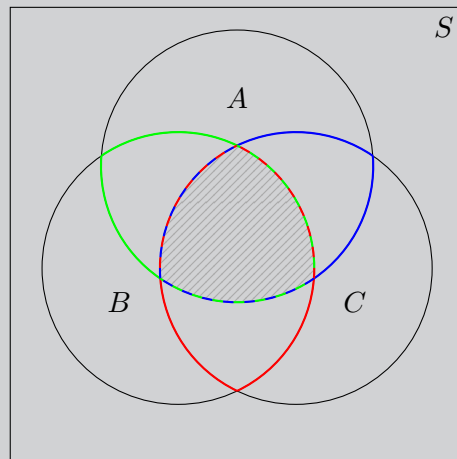
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In the rest of this chapter we will look at tools and techniques for working with probability problems.

When working with more complex problems, we can have three or more events that intersect in various ways. To solve these problems, we usually want to count the number (or percentage) of outcomes in an event, or a combination of events. Venn diagrams are a useful tool for recording and visualising the counts.

Investigation: Venn diagram for 3 events

The diagram below shows a general Venn diagram for 3 events.



Write down the sets corresponding to each of the three coloured regions and also to the shaded region. Remember that the intersections between circles represent the intersections between the different events.

What is the event for

- the red region;
- the green region;
- the blue region; and
- the shaded region?

Worked example 8: Venn diagram for 3 events

QUESTION

Draw a Venn diagram that shows the following sample space and events:

- S : all the integers from 1 to 30
- P : prime numbers
- M : multiples of 3
- F : factors of 30

SOLUTION

Step 1: Write down the sample space and event sets

The sample space contains all the positive integers up to 30.

$$S = \{1; 2; 3; \dots; 30\}$$

The prime numbers between 1 and 30 are

$$P = \{2; 3; 5; 7; 11; 13; 17; 19; 23; 29\}$$

The multiples of 3 between 1 and 30 are

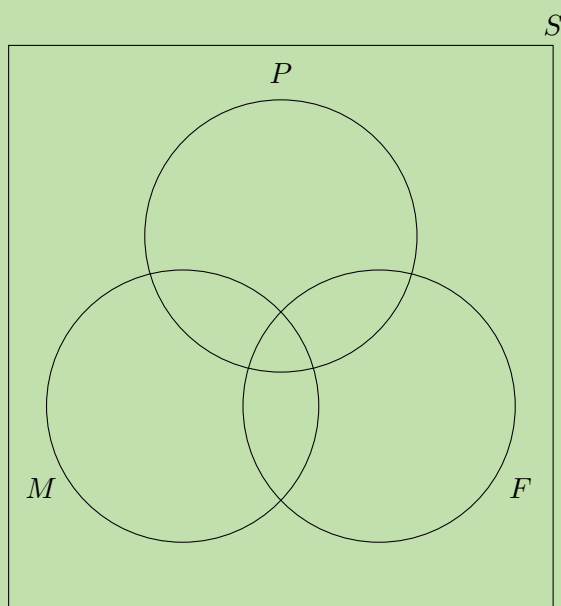
$$M = \{3; 6; 9; 12; 15; 18; 21; 24; 27; 30\}$$

The factors of 30 are

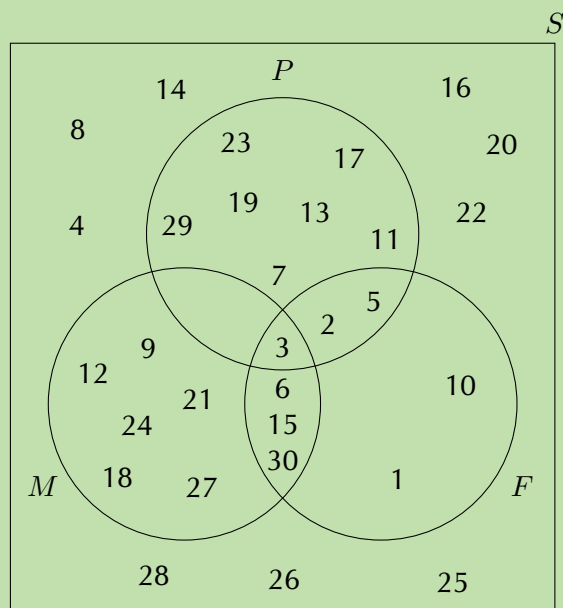
$$F = \{1; 2; 3; 5; 6; 10; 15; 30\}$$

Step 2: Draw the outline of the Venn diagram

There are 3 events, namely P , M and F , and the sample space, S . Put this information on a Venn diagram:



Step 3: Place the outcomes in the appropriate event sets



Worked example 9: Venn diagram for 3 events

QUESTION

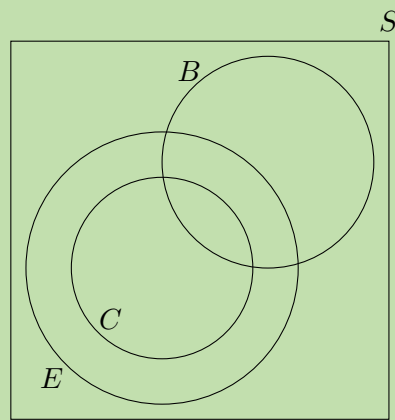
At Dawnview High there are 400 Grade 11 learners. 270 do Computer Science, 300 do English and 50 do Business studies. All those doing Computer Science do English, 20 take Computer Science and Business studies and 35 take English and Business studies. Using a Venn diagram, calculate the probability that a pupil drawn at random will take:

1. English, but not Business studies or Computer Science
2. English but not Business studies
3. English or Business studies but not Computer Science
4. English or Business studies

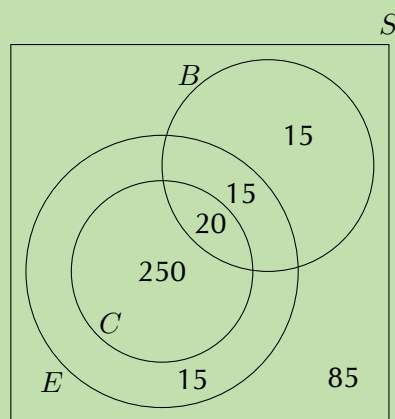
SOLUTION

Step 1: Draw the outline of the Venn diagram

We need to be careful with this problem. In the question statement we are told that all the learners who do Computer Science also do English. This means that the circle for Computer Science on the Venn diagram needs to be inside the circle for English.

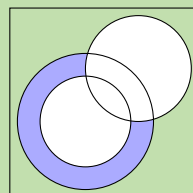


Step 2: Fill in the counts on the Venn diagram



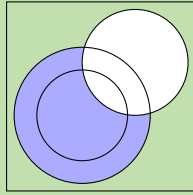
Step 3: Compute probabilities

To find the number of learners taking English, but not Business studies or Computer Science, we need to look at this region of the Venn diagram:



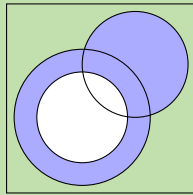
The count in this region is 15 and there are a total of 400 learners in the grade. Therefore the probability that a learner will take English but not Business studies or Computer Science is $\frac{15}{400} = \frac{3}{80}$.

To find the number of learners taking English but not Business studies, we need to look at this region of the Venn diagram:



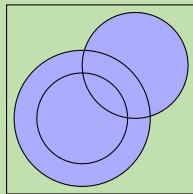
The count in this region is 265. Therefore the probability that a learner will take English but not Business studies is $\frac{265}{400} = \frac{53}{80}$.

To find the number of learners taking English or Business studies but not Computer Science, we need to look at this region of the Venn diagram:



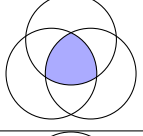
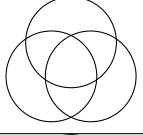
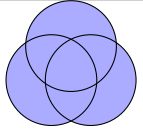
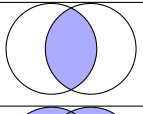
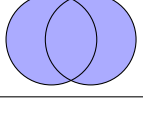
The count in this region is 45. Therefore the probability that a learner will take English or Business studies but not Computer Science is $\frac{45}{400} = \frac{9}{80}$.

To find the number of learners taking English or Business studies, we need to look at this region of the Venn diagram:



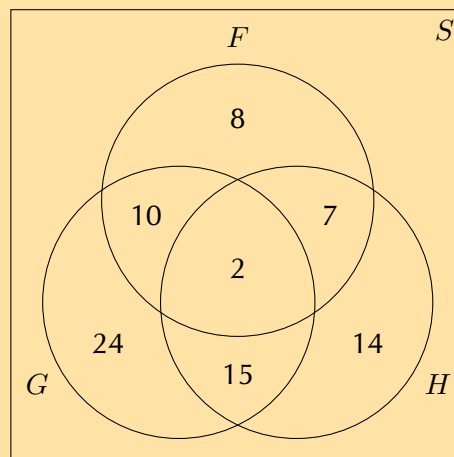
The count in this region is 315. Therefore the probability that a learner will take English or Business studies is $\frac{315}{400} = \frac{63}{80}$.

There are some words that tell you which part of the Venn diagram should be filled in. The following table summarises the most important ones:

Words	Symbols	Venn diagram
"all"	$A \text{ and } B \text{ and } C / A \cap B \cap C$	
"none"		
"at least one"	$A \text{ or } B \text{ or } C / A \cup B \cup C$	
"both A and B"	$A \text{ and } B / A \cap B$	
"A or B"	$A \text{ or } B / A \cup B$	

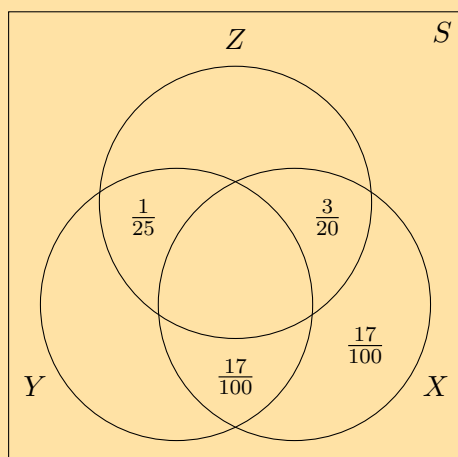
Exercise 10 – 4: Venn diagrams

1. Use the Venn diagram below to answer the following questions. Also given: $n(S) = 120$.



- Compute $P(F)$.
- Compute $P(G \text{ or } H)$.
- Compute $P(F \text{ and } G)$.
- Are F and G dependent or independent?

2. The Venn diagram below shows the probabilities of 3 events. Complete the Venn diagram using the additional information provided.



- $P(Z \text{ and (not } Y)) = \frac{31}{100}$
- $P(Y \text{ and } X) = \frac{23}{100}$
- $P(Y) = \frac{39}{100}$

After completing the Venn diagram, compute the following:

$$P(Z \text{ and not } (X \text{ or } Y))$$

3. There are 79 Grade 10 learners at school. All of these take some combination of Maths, Geography and History. The number who take Geography is 41; those who take History is 36; and 30 take Maths. The number who take Maths and History is 16; the number who take Geography and History is 6, and there are 8 who take Maths only and 16 who take History only.
- Draw a Venn diagram to illustrate all this information.
 - How many learners take Maths and Geography but not History?
 - How many learners take Geography only?
 - How many learners take all three subjects?

4. Draw a Venn diagram with 3 mutually exclusive events. Use the diagram to show that for 3 mutually exclusive events, A , B and C , the following is true:

$$P(A \text{ or } B \text{ or } C) = P(A) + P(B) + P(C)$$

This is the addition rule for 3 mutually exclusive events.

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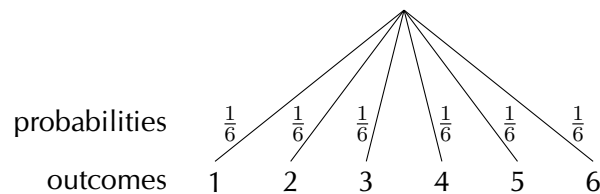
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Tree diagrams are useful for organising and visualising the different possible outcomes of a sequence of events. For each possible outcome of the first event, we draw a line where we write down the probability of that outcome and the state of the world if that outcome happened. Then, for each possible outcome of the second event we do the same thing.

Below is an example of a simple tree diagram, showing the possible outcomes of rolling a 6-sided die.



Note that each outcome (the numbers 1 to 6) is shown at the end of a line; and that the probability of each outcome (all $\frac{1}{6}$ in this case) is shown on a line. The probabilities have to add up to 1 in order to cover all of the possible outcomes. In the examples below, we will see how to draw tree diagrams with multiple events and how to compute probabilities using the diagrams.

Earlier in this chapter you learned about dependent and independent events. Tree diagrams are very helpful for analysing dependent events. A tree diagram allows you to show how each possible outcome of one event affects the probabilities of the other events.

Tree diagrams are not so useful for independent events since we can just multiply the probabilities of separate events to get the probability of the combined event. Remember that for independent events:

$$P(A \text{ and } B) = P(A) \times P(B)$$

So if you already know that events are independent, it is usually easier to solve a problem without using tree diagrams. But if you are uncertain about whether events are independent or if you know that they are not, you should use a tree diagram.

Worked example 10: Drawing a tree diagram

QUESTION

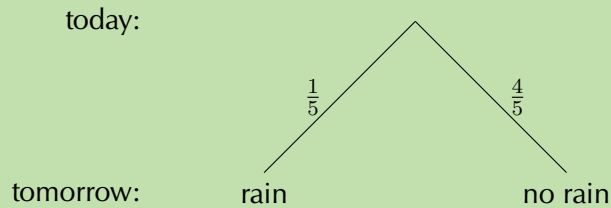
If it rains on a given day, the probability that it rains the next day is $\frac{1}{3}$. If it does not rain on a given day, the probability that it rains the next day is $\frac{1}{6}$. The probability that it will rain tomorrow is $\frac{1}{5}$. What is the probability that it will rain the day after tomorrow? Draw a tree diagram of all the possibilities to determine the answer.

SOLUTION

Step 1: Draw the first level of the tree diagram

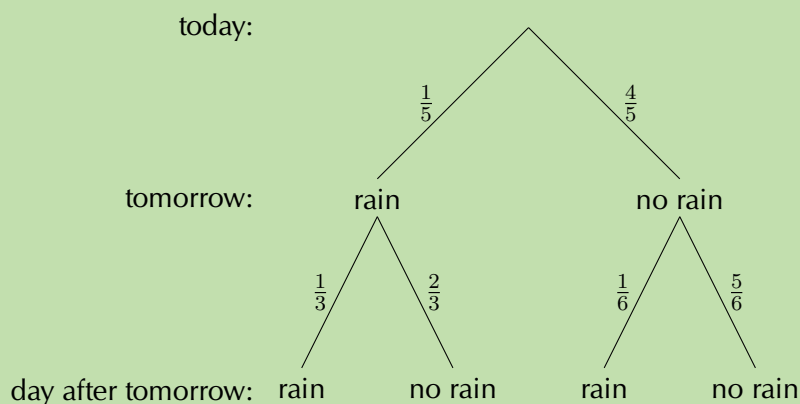
Before we can determine what happens on the day after tomorrow, we first have to determine what might happen tomorrow. We are told that there is a $\frac{1}{5}$ probability that

it will rain tomorrow. Here is how to represent this information using a tree diagram:



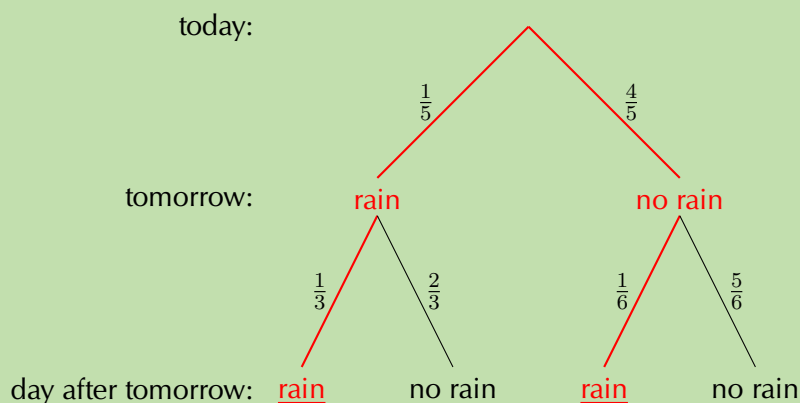
Step 2: Draw the second level of the tree diagram

We are also told that if it **does** rain on one day, there is a $\frac{1}{3}$ probability that it will also rain on the following day. On the other hand, if it **does not** rain on one day, there is only a $\frac{1}{6}$ probability that it will also rain on the following day. Using this information we complete the tree diagram:



Step 3: Compute the probability

We are asked what the probability is that it will rain the day after tomorrow. On the tree diagram above we can see that there are 2 situations where it rains on the day after tomorrow. They are marked in red below.



To get the probability for the first situation (that it rains tomorrow and the day after tomorrow) we have to multiply the probabilities along the first red line.

$$\begin{aligned}
 &P(\text{rain tomorrow and rain day after tomorrow}) \\
 &= \frac{1}{5} \times \frac{1}{3} \\
 &= \frac{1}{15}
 \end{aligned}$$

To get the probability for the second situation (that it does not rain tomorrow, but it does rain the day after tomorrow) we have to multiply the probabilities along the second red line.

$$\begin{aligned} &P(\text{not rain tomorrow and rain day after tomorrow}) \\ &= \frac{4}{5} \times \frac{1}{6} \\ &= \frac{2}{15} \end{aligned}$$

Therefore the total probability that it will rain the day after tomorrow is the sum of the probabilities along the two red paths, namely

$$\frac{1}{15} + \frac{2}{15} = \frac{1}{5}$$

Worked example 11: Drawing a tree diagram

QUESTION

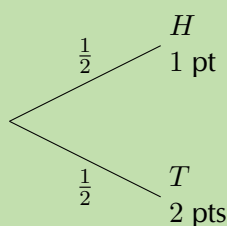
You play the following game. You flip a coin. If it comes up tails, you get 2 points and your turn ends. If it comes up heads, you get only 1 point, but you can flip the coin again. If you flip the coin multiple times in one turn, you add up the points. You can flip the coin at most 3 times in one turn. What is the probability that you will get exactly 3 points in one turn? Draw a tree diagram to visualise the different possibilities.

SOLUTION

Step 1: Write down the events and their symbols

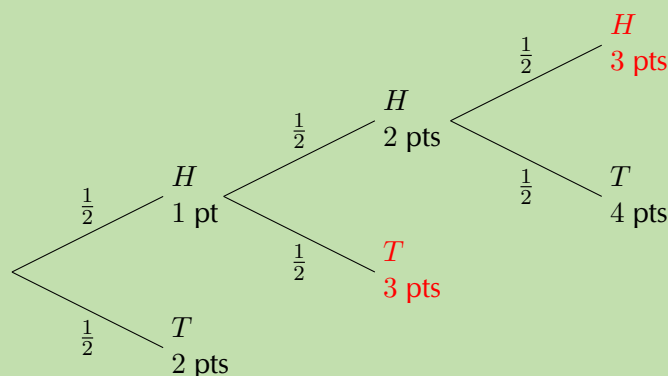
Each coin toss has one of two possible outcomes, namely heads (H) and tails (T). Each outcome has a probability of $\frac{1}{2}$. We are asked to count the number of points, so we will also indicate how many points we have for each outcome.

Step 2: Draw the first level of the tree diagram



This tree diagram shows the possible outcomes after 1 flip of the coin. Remember that we can have up to 3 flips, so the diagram is not complete yet. If the coin comes up heads, we flip the coin again. If the coin comes up tails, we stop.

Step 3: Draw the second and third level of the tree diagram



In this tree diagram you can see that we add up the points we get with each coin flip. After three coin flips, the game is over.

Step 4: Find the relevant outcomes and compute the probability

We are interested in getting exactly 3 points during the game. To find these outcomes we look only at the tips of the tree. We end with exactly 3 points when the coin flips are

- (H; T) with probability $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$;
- (H; H; H) with probability $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$.

Notice that we compute the probability of an outcome by multiplying all the probabilities along the path from the start of the tree to the tip where the outcome is. We add the above two probabilities to obtain the final probability of getting exactly 3 points as $\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$.

Worked example 12: Drawing a tree diagram

QUESTION

A person takes part in a medical trial that tests the effect of a medicine on a disease. Half the people are given medicine and the other half are given a sugar pill, which has no effect on the disease. The medicine has a 60% chance of curing someone. But, people who do not get the medicine still have a 10% chance of getting well. There are 50 people in the trial and they all have the disease. Talwar takes part in the trial, but we do not know whether he got the medicine or the sugar pill. Draw a tree diagram of all the possible cases. What is the probability that Talwar gets cured?

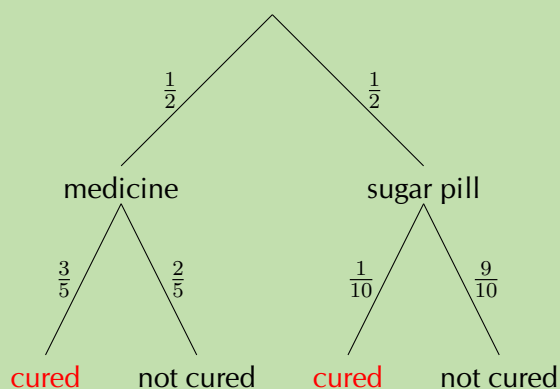
SOLUTION

Step 1: Summarise the information in the problem

There are two uncertain events in this problem. Each person either receives medicine (probability $\frac{1}{2}$) or a sugar pill (probability $\frac{1}{2}$). Each person also gets cured (probability

$\frac{3}{5}$ with medicine and $\frac{1}{10}$ without) or stays ill (probability $\frac{2}{5}$ with medicine and $\frac{9}{10}$ without).

Step 2: Draw the tree diagram



In the first level of the tree diagram we show that Talwar either gets the medicine or the sugar pill. The second level of the tree diagram shows whether Talwar is cured or not, depending on which one of the pills he got.

Step 3: Compute the required probability

We multiply the probabilities along each path in the tree diagram that leads to Talwar being cured:

$$\frac{1}{2} \times \frac{3}{5} = \frac{3}{10}$$

$$\frac{1}{2} \times \frac{1}{10} = \frac{1}{20}$$

We then add these probabilities to get the final answer. The probability that Talwar is cured is $\frac{7}{20}$.

Exercise 10 – 5: Tree diagrams

1. You roll a die twice and add up the dots to get a score. Draw a tree diagram to represent this experiment. What is the probability that your score is a multiple of 5?
2. What is the probability of throwing at least one five in four rolls of a regular 6-sided die? Hint: do not show all possible outcomes of each roll of the die. We are interested in whether the outcome is 5 or not 5 only.
3. You flip one coin 4 times.
 - a) What is the probability of getting exactly 3 heads?
 - b) What is the probability of getting at least 3 heads?

4. You flip 4 different coins at the same time.
- a) What is the probability of getting exactly 3 heads?
 - b) What is the probability of getting at least 3 heads?

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1. 23BK 2. 23BM 3. 23BN 4. 23BP



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10.5 Contingency tables

EMBJX

A contingency table is another tool for keeping a record of the counts or percentages in a probability problem. Contingency tables are especially helpful for figuring out whether events are dependent or independent.

We will be studying two-way contingency tables, where we count the number of outcomes for 2 events and their complements, making 4 events in total. A two-way contingency table always shows the counts for the 4 possible combinations of events, as well as the totals for each event and its complement. We can use a contingency table to compute the probabilities of various events by computing the ratios between counts, and to determine whether the events are dependent or independent. The example below shows a two-way contingency table, representing the outcome of a medical study.

Worked example 13: Contingency tables

QUESTION

A medical trial into the effectiveness of a new medication was carried out. 120 females and 90 males took part in the trial. Out of those people, 50 females and 30 males responded positively to the medication. Given below is a contingency table with the given information filled in.

	Female	Male	Totals
Positive	50	30	
Negative			
Totals	120	90	

- 1. What is the probability that the medicine gives a positive result for females?
- 2. What is the probability that the medicine gives a negative result for males?
- 3. Was the medication's success independent of gender? Explain.

SOLUTION

Step 1: Complete the contingency table

The best place to start is always to complete the contingency table. Because each column has to sum up to its total, we can work out the number of females and males who responded negatively to the medication. Then we can add each row to get the totals on the right hand side of the table.

	Female	Male	Totals
Positive	50	30	80
Negative	70	60	130
Totals	120	90	210

Step 2: Compute the required probabilities

The way the first question is phrased, we need to determine the probability that a person responds positively if she is female. This means that we do not include males in this calculation. So, the probability that the medicine gives a positive result for females is the ratio between the number of females who got a positive response and the total number of females.

$$\begin{aligned}P(\text{positive if female}) &= \frac{n(\text{positive and female})}{n(\text{female})} \\&= \frac{50}{120} \\&= \frac{5}{12}\end{aligned}$$

Similarly, the probability that the medicine gives a negative result for males is:

$$\begin{aligned}P(\text{negative if male}) &= \frac{n(\text{negative and male})}{n(\text{male})} \\&= \frac{60}{90} \\&= \frac{2}{3}\end{aligned}$$

Step 3: Independence

We need to determine whether the effect of the medicine and the gender of a participant are dependent or independent. According to the definition, two events are independent if and only if

$$P(A \text{ and } B) = P(A) \times P(B)$$

We will look at the events that a participant is female and that the participant responded positively to the trial.

$$\begin{aligned}P(\text{female}) &= \frac{n(\text{female})}{n(\text{total trials})} \\&= \frac{120}{210} \\&= \frac{4}{7}\end{aligned}$$

$$\begin{aligned}
 P(\text{positive}) &= \frac{n(\text{positive})}{n(\text{total trials})} \\
 &= \frac{80}{210} \\
 &= \frac{8}{21}
 \end{aligned}$$

$$\begin{aligned}
 P(\text{female and positive}) &= \frac{n(\text{female and positive})}{n(\text{total trials})} \\
 &= \frac{50}{210} \\
 &= \frac{5}{21}
 \end{aligned}$$

From these probabilities we can see that

$$P(\text{female and positive}) \neq P(\text{female}) \times P(\text{positive})$$

and therefore the gender of a participant and the outcome of a trial are dependent events.

Worked example 14: Contingency tables

QUESTION

Use the contingency table below to answer the following questions.

	Grade 11	Grade 12	Totals
Has cellphone	59	50	109
No cellphone	6	3	9
Totals	65	53	118

1. What is the probability that a learner from Grade 11 has a cellphone?
2. What is the probability that a learner who does not have a cellphone is from Grade 11.
3. Are the grade of a learner and whether he has a cellphone or not independent events? Explain your answer.

SOLUTION

1. There are 65 learners in Grade 11 and 59 of them have a cellphone. Therefore the probability that a learner from Grade 11 has a cellphone is $\frac{59}{65}$.
2. There are 9 learners who do not have a cellphone and 6 of them are in Grade 11. Therefore the probability that a learner who does not have a cellphone is from from Grade 11 is $\frac{6}{9} = \frac{2}{3}$.

3. To test for independence, we will consider whether a learner is in Grade 11 and whether a learner has a cellphone. The probability that a learner is in Grade 11 is $\frac{65}{118}$. The probability that a learner has a cellphone is $\frac{109}{118}$. The probability that a learner is in Grade 11 and has a cellphone is $\frac{59}{118} = \frac{1}{2}$. Since $\frac{1}{2} \neq \frac{65}{118} \times \frac{109}{118}$ the grade of a learner and whether he has a cellphone are dependent.

Exercise 10 – 6: Contingency tables

1. Use the contingency table below to answer the following questions.

	Brown eyes	Not brown eyes	Totals
Black hair	50	30	80
Red hair	70	80	150
Totals	120	110	230

- What is the probability that someone with black hair has brown eyes?
 - What is the probability that someone has black hair?
 - What is the probability that someone has brown eyes?
 - Are having black hair and having brown eyes dependent or independent events?
2. Given the following contingency table, identify the events and determine whether they are dependent or independent.

	Location A	Location B	Totals
Buses left late	15	40	55
Buses left on time	25	20	45
Totals	40	60	100

3. You are given the following information.

- Events A and B are independent.
- $P(\text{not } A) = 0,3$.
- $P(B) = 0,4$.

Complete the contingency table below.

	A	not A	Totals
B			
not B			
Totals			50

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1. 23BQ 2. 23BR 3. 23BS



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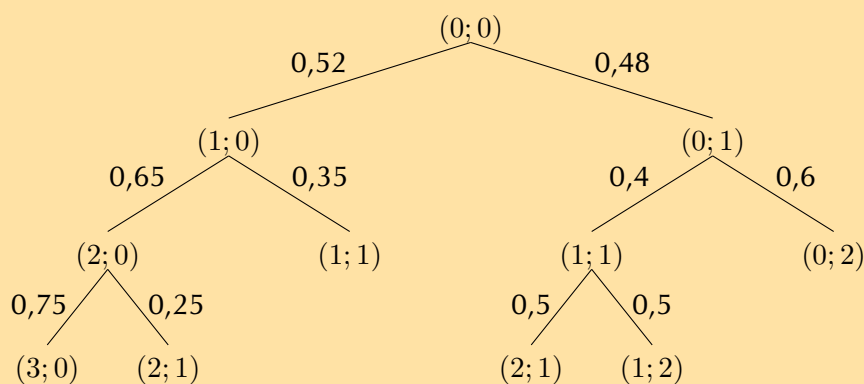
▶ See presentation: 23BT at www.everythingmaths.co.za

- Terminology:
 - **Outcome:** a single observation of an experiment.
 - **Sample space** of an experiment: the set of all possible outcomes of the experiment.
 - **Event:** a set of outcomes of an experiment.
 - **Probability** of an event: a real number between 0 and 1 that describes how likely it is that the event will occur.
 - **Relative frequency** of an event: the number of times that the event occurs during experimental trials, divided by the total number of trials conducted.
 - **Union** of events: the set of all outcomes that occur in at least one of the events, written as “ A or B ”.
 - **Intersection** of events: the set of all outcomes that occur in all of the events, written as “ A and B ”.
 - **Mutually exclusive events:** events with no outcomes in common, that is $(A \text{ and } B) = \emptyset$.
 - **Complementary events:** two mutually exclusive events that together contain all the outcomes in the sample space. We write the complement as “not A ”.
 - **Independent events:** two events where knowing the outcome of one event does not affect the probability of the other event. Events are independent if and only if $P(A \text{ and } B) = P(A) \times P(B)$.
- Identities:
 - The **addition rule:** $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
 - The **addition rule for 2 mutually exclusive events:** $P(A \text{ or } B) = P(A) + P(B)$
 - The **complementary rule:** $P(\text{not } A) = 1 - P(A)$
- A **Venn diagram** is a visual tool used to show how events overlap. Each region in a Venn diagram represents an event and could contain either the outcomes in the event, the number of outcomes in the event or the probability of the event.
- A **tree diagram** is a visual tool that helps with computing probabilities for dependent events. The outcomes of each event are shown along with the probability of each outcome. For each event that depends on a previous event, we go one level deeper into the tree. To compute the probability of some combination of outcomes, we
 - find all the paths that contain the outcome of interest;
 - multiply the probabilities along each path;
 - add the probabilities between different paths.
- A **2-way contingency table** is a tool for organising data, especially when we want to determine whether two events, each with only two outcomes, are dependent or independent. The counts for each possible combination of outcomes are entered into the table, along with the totals of each row and column.

Exercise 10 – 7: End of chapter exercises

1. Jane invested in the stock market. The probability that she will not lose all her money is 0,32. What is the probability that she will lose all her money? Explain.
2. If D and F are mutually exclusive events, with $P(\text{not } D) = 0,3$ and $P(D \text{ or } F) = 0,94$, find $P(F)$.
3. A car sales person has pink, lime-green and purple models of car A and purple, orange and multicolour models of car B . One dark night a thief steals a car.
 - a) What is the experiment and sample space?
 - b) What is the probability of stealing either a model of A or a model of B ?
 - c) What is the probability of stealing both a model of A and a model of B ?
4. The probability of event X is 0,43 and the probability of event Y is 0,24. The probability of both occurring together is 0,10. What is the probability that X or Y will occur?
5. $P(H) = 0,62$; $P(J) = 0,39$ and $P(H \text{ and } J) = 0,31$. Calculate:
 - a) $P(H')$
 - b) $P(H \text{ or } J)$
 - c) $P(H' \text{ or } J')$
 - d) $P(H' \text{ or } J)$
 - e) $P(H' \text{ and } J')$
6. The last ten letters of the alphabet are placed in a hat and people are asked to pick one of them. Event D is picking a vowel, event E is picking a consonant and event F is picking one of the last four letters. Draw a Venn diagram showing the outcomes in the sample space and the different events. Then calculate the following probabilities:
 - a) $P(\text{not } F)$
 - b) $P(F \text{ or } D)$
 - c) $P(\text{neither } E \text{ nor } F)$
 - d) $P(D \text{ and } E)$
 - e) $P(E \text{ and } F)$
 - f) $P(E \text{ and } D')$
7. Thobeka compares three neighbourhoods (we'll call them A , B and C) to see where the best place is to live. She interviews 80 people and asks them whether they like each of the neighbourhoods, or not.
 - 40 people like neighbourhood A .
 - 35 people like neighbourhood B .
 - 40 people like neighbourhood C .
 - 21 people like both neighbourhoods A and C .
 - 18 people like both neighbourhoods B and C .
 - 68 people like at least one neighbourhood.
 - 7 people like all three neighbourhoods.

- Use this information to draw a Venn diagram.
 - How many people like none of the neighbourhoods?
 - How many people like neighbourhoods A and B , but not C ?
 - What is the probability that a randomly chosen person from the survey likes at least one of the neighbourhoods?
8. Let G and H be two events in a sample space. Suppose that $P(G) = 0,4$; $P(H) = h$; and $P(G \text{ or } H) = 0,7$.
- For what value of h are G and H mutually exclusive?
 - For what value of h are G and H independent?
9. The following tree diagram represents points scored by two teams in a soccer game. At each level in the tree, the points are shown as (points for Team 1; points for Team 2).



- Use this diagram to determine the probability that:
- Team 1 will win
 - The game will be a draw
 - The game will end with an even number of total points
10. A bag contains 10 orange balls and 7 black balls. You draw 3 balls from the bag **without replacement**. What is the probability that you will end up with exactly 2 orange balls? Represent this experiment using a tree diagram.
11. Complete the following contingency table and determine whether the events are dependent or independent.

	Durban	Bloemfontein	Totals
Liked living there	130	30	
Did not like living there	140		340
Totals		230	500

12. Summarise the following information about a medical trial with 2 types of multivitamin in a contingency table and determine whether the events are dependent or independent.
- 960 people took part in the medical trial.
 - 540 people used multivitamin A for a month and 400 of those people showed an improvement in their health.

- 300 people showed an improvement in health when using multivitamin B for a month.

If the events are independent, it means that the two multivitamins have the same effect on people. If the events are dependent, it means that one multivitamin is better than the other. Which multivitamin is better than the other, or are the both equally effective?

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1. [23BV](#) 2. [23BW](#) 3. [23BX](#) 4. [23BY](#) 5. [23BZ](#) 6. [23C2](#)
7. [23C3](#) 8. [23C4](#) 9. [23C5](#) 10. [23C6](#) 11. [23C7](#) 12. [23C8](#)



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Statistics

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11.1 Revision

EMBJZ

Measures of central tendency

EMBK2

The mean and median of a data set both give an indication where the centre of the data distribution is located. The **mean**, or average, is calculated as

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

where the x_i are the data and n is the number of data. We read $\bar{x}$ as “x bar”.

The **median** is the middle value of an ordered data set. To find the median, we first sort the data and then pick out the value in the middle of the sorted list. If the middle is in between two values, the median is the average of those two values.

► See video: 23C9 at www.everythingmaths.co.za

Worked example 1: Computing measures of central tendency

QUESTION

Compute the mean and median of the following data set:

72,5 ; 92,6 ; 15,6 ; 53,0 ; 86,4 ; 89,9 ; 90,9 ; 21,7 ; 46,0 ; 4,1 ; 51,7 ; 2,2

SOLUTION**Step 1: Compute the mean**

Using the formula for the mean, we first compute the sum of the values and then divide by the number of values.

$$\begin{aligned}\bar{x} &= \frac{626,6}{12} \\ &\approx 52,22\end{aligned}$$

Step 2: Compute the median

To find the median, we first have to sort the data:

2,2 ; 4,1 ; 15,6 ; 21,7 ; 46,0 ; 51,7 ; 53,0 ; 72,5 ; 86,4 ; 89,9 ; 90,9 ; 92,6

Since there are an even number of values, the median will lie between two values. In this case, the two values in the middle are 51,7 and 53,0. Therefore the median is 52,35.

Measures of dispersion tell us how spread out a data set is. If a measure of dispersion is small, the data are clustered in a small region. If a measure of dispersion is large, the data are spread out over a large region.

The **range** is the difference between the maximum and minimum values in the data set.

The **inter-quartile range** is the difference between the first and third quartiles of the data set. The quartiles are computed in a similar way to the median. The median is halfway into the ordered data set and is sometimes also called the second quartile. The first quartile is one quarter of the way into the ordered data set; whereas the third quartile is three quarters of the way into the ordered data set.

▶ See video: 23CB at www.everythingmaths.co.za

Worked example 2: Range and inter-quartile range

QUESTION

Determine the range and the inter-quartile range of the following data set.

14 ; 17 ; 45 ; 20 ; 19 ; 36 ; 7 ; 30 ; 8

SOLUTION

Step 1: Sort the values in the data set

To determine the range we need to find the minimum and maximum values in the data set. To determine the inter-quartile range we need to compute the first and third quartiles of the data set. For both of these requirements, it is easier to order the data set first.

The sorted data set is

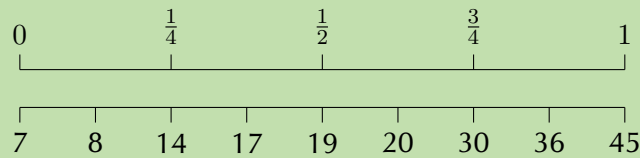
7 ; 8 ; 14 ; 17 ; 19 ; 20 ; 30 ; 36 ; 45

Step 2: Find the minimum, maximum and range

The minimum value is the first value in the ordered data set, namely 7. The maximum is the last value in the ordered data set, namely 45. The range is the difference between the minimum and maximum: $45 - 7 = 38$.

Step 3: Find the quartiles and inter-quartile range

The diagram below shows how we find the quartiles one quarter, one half and three quarters of the way into the ordered list of values.



From this diagram we can see that the first quartile is at a value of 14, the second quartile (median) is at a value of 19 and the third quartile is at a value of 30.

The inter-quartile range is the difference between the first and third quartiles. The first quartile is 14 and the third quartile is 30. Therefore the inter-quartile range is $30 - 14 = 16$.

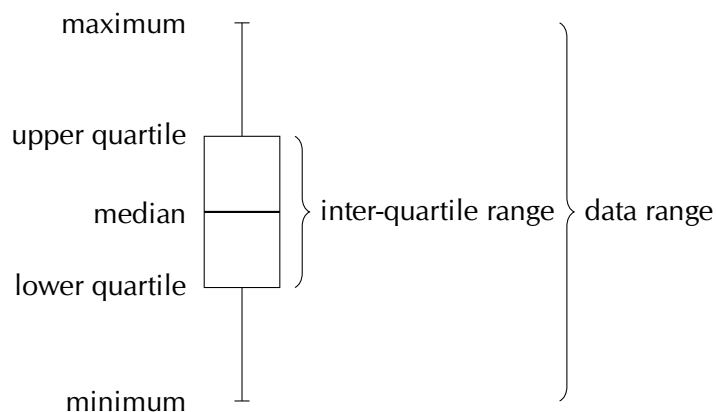
Five number summary

EMBK4

The **five number summary** combines a measure of central tendency, namely the median, with measures of dispersion, namely the range and the inter-quartile range. This gives a good overview of the overall data distribution. More precisely, the five number summary is written in the following order:

- minimum;
- first quartile;
- median;
- third quartile;
- maximum.

The five number summary is often presented visually using a **box and whisker diagram**. A box and whisker diagram is shown below, with the positions of the five relevant numbers labelled. Note that this diagram is drawn vertically, but that it may also be drawn horizontally.



▶ See video: 23CC at www.everythingmaths.co.za

Worked example 3: Five number summary

QUESTION

Draw a box and whisker diagram for the following data set:

1,25 ; 1,5 ; 2,5 ; 2,5 ; 3,1 ; 3,2 ; 4,1 ; 4,25 ; 4,75 ; 4,8 ; 4,95 ; 5,1

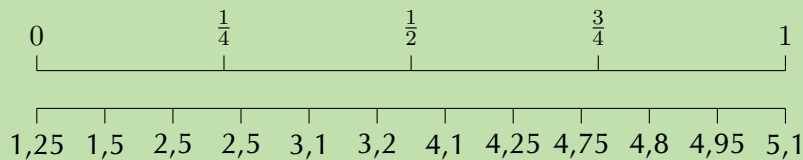
SOLUTION

Step 1: Determine the minimum and maximum

Since the data set is already ordered, we can read off the minimum as the first value (1,25) and the maximum as the last value (5,1).

Step 2: Determine the quartiles

There are 12 values in the data set.



Using the figure above we can see that the median is between the sixth and seventh values, making it.

$$\frac{3,2 + 4,1}{2} = 3,65$$

The first quartile lies between the third and fourth values, making it

$$Q_1 = \frac{2,5 + 2,5}{2} = 2,5$$

The third quartile lies between the ninth and tenth values, making it

$$Q_3 = \frac{4,75 + 4,8}{2} = 4,775$$

Step 3: Draw the box and whisker diagram

We now have the five number summary as (1,25; 2,5; 3,65; 4,775; 5,1). The box and whisker diagram representing the five number summary is given below.

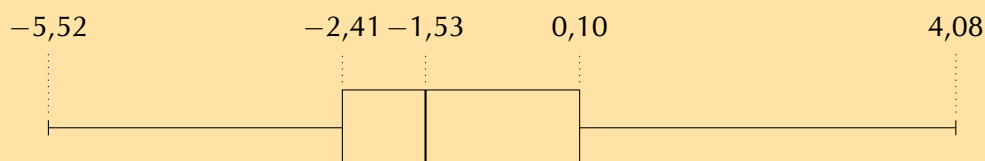


Exercise 11 – 1: Revision

1. For each of the following data sets, compute the mean and all the quartiles. Round your answers to one decimal place.

- a) $-3,4 ; -3,1 ; -6,1 ; -1,5 ; -7,8 ; -3,4 ; -2,7 ; -6,2$
b) $-6 ; -99 ; 90 ; 81 ; 13 ; -85 ; -60 ; 65 ; -49$
c) $7 ; 45 ; 11 ; 3 ; 9 ; 35 ; 31 ; 7 ; 16 ; 40 ; 12 ; 6$

2. Use the following box and whisker diagram to determine the range and inter-quartile range of the data.



3. Draw the box and whisker diagram for the following data.

$0,2 ; -0,2 ; -2,7 ; 2,9 ; -0,2 ; -4,2 ; -1,8 ; 0,4 ; -1,7 ; -2,5 ; 2,7 ; 0,8 ; -0,5$

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1a. [23CD](#) 1b. [23CF](#) 1c. [23CG](#) 2. [23CH](#) 3. [23CJ](#)



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11.2 Histograms

EMBK5

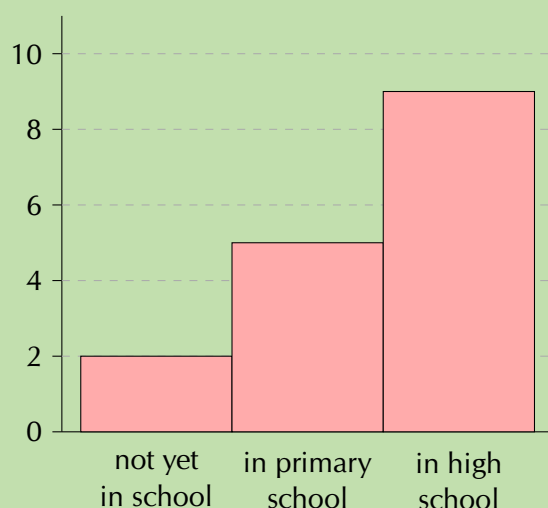
A histogram is a graphical representation of how many times different, mutually exclusive events are observed in an experiment. To interpret a histogram, we find the events on the x -axis and the counts on the y -axis. Each event has a rectangle that shows what its count (or frequency) is.

▶ See video: [23CK](#) at www.everythingmaths.co.za

Worked example 4: Reading histograms

QUESTION

Use the following histogram to determine the events that were recorded and the relative frequency of each event. Summarise your answer in a table.



SOLUTION

Step 1: Determine the events

The events are shown on the x -axis. In this example we have “not yet in school”, “in primary school” and “in high school”.

Step 2: Read off the count for each event

The counts are shown on the y -axis and the height of each rectangle shows the frequency for each event.

- not yet in school: 2
- in primary school: 5
- in high school: 9

Step 3: Calculate relative frequency

The relative frequency of an event in an experiment is the number of times that the event occurred divided by the total number of times that the experiment was completed. In this example we add up the frequencies for all the events to get a total frequency of 16. Therefore the relative frequencies are:

- not yet in school: $\frac{2}{16} = \frac{1}{8}$
- in primary school: $\frac{5}{16}$
- in high school: $\frac{9}{16}$

Step 4: Summarise

Event	Count	Relative frequency
not yet in school	2	$\frac{1}{8}$
in primary school	5	$\frac{5}{16}$
in high school	9	$\frac{9}{16}$

To draw a histogram of a data set containing numbers, the numbers first have to be grouped. Each group is defined by an interval. We then count how many times numbers from each group appear in the data set and draw a histogram using the counts.

Worked example 5: Draw a histogram

QUESTION

The following data represent the heights of 16 adults in centimetres.

162 ; 168 ; 177 ; 147 ; 189 ; 171 ; 173 ; 168
178 ; 184 ; 165 ; 173 ; 179 ; 166 ; 168 ; 165

Divide the data into 5 equal length intervals between 140 cm and 190 cm and draw a histogram.

SOLUTION

Step 1: Determine intervals

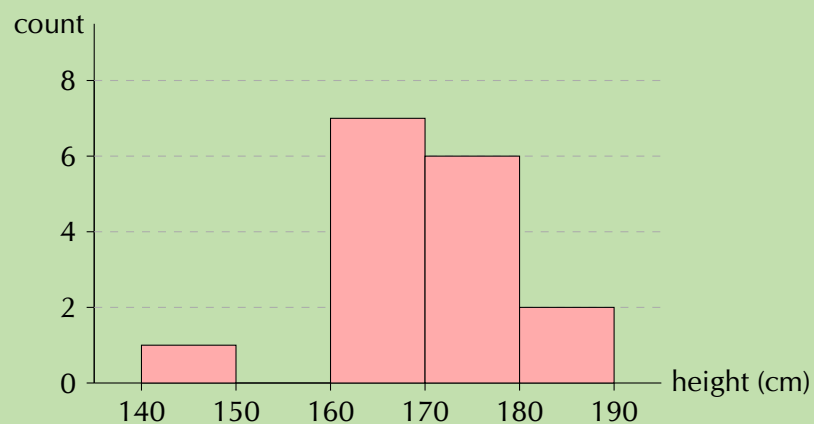
To have 5 intervals of the same length between 140 and 190, we need an interval length of 10. Therefore the intervals are (140; 150]; (150; 160]; (160; 170]; (170; 180]; and (180; 190].

Step 2: Count data

The following table summarises the number of data values in each of the intervals.

Interval	(140; 150]	(150; 160]	(160; 170]	(170; 180]	(180; 190]
Count	1	0	7	6	2

Step 3: Draw the histogram



A frequency polygon is sometimes used to represent the same information as in a histogram. A frequency polygon is drawn by using line segments to connect the middle of the top of each bar in the histogram. This means that the frequency polygon connects the coordinates at the centre of each interval and the count in each interval.

Worked example 6: Drawing a frequency polygon

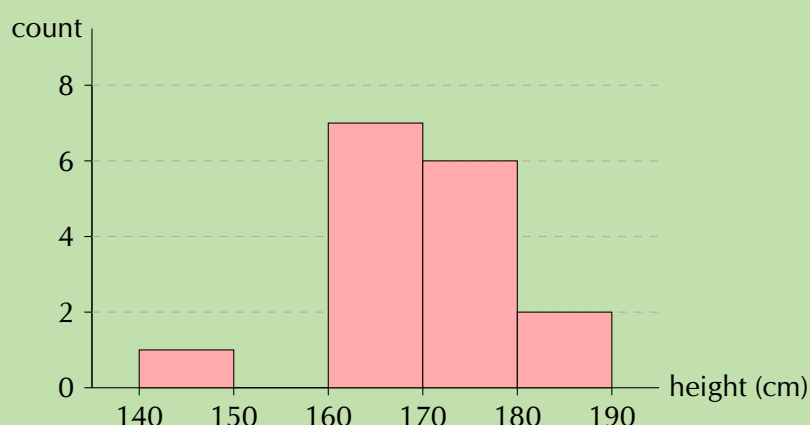
QUESTION

Use the histogram from the previous example to draw a frequency polygon of the same data.

SOLUTION

Step 1: Draw the histogram

We already know that the histogram looks like this:



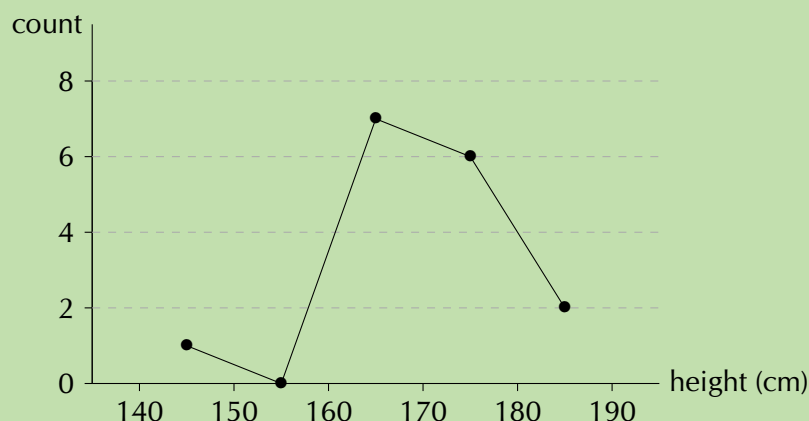
Step 2: Connect the tops of the rectangles

When we draw line segments between the tops of the rectangles in the histogram, we get the following picture:



Step 3: Draw final frequency polygon

Finally, we remove the histogram to show only the frequency polygon.



Frequency polygons are particularly useful for comparing two data sets. Comparing two histograms would be more difficult since we would have to draw the rectangles of the two data sets on top of each other. Because frequency polygons are just lines, they do not pose the same problem.

Worked example 7: Drawing frequency polygons

QUESTION

Here is another data set of heights, this time of Grade 11 learners.

132 ; 132 ; 156 ; 147 ; 162 ; 168 ; 152 ; 174
141 ; 136 ; 161 ; 148 ; 140 ; 174 ; 174 ; 162

Draw the frequency polygon for this data set using the same interval length as in the previous example. Then compare the two frequency polygons on one graph to see the differences between the distributions.

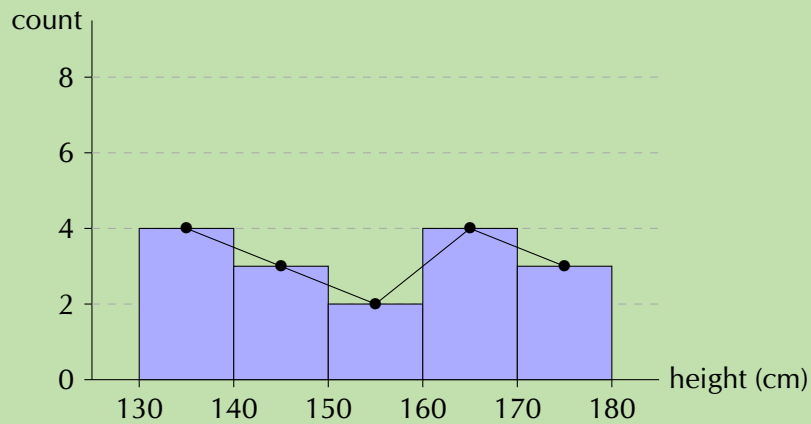
SOLUTION

Step 1: Frequency table

We first create the table of counts for the new data set.

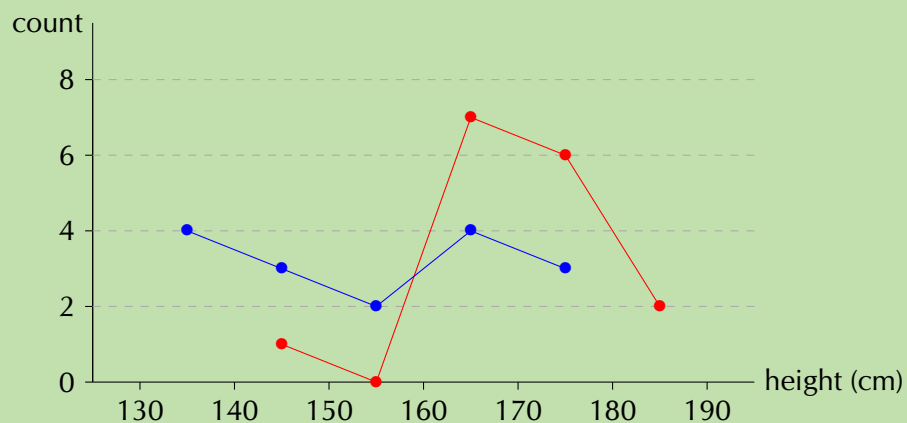
Interval	(130; 140]	(140; 150]	(150; 160]	(160; 170]	(170; 180]
Count	4	3	2	4	3

Step 2: Draw histogram and frequency polygon



Step 3: Compare frequency polygons

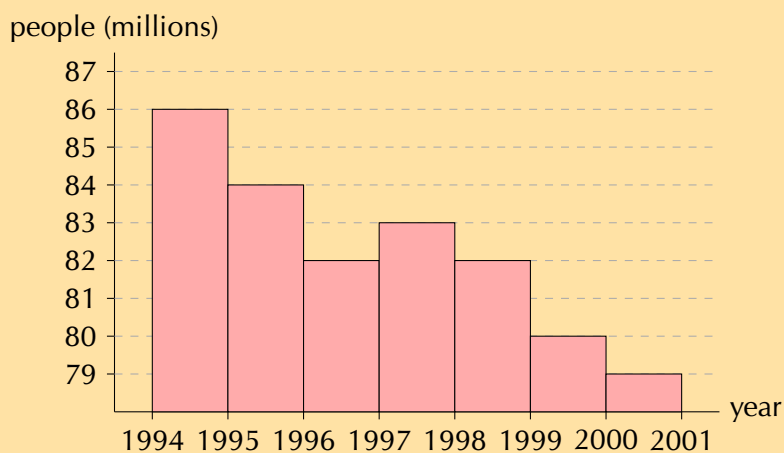
We draw the two frequency polygons on the same axes. The red line indicates the distribution over heights for adults and the blue line, for Grade 11 learners.



From this plot we can easily see that the heights for Grade 11 learners are distributed more towards the left (shorter) than adults. The learner heights also seem to be more evenly distributed between 130 and 180 cm, whereas the adult heights are mostly between 160 and 180 cm.

Exercise 11 – 2: Histograms

1. Use the histogram below to answer the following questions. The histogram shows the number of people born around the world each year. The ticks on the x -axis are located at the start of each year.



- How many people were born between the beginning of 1994 and the beginning of 1996?
 - Is the number people in the world population increasing or decreasing? (Ignore the rate at which people are dying for this question.)
 - How many more people were born in 1994 than in 1997?
2. In a traffic survey, a random sample of 50 motorists were asked the distance (d) they drove to work daily. The results of the survey are shown in the table below. Draw a histogram to represent the data.

d	$0 < d \leq 10$	$10 < d \leq 20$	$20 < d \leq 30$	$30 < d \leq 40$	$40 < d \leq 50$
f	9	19	15	5	4

3. Below is data for the prevalence of HIV in South Africa. HIV prevalence refers to the percentage of people between the ages of 15 and 49 who are infected with HIV.

year	2002	2003	2004	2005	2006	2007	2008	2009
prevalence (%)	17,7	18,0	18,1	18,1	18,1	18,0	17,9	17,9

Draw a frequency polygon of this data set.

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Cumulative histograms, also known as ogives, are graphs that can be used to determine how many data values lie above or below a particular value in a data set. The cumulative frequency is calculated from a frequency table, by adding each frequency to the total of the frequencies of all data values before it in the data set. The last value for the cumulative frequency will always be equal to the total number of data values, since all frequencies will already have been added to the previous total.

An ogive is drawn by

- plotting the beginning of the first interval at a y -value of zero;
- plotting the end of every interval at the y -value equal to the cumulative count for that interval; and
- connecting the points on the plot with straight lines.

In this way, the end of the final interval will always be at the total number of data since we will have added up across all intervals.

Worked example 8: Cumulative frequencies and ogives

QUESTION

Determine the cumulative frequencies of the following grouped data and complete the table below. Use the table to draw an ogive of the data.

Interval	Frequency	Cumulative frequency
$10 < n \leq 20$	5	
$20 < n \leq 30$	7	
$30 < n \leq 40$	12	
$40 < n \leq 50$	10	
$50 < n \leq 60$	6	

SOLUTION

Step 1: Compute cumulative frequencies

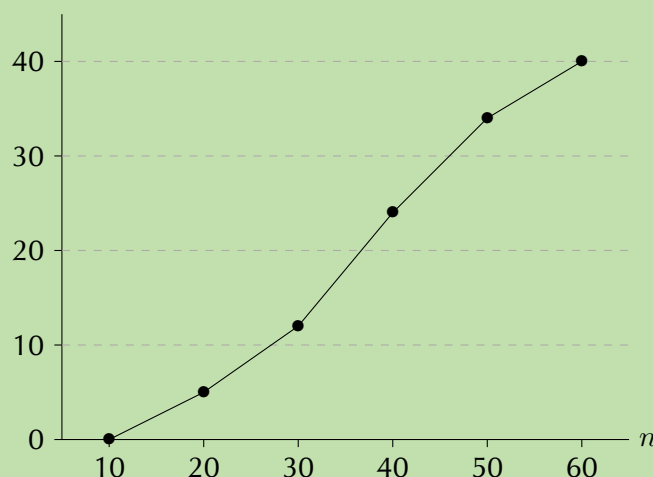
To determine the cumulative frequency, we add up the frequencies going down the table. The first cumulative frequency is just the same as the frequency, because we are adding it to zero. The final cumulative frequency is always equal to the sum of all the frequencies. This gives the following table:

Interval	Frequency	Cumulative frequency
$10 < n \leq 20$	5	5
$20 < n \leq 30$	7	12
$30 < n \leq 40$	12	24
$40 < n \leq 50$	10	34
$50 < n \leq 60$	6	40

Step 2: Plot the ogive

The first coordinate in the plot always starts at a y -value of 0 because we always start from a count of zero. So, the first coordinate is at $(10; 0)$ — at the beginning of the first interval. The second coordinate is at the end of the first interval (which is also the beginning of the second interval) and at the first cumulative count, so $(20; 5)$. The third coordinate is at the end of the second interval and at the second cumulative count, namely $(30; 12)$, and so on.

Computing all the coordinates and connecting them with straight lines gives the following ogive.



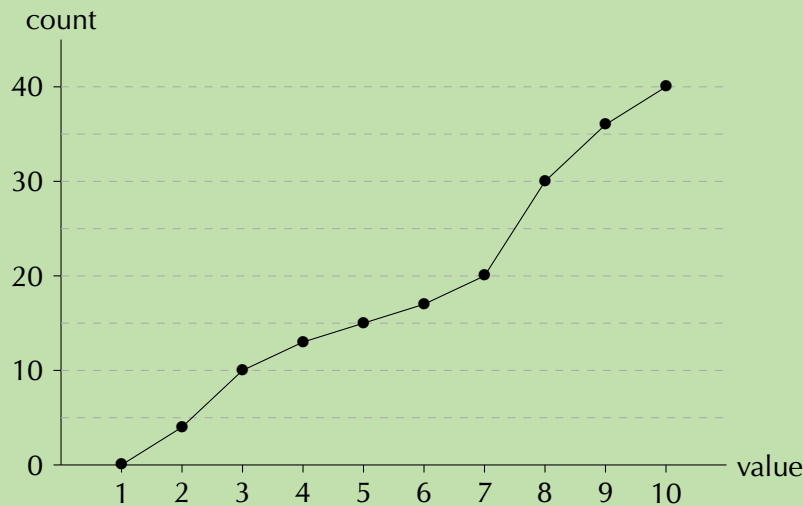
Ogives do look similar to frequency polygons, which we saw earlier. The most important difference between them is that an ogive is a plot of **cumulative** values, whereas a frequency polygon is a plot of the values themselves. So, to get from a frequency polygon to an ogive, we would add up the counts as we move from left to right in the graph.

Ogives are useful for determining the median, percentiles and five number summary of data. Remember that the median is simply the value in the middle when we order the data. A quartile is simply a quarter of the way from the beginning or the end of an ordered data set. With an ogive we already know how many data values are above or below a certain point, so it is easy to find the middle or a quarter of the data set.

Worked example 9: Ogives and the five number summary

QUESTION

Use the following ogive to compute the five number summary of the data. Remember that the five number summary consists of the minimum, all the quartiles (including the median) and the maximum.



SOLUTION

Step 1: Find the minimum and maximum

The minimum value in the data set is 1 since this is where the ogive starts on the horizontal axis. The maximum value in the data set is 10 since this is where the ogive stops on the horizontal axis.

Step 2: Find the quartiles

The quartiles are the values that are $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{3}{4}$ of the way into the ordered data set. Here the counts go up to 40, so we can find the quartiles by looking at the values corresponding to counts of 10, 20 and 30. On the ogive a count of

- 10 corresponds to a value of 3 (first quartile);
- 20 corresponds to a value of 7 (second quartile); and
- 30 corresponds to a value of 8 (third quartile).

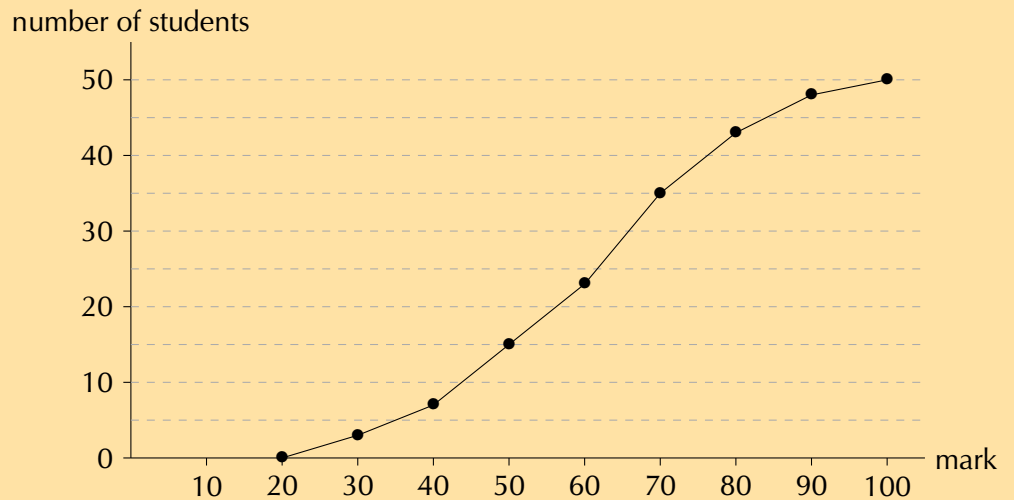
Step 3: Write down the five number summary

The five number summary is (1; 3; 7; 8; 10). The box-and-whisker plot of this data set is given below.

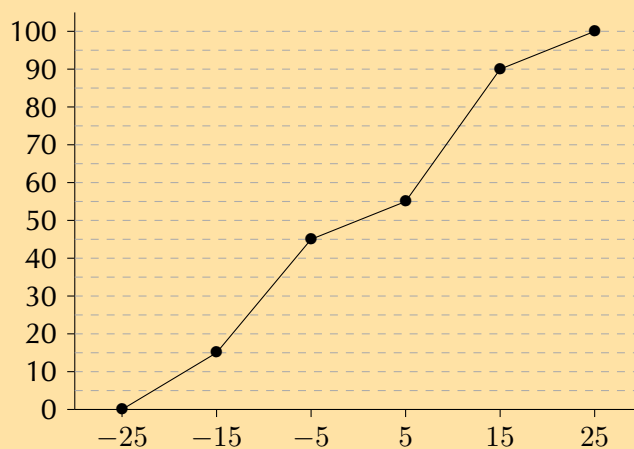


Exercise 11 – 3: Ogives

1. Use the ogive to answer the questions below. Note that marks are given as a percentage.



- How many students got between 50% and 70%?
 - How many students got at least 70%?
 - Compute the average mark for this class, rounded to the nearest integer.
2. Draw the histogram corresponding to this ogive.



3. The following data set lists the ages of 24 people.

2; 5; 1; 76; 34; 23; 65; 22; 63; 45; 53; 38

4; 28; 5; 73; 79; 17; 15; 5; 34; 37; 45; 56

Use the data to answer the following questions.

- Using an interval width of 8 construct a cumulative frequency plot.
- How many are below 30?
- How many are below 60?
- Giving an explanation state below what value the bottom 50% of the ages fall.
- Below what value do the bottom 40% fall?
- Construct a frequency polygon.

4. The weights of bags of sand in grams is given below (rounded to the nearest tenth):

50,1; 40,4; 48,5; 29,4; 50,2; 55,3; 58,1; 35,3; 54,2; 43,5

60,1; 43,9; 45,3; 49,2; 36,6; 31,5; 63,1; 49,3; 43,4; 54,1

- Decide on an interval width and state what you observe about your choice.
- Give your lowest interval.
- Give your highest interval.
- Construct a cumulative frequency graph and a frequency polygon.
- Below what value do 53% of the cases fall?
- Below what value of 60% of the cases fall?

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11.4 Variance and standard deviation

EMBK8

Measures of central tendency (mean, median and mode) provide information on the data values at the centre of the data set. Measures of dispersion (quartiles, percentiles, ranges) provide information on the spread of the data around the centre. In this section we will look at two more measures of dispersion called the **variance** and the **standard deviation**.

▶ See video: 23CV at www.everythingmaths.co.za

Variance

EMBK9

DEFINITION: Variance

Let a population consist of n elements, $\{x_1; x_2; \dots; x_n\}$. Write the mean of the data as $\bar{x}$.

The variance of the data is the average squared distance between the mean and each data value.

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

NOTE:

The variance is written as σ^2 . It might seem strange that it is written in squared form, but you will see why soon when we discuss the standard deviation.

The variance has the following properties.

- It is never negative since every term in the variance sum is squared and therefore either positive or zero.
- It has squared units. For example, the variance of a set of heights measured in centimetres will be given in centimeters squared. Since the population variance is squared, it is not directly comparable with the mean or the data themselves. In the next section we will describe a different measure of dispersion, the standard deviation, which has the same units as the data.

Worked example 10: Variance

QUESTION

You flip a coin 100 times and it lands on heads 44 times. You then use the same coin and do another 100 flips. This time it lands on heads 49 times. You repeat this experiment a total of 10 times and get the following results for the number of heads.

$$\{44; 49; 52; 62; 53; 48; 54; 49; 46; 51\}$$

Compute the mean and variance of this data set.

SOLUTION

Step 1: Compute the mean

The formula for the mean is

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

In this case, we sum the data and divide by 10 to get $\bar{x} = 50,8$.

Step 2: Compute the variance

The formula for the variance is

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

We first subtract the mean from each datum and then square the result.

x_i	44	49	52	62	53	48	54	49	46	51
$x_i - \bar{x}$	-6,8	-1,8	1,2	11,2	2,2	-2,8	3,2	-1,8	-4,8	0,2
$(x_i - \bar{x})^2$	46,24	3,24	1,44	125,44	4,84	7,84	10,24	3,24	23,04	0,04

The variance is the sum of the last row in this table divided by 10, so $\sigma^2 = 22,56$.

Since the variance is a squared quantity, it cannot be directly compared to the data values or the mean value of a data set. It is therefore more useful to have a quantity which is the square root of the variance. This quantity is known as the standard deviation.

DEFINITION: *Standard deviation*

Let a population consist of n elements, $\{x_1; x_2; \dots; x_n\}$, with a mean of $\bar{x}$. The standard deviation of the data is

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$$

In statistics, the standard deviation is a very common measure of dispersion. Standard deviation measures how spread out the values in a data set are around the mean. More precisely, it is a measure of the average distance between the values of the data in the set and the mean. If the data values are all similar, then the standard deviation will be low (closer to zero). If the data values are highly variable, then the standard deviation is high (further from zero).

The standard deviation is always a positive number and is always measured in the same units as the original data. For example, if the data are distance measurements in kilograms, the standard deviation will also be measured in kilograms.

The mean and the standard deviation of a set of data are usually reported together. In a certain sense, the standard deviation is a natural measure of dispersion if the centre of the data is taken as the mean.

Investigation: Tabulating results

It is often useful to set your data out in a table so that you can apply the formulae easily. Complete the table below to calculate the standard deviation of $\{57; 53; 58; 65; 48; 50; 66; 51\}$.

- Firstly, remember to calculate the mean, $\bar{x}$.
- Complete the following table.

index: i	datum: x_i	deviation: $x_i - \bar{x}$	deviation squared: $(x_i - \bar{x})^2$
1	57		
2	53		
3	58		
4	65		
5	48		
6	50		
7	66		
8	51		
	$\sum x_i = \dots$	$\sum (x_i - \bar{x}) = \dots$	$\sum (x_i - \bar{x})^2 = \dots$

- The sum of the deviations is always zero. Why is this? Find out.
- Calculate the variance using the completed table.
- Then calculate the standard deviation.

Worked example 11: Variance and standard deviation

QUESTION

What is the variance and standard deviation of the possibilities associated with rolling a fair die?

SOLUTION

Step 1: Determine all the possible outcomes

When rolling a fair die, the sample space consists of 6 outcomes. The data set is therefore $x = \{1; 2; 3; 4; 5; 6\}$ and $n = 6$.

Step 2: Calculate the mean

The mean is:

$$\begin{aligned}\bar{x} &= \frac{1}{6} (1 + 2 + 3 + 4 + 5 + 6) \\ &= 3,5\end{aligned}$$

Step 3: Calculate the variance

The variance is:

$$\begin{aligned}\sigma^2 &= \frac{\sum (x - \bar{x})^2}{n} \\ &= \frac{1}{6} (6,25 + 2,25 + 0,25 + 0,25 + 2,25 + 6,25) \\ &= 2,917\end{aligned}$$

Step 4: Calculate the standard deviation

The standard deviation is:

$$\begin{aligned}\sigma &= \sqrt{2,917} \\ &= 1,708\end{aligned}$$

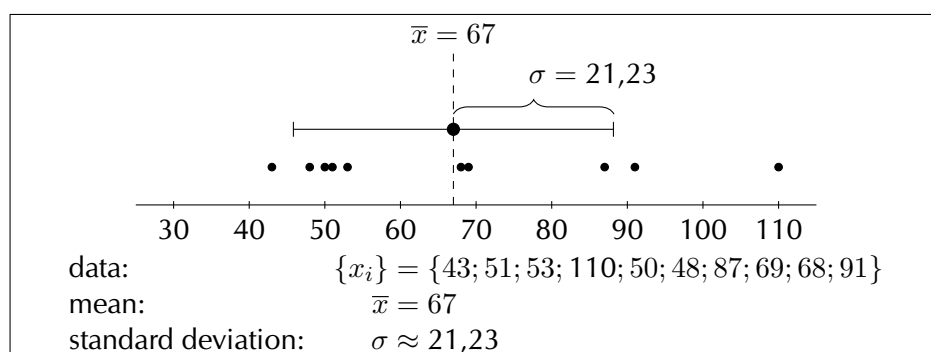
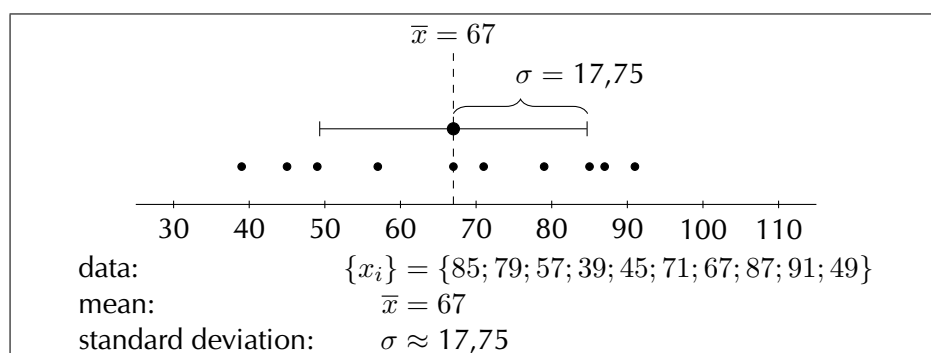
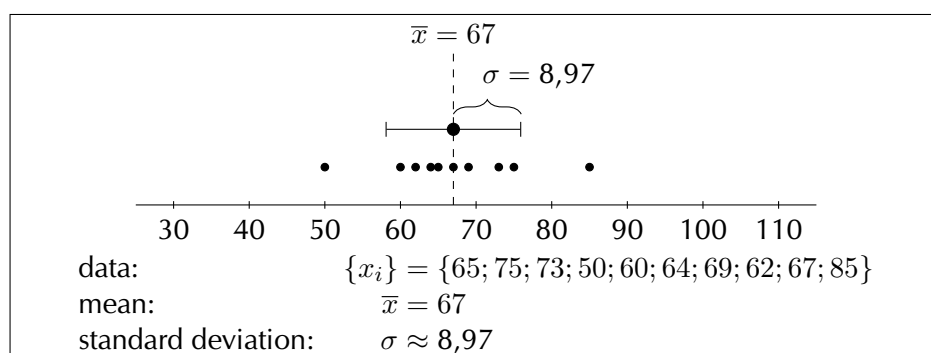
▶ See video: 23CW at www.everythingmaths.co.za

A large standard deviation indicates that the data values are far from the mean and a small standard deviation indicates that they are clustered closely around the mean.

For example, consider the following three data sets:

$$\begin{aligned} &\{65; 75; 73; 50; 60; 64; 69; 62; 67; 85\} \\ &\{85; 79; 57; 39; 45; 71; 67; 87; 91; 49\} \\ &\{43; 51; 53; 110; 50; 48; 87; 69; 68; 91\} \end{aligned}$$

Each of these data sets has the same mean, namely 67. However, they have different standard deviations, namely 8,97, 17,75 and 21,23. The following figures show plots of the data sets with the mean and standard deviation indicated on each. You can see how the standard deviation is larger when the data are more spread out.



The standard deviation may also be thought of as a measure of uncertainty. In the physical sciences, for example, the reported standard deviation of a group of repeated measurements represents the precision of those measurements. When deciding whether

measurements agree with a theoretical prediction, the standard deviation of those measurements is very important: if the mean of the measurements is too far away from the prediction (with the distance measured in standard deviations), then we consider the measurements as contradicting the prediction. This makes sense since they fall outside the range of values that could reasonably be expected to occur if the prediction were correct.

Exercise 11 – 4: Variance and standard deviation

1. Bridget surveyed the price of petrol at petrol stations in Cape Town and Durban. The data, in rands per litre, are given below.

Cape Town	3,96	3,76	4,00	3,91	3,69	3,72
Durban	3,97	3,81	3,52	4,08	3,88	3,68

- a) Find the mean price in each city and then state which city has the lowest mean.
 - b) Find the standard deviation of each city's prices.
 - c) Which city has the more consistently priced petrol? Give reasons for your answer.
2. Compute the mean and variance of the following set of values.
150 ; 300 ; 250 ; 270 ; 130 ; 80 ; 700 ; 500 ; 200 ; 220 ; 110 ; 320 ; 420 ; 140
 3. Compute the mean and variance of the following set of values.
−6,9 ; −17,3 ; 18,1 ; 1,5 ; 8,1 ; 9,6 ; −13,1 ; −14,0 ; 10,5 ; −14,8 ; −6,5 ; 1,4
 4. The times for 8 athletes who ran a 100 m sprint on the same track are shown below. All times are in seconds.
10,2 ; 10,8 ; 10,9 ; 10,3 ; 10,2 ; 10,4 ; 10,1 ; 10,4
 - a) Calculate the mean time.
 - b) Calculate the standard deviation for the data.
 - c) How many of the athletes' times are more than one standard deviation away from the mean?
 5. The following data set has a mean of 14,7 and a variance of 10,01.

18 ; 11 ; 12 ; a ; 16 ; 11 ; 19 ; 14 ; b ; 13

Compute the values of a and b .

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1. 23CX 2. 23CY 3. 23CZ 4. 23D2 5. 23D3



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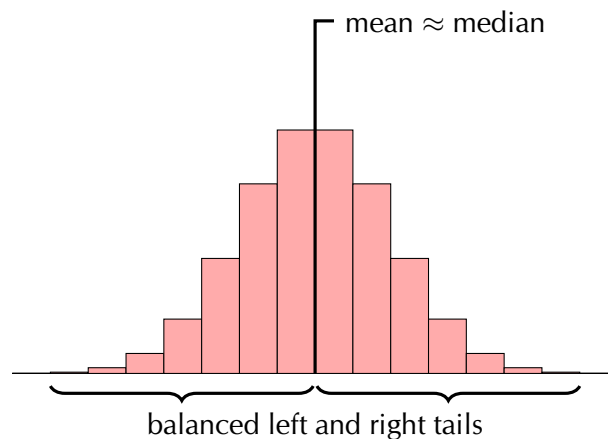
We are now going to classify data sets into 3 categories that describe the shape of the data distribution: symmetric, left skewed, right skewed. We can use this classification for any data set, but here we will look only at distributions with one peak. Most of the data distributions that you have seen so far have only one peak, so the plots in this section should look familiar.

Distributions with one peak are called **unimodal distributions**. Unimodal literally means having one mode. (Remember that a mode is a maximum in the distribution.)

Symmetric distributions

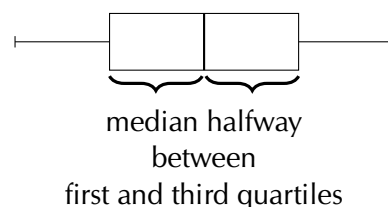
EMBK F

A symmetric distribution is one where the left and right hand sides of the distribution are roughly equally balanced around the mean. The histogram below shows a typical symmetric distribution.



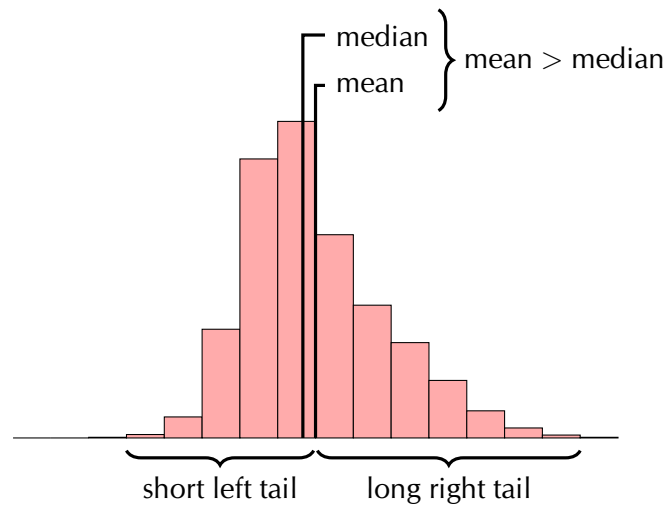
For symmetric distributions, the mean is approximately equal to the median. The **tails** of the distribution are the parts to the left and to the right, away from the mean. The tail is the part where the counts in the histogram become smaller. For a symmetric distribution, the left and right tails are equally balanced, meaning that they have about the same length.

The figure below shows the box and whisker diagram for a typical symmetric data set.

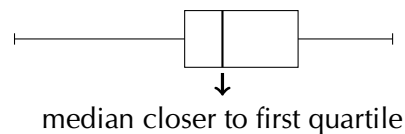


Another property of a symmetric distribution is that its median (second quartile) lies in the middle of its first and third quartiles. Note that the whiskers of the plot (the minimum and maximum) do not have to be equally far away from the median. In the next section on outliers, you will see that the minimum and maximum values do not necessarily match the rest of the data distribution well.

A distribution that is **skewed right** (also known as **positively skewed**) is shown below.



Now the picture is not symmetric around the mean anymore. For a right skewed distribution, the mean is typically greater than the median. Also notice that the tail of the distribution on the right hand (positive) side is longer than on the left hand side.

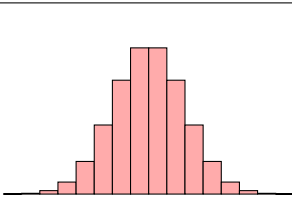
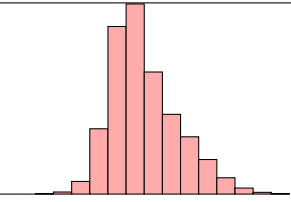
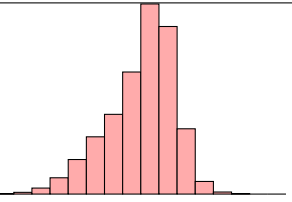
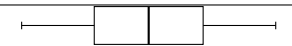


From the box and whisker diagram we can also see that the median is closer to the first quartile than the third quartile. The fact that the right hand side tail of the distribution is longer than the left can also be seen.

A distribution that is skewed left has exactly the opposite characteristics of one that is skewed right:

- the mean is typically less than the median;
- the tail of the distribution is longer on the left hand side than on the right hand side; and
- the median is closer to the third quartile than to the first quartile.

The table below summarises the different categories visually.

Symmetric	Skewed right (positive)	Skewed left (negative)
		
		

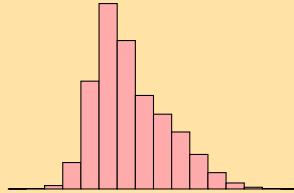
Exercise 11 – 5: Symmetric and skewed data

1. Is the following data set symmetric, skewed right or skewed left? Motivate your answer.

27 ; 28 ; 30 ; 32 ; 34 ; 38 ; 41 ; 42 ; 43 ; 44 ; 46 ; 53 ; 56 ; 62

2. State whether each of the following data sets are symmetric, skewed right or skewed left.

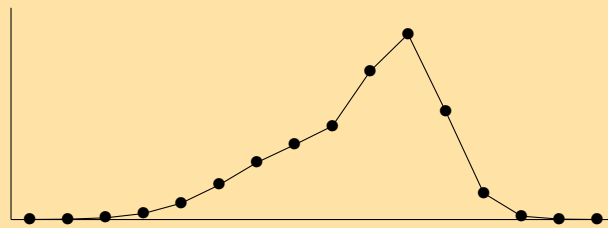
- a) A data set with this histogram:



- b) A data set with this box and whisker plot:



- c) A data set with this frequency polygon:



- d) The following data set:

11,2 ; 5 ; 9,4 ; 14,9 ; 4,4 ; 18,8 ; -0,4 ; 10,5 ; 8,3 ; 17,8

3. Two data sets have the same range and interquartile range, but one is skewed right and the other is skewed left. Sketch the box and whisker plot for each of these data sets. Then, invent data (6 points in each data set) that matches the descriptions of the two data sets.

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1. 23D4 2a. 23D5 2b. 23D6 2c. 23D7 2d. 23D8 3. 23D9



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An outlier in a data set is a value that is far away from the rest of the values in the data set. In a box and whisker diagram, outliers are usually close to the whiskers of the diagram. This is because the centre of the diagram represents the data between the first and third quartiles, which is where 50% of the data lie, while the whiskers represent the extremes — the minimum and maximum — of the data.

Worked example 12: Identifying outliers

QUESTION

Find the outliers in the following data set by drawing a box and whisker diagram and locating the data values on the diagram.

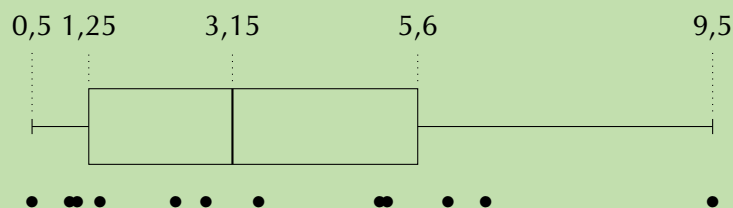
0,5 ; 1 ; 1,1 ; 1,4 ; 2,4 ; 2,8 ; 3,5 ; 5,1 ; 5,2 ; 6 ; 6,5 ; 9,5

SOLUTION

Step 1: Determine the five number summary

The minimum of the data set is 0,5. The maximum of the data set is 9,5. Since there are 12 values in the data set, the median lies between the sixth and seventh values, making it equal to $\frac{2,8+3,5}{2} = 3,15$. The first quartile lies between the third and fourth values, making it equal to $\frac{1,1+1,4}{2} = 1,25$. The third quartile lies between the ninth and tenth values, making it equal to $\frac{5,2+6}{2} = 5,6$.

Step 2: Draw the box and whisker diagram



In the figure above, each value in the data set is shown with a black dot.

Step 3: Find the outliers

From the diagram we can see that most of the values are between 1 and 6. The only value that is very far away from this range is the maximum at 9,5. Therefore 9,5 is the only outlier in the data set.

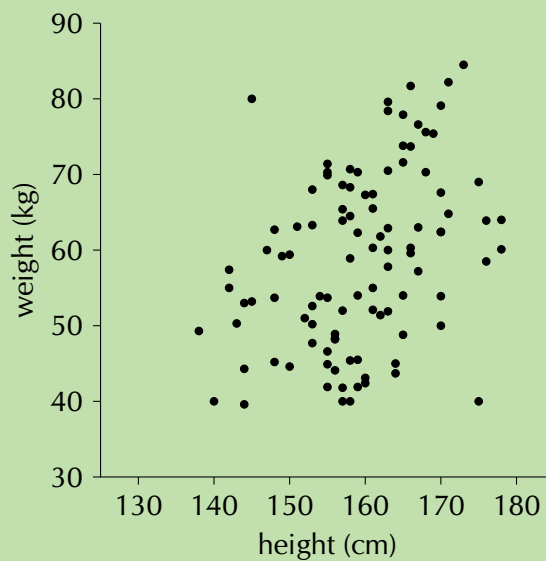
You should also be able to identify outliers in plots of two variables. A **scatter plot** is a graph that shows the relationship between two random variables. We call these data **bivariate** (literally meaning two variables) and we plot the data for two different variables on one set of axes. The following example shows what a typical scatter plot looks like. For Grade 11 you do not need to learn how to draw these 2-dimensional

scatter plots, but you should be able to identify outliers on them. As before, an outlier is a value that is far removed from the main distribution of data.

Worked example 13: Scatter plot

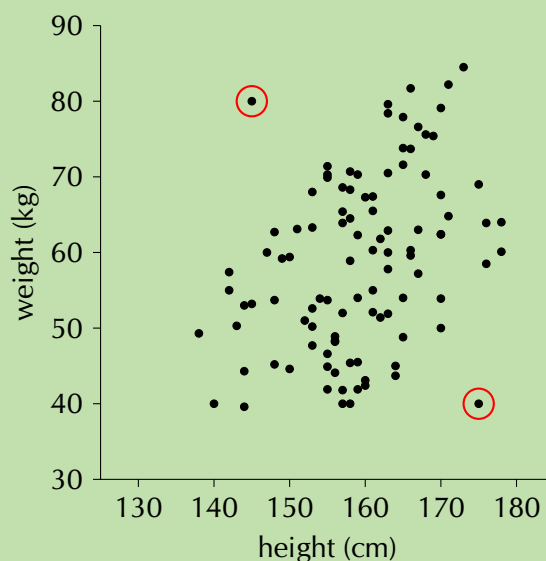
QUESTION

We have a data set that relates the heights and weights of a number of people. The height is the first variable and its value is plotted along the horizontal axis. The weight is the second variable and its value is plotted along the vertical axis. The data values are shown on the plot below. Identify any outliers on the scatter plot.



SOLUTION

We inspect the plot visually and notice that there are two points that lie far away from the main data distribution. These two points are circled in the plot below.



Exercise 11 – 6: Outliers

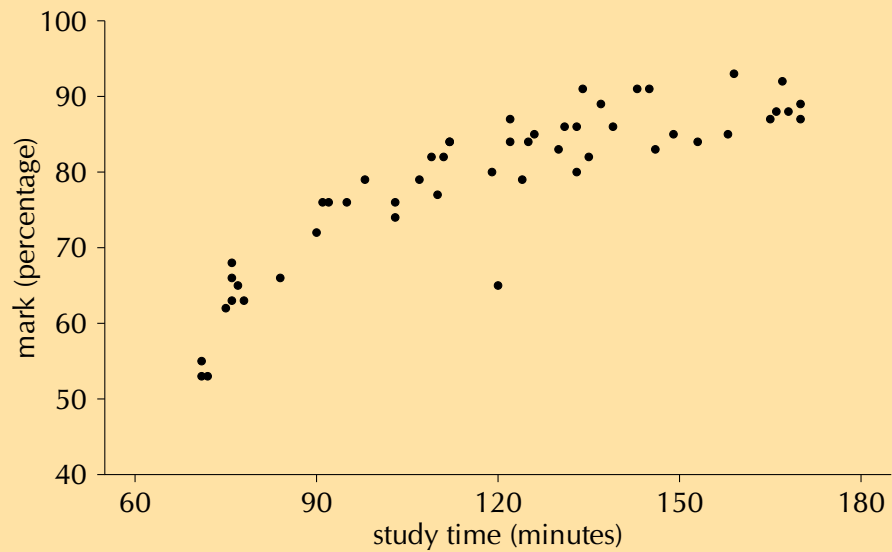
1. For each of the following data sets, draw a box and whisker diagram and determine whether there are any outliers in the data.

a) 30 ; 21,4 ; 39,4 ; 33,4 ; 21,1 ; 29,3 ; 32,8 ; 31,6 ; 36 ;
27,9 ; 27,3 ; 29,4 ; 29,1 ; 38,6 ; 33,8 ; 29,1 ; 37,1

b) 198 ; 166 ; 175 ; 147 ; 125 ; 194 ; 119 ; 170 ; 142 ; 148

c) 7,1 ; 9,6 ; 6,3 ; -5,9 ; 0,7 ; -0,1 ; 4,4 ; -11,7 ; 10 ; 2,3 ; -3,7 ; 5,8 ; -1,4
; 1,7 ; -0,7

2. A class's results for a test were recorded along with the amount of time spent studying for it. The results are given below. Identify any outliers in the data.



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1a. 23DB 1b. 23DC 1c. 23DD 2. 23DF



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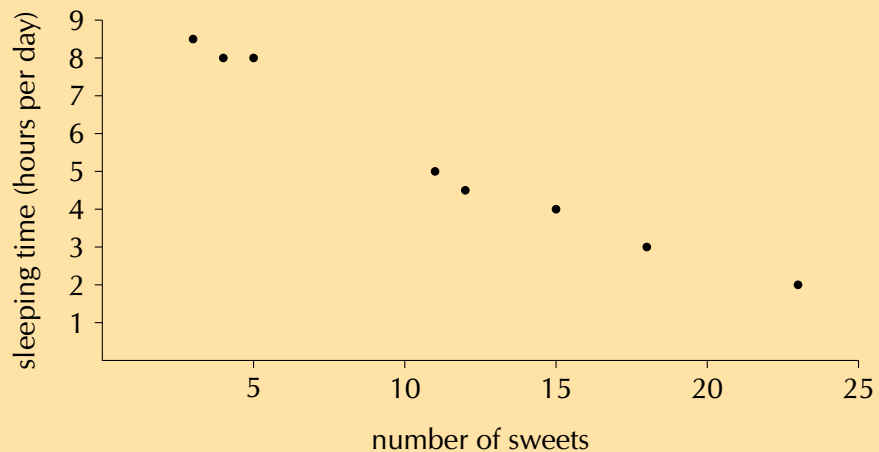
🔗 See presentation: 23DG at www.everythingmaths.co.za

- Histograms visualise how many times different events occurred. Each rectangle in a histogram represents one event and the height of the rectangle is relative to the number of times that the event occurred.
- Frequency polygons represent the same information as histograms, but using lines and points rather than rectangles. A frequency polygon connects the middle of the top edge of each rectangle in a histogram.
- Ogives (also known as cumulative histograms) show the total number of times that a value or anything less than that value appears in the data set. To draw an ogive you need to add up all the counts in a histogram from left to right.
 - The first count in an ogive is always zero.
 - The last count in an ogive is always the sum of all the counts in the data set.
- The variance and standard deviation are measures of dispersion.
 - The standard deviation is the square root of the variance.
 - Variance: $\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$
 - Standard deviation: $\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$
 - The standard deviation is measured in the same units as the mean and the data, but the variance is not. The variance is measured in the square of the data units.
- In a symmetric distribution
 - the mean is approximately equal to the median; and
 - the tails of the distribution are balanced.
- In a right (positively) skewed distribution
 - the mean is greater than the median;
 - the tail on the right hand side is longer than the tail on the left hand side; and
 - the median is closer to the first quartile than the third quartile.
- In a left (negatively) skewed distribution
 - the mean is less than the median;
 - the tail on the left hand side is longer than the tail on the right hand side; and
 - the median is closer to the third quartile than the first quartile.
- An outlier is a value that is far away from the rest of the data.

Exercise 11 – 7: End of chapter exercises

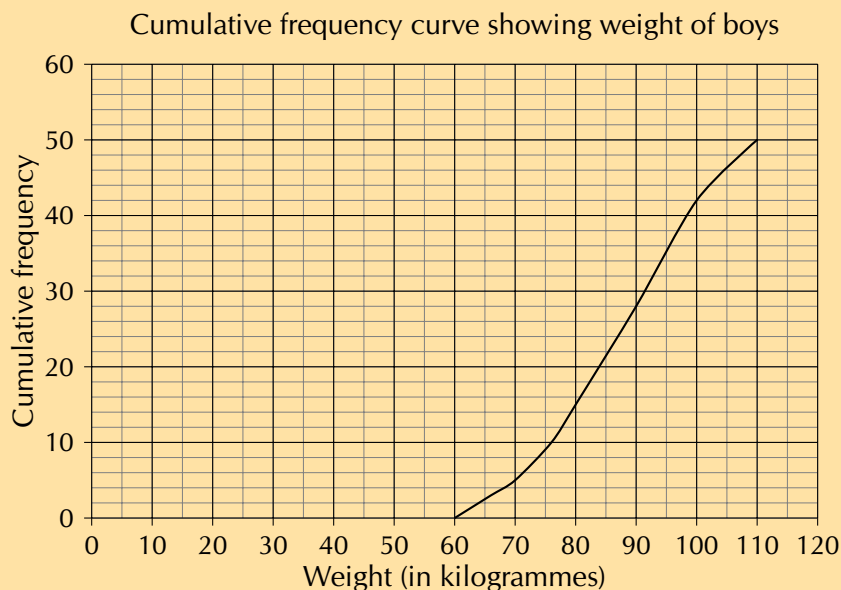
1. Draw a histogram, frequency polygon and ogive of the following data set. To count the data, use intervals with a width of 1, starting from 0.
 $0,4 ; 3,1 ; 1,1 ; 2,8 ; 1,5 ; 1,3 ; 2,8 ; 3,1 ; 1,8 ; 1,3 ;$
 $2,6 ; 3,7 ; 3,3 ; 5,7 ; 3,7 ; 7,4 ; 4,6 ; 2,4 ; 3,5 ; 5,3$
2. Draw a box and whisker diagram of the following data set and explain whether it is symmetric, skewed right or skewed left.
 $-4,1 ; -1,1 ; -1 ; -1,2 ; -1,5 ; -3,2 ; -4 ; -1,9 ; -4 ;$
 $-0,8 ; -3,3 ; -4,5 ; -2,5 ; -4,4 ; -4,6 ; -4,4 ; -3,3$
3. Eight children's sweet consumption and sleeping habits were recorded. The data are given in the following table and scatter plot.

Number of sweets per week	15	12	5	3	18	23	11	4
Average sleeping time (hours per day)	4	4,5	8	8,5	3	2	5	8



- a) What is the mean and standard deviation of the number of sweets eaten per day?
 - b) What is the mean and standard deviation of the number of hours slept per day?
 - c) Make a list of all the outliers in the data set.
4. The monthly incomes of eight teachers are as follows:
 $R\ 10\ 050 ; R\ 14\ 300 ; R\ 9800 ; R\ 15\ 000 ; R\ 12\ 140 ; R\ 13\ 800 ; R\ 11\ 990 ; R\ 12\ 900.$
 - a) What is the mean and standard deviation of their incomes?
 - b) How many of the salaries are less than one standard deviation away from the mean?
 - c) If each teacher gets a bonus of R 500 added to their pay what is the new mean and standard deviation?
 - d) If each teacher gets a bonus of 10% on their salary what is the new mean and standard deviation?
 - e) Determine for both of the above, how many salaries are less than one standard deviation away from the mean.

- f) Using the above information work out which bonus is more beneficial financially for the teachers.
5. The weights of a random sample of boys in Grade 11 were recorded. The cumulative frequency graph (ogive) below represents the recorded weights.



- a) How many of the boys weighed between 90 and 100 kilogrammes?
- b) Estimate the median weight of the boys.
- c) If there were 250 boys in Grade 11, estimate how many of them would weigh less than 80 kilogrammes?
6. Three sets of 12 learners each had their test scores recorded. The test was out of 50. Use the given data to answer the following questions.

Set A	Set B	Set C
25	32	43
47	34	47
15	35	16
17	32	43
16	25	38
26	16	44
24	38	42
27	47	50
22	43	50
24	29	44
12	18	43
31	25	42

- a) For each of the sets calculate the mean and the five number summary.
- b) Make box and whisker plots of the three data sets on the same set of axes.
- c) State, with reasons, whether each of the three data sets are symmetric or skewed (either right or left).

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1. 23DH 2. 23DJ 3. 23DK 4. 23DM 5. 23DN 6. 23DP



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Linear programming

12.1 *Introduction*

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12.1 Introduction

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In everyday life people are interested in knowing the most efficient way of carrying out a task or achieving a goal. For example, a farmer wants to know how many hectares to plant during a season in order to maximise the yield (produce), a stock broker wants to know how much to invest in stocks in order to maximise profit, an entrepreneur wants to know how many people to employ to minimise expenditure. These are optimisation problems; we want to determine either the maximum or the minimum in a specific situation.

To describe this mathematically, we assign variables to represent the different factors that influence the situation. Optimisation means finding the combination of variables that gives the best result.

▶ See video: 23DQ at www.everythingmaths.co.za

Worked example 1: Mountees and Roadees**QUESTION**

Investigate the following situation and use your knowledge of mathematics to solve the problem:

Mr. Hunter manufactures bicycles. He produces two different models of bicycles; strong mountain bikes called Mountees and fast road bikes called Roadees. He cannot produce more than 5 Mountees on a day and he can manufacture a maximum of 3 Roadees a day. He needs 1 technician to assemble a Mountee and 2 technicians to assemble a Roadee. The company has 8 technicians in the assembly department. The profit on a Mountee is R 800 and R 2400 on a Roadee. The demand is such that he can sell all the bikes he manufactures.

Determine the number of each model of bicycle that must be manufactured in order to make the maximum profit.

SOLUTION**Step 1: Assign variables**

In this situation there are two variables that we need to consider: let the number of Mountees produced be M and let the number of Roadees produced be R .

Step 2: Organise the information given

Write down a summary of the information given in the problem so that we consider

all the different components in the situation.

maximum number for M	$= 5$
maximum number for R	$= 3$
number of technicians needed for $M = 1$	
number of technicians needed for $R = 2$	
total number of technicians	$= 8$
profit per M	$= 800$
profit per R	$= 2400$

Step 3: Draw up a table

Use the summary to draw up a table of all the possible combinations of the number of Mountees and Roadees that can be manufactured per day:

M	R			
	0	1	2	3
0	(0; 0)	(0; 1)	(0; 2)	(0; 3)
1	(1; 0)	(1; 1)	(1; 2)	(1; 3)
2	(2; 0)	(2; 1)	(2; 2)	(2; 3)
3	(3; 0)	(3; 1)	(3; 2)	(3; 3)
4	(4; 0)	(4; 1)	(4; 2)	(4; 3)
5	(5; 0)	(5; 1)	(5; 2)	(5; 3)

Note that there are 24 possible combinations.

Step 4: Consider the limitation of the number of technicians

It takes 1 technician to assemble a Mountee and 2 technicians to assemble a Roadee. There are a total of 8 technicians in the assembly department, therefore we can write that $1(M) + 2(R) \leq 8$.

With this limitation, we are able to eliminate some of the combinations in the table where $M + 2R > 8$:

M	R			
	0	1	2	3
0	(0; 0)	(0; 1)	(0; 2)	(0; 3)
1	(1; 0)	(1; 1)	(1; 2)	(1; 3)
2	(2; 0)	(2; 1)	(2; 2)	(2; 3)
3	(3; 0)	(3; 1)	(3; 2)	(3; 3)
4	(4; 0)	(4; 1)	(4; 2)	(4; 3)
5	(5; 0)	(5; 1)	(5; 2)	(5; 3)

These combinations have been excluded as possible answers. For example, (5; 3) gives $5 + 2(3) = 11$ technicians.

Step 5: Consider the profit on the bicycles

We can express the profit (P) per day as: $P = 800(M) + 2400(R)$. Notice that a higher profit is made on a Roadee.

By substituting the different combinations for M and R , we can find the values that give the maximum profit:

$$\begin{aligned}\text{For } (5; 0) \quad P &= 800(5) + 2400(0) \\ &= \text{R } 4000\end{aligned}$$

$$\begin{aligned}\text{For } (3; 1) \quad P &= 800(3) + 2400(1) \\ &= \text{R } 4800\end{aligned}$$

M	R			
	0	1	2	3
0	$(0; 0) \Rightarrow \text{R } 0$	$(0; 1) \Rightarrow \text{R } 2400$	$(0; 2) \Rightarrow \text{R } 4800$	$(0; 3) \Rightarrow \text{R } 7200$
1	$(1; 0) \Rightarrow \text{R } 800$	$(1; 1) \Rightarrow \text{R } 3200$	$(1; 2) \Rightarrow \text{R } 5600$	$(1; 3) \Rightarrow \text{R } 8000$
2	$(2; 0) \Rightarrow \text{R } 1600$	$(2; 1) \Rightarrow \text{R } 4000$	$(2; 2) \Rightarrow \text{R } 6400$	$(2; 3) \Rightarrow \text{R } 8800$
3	$(3; 0) \Rightarrow \text{R } 2400$	$(3; 1) \Rightarrow \text{R } 4800$	$(3; 2) \Rightarrow \text{R } 7200$	$(3; 3)$
4	$(4; 0) \Rightarrow \text{R } 3200$	$(4; 1) \Rightarrow \text{R } 5600$	$(4; 2) \Rightarrow \text{R } 8000$	$(4; 3)$
5	$(5; 0) \Rightarrow \text{R } 4000$	$(5; 1) \Rightarrow \text{R } 6400$	$(5; 2)$	$(5; 3)$

Step 6: Write the final answer

Therefore the maximum profit of R 8800 is obtained if 2 Mountees and 3 Roadees are manufactured per day.

Exercise 12 – 1: Optimisation

1. Furniture store opening special:

As part of their opening special, a furniture store has promised to give away at least 40 prizes with a total value of at least R 4000. They intend to give away kettles and toasters. They decide there will be at least 10 units of each prize. A kettle costs the company R 120 and a toaster costs R 100.

Determine how many of each prize will represent the cheapest option for the company. Calculate how much this combination of kettles and toasters will cost.

Use a suitable strategy to organise the information and solve the problem.

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1. [23DR](#)



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Optimisation using graphs

A more efficient way to solve optimisation problems is using graphs.

We write the limitations in the situation, called constraints, as inequalities. Some constraints can be modelled by an equation, which needs to be maximised or minimized. We sketch the inequalities and indicate the region above or below the line that is to be considered in determining the solution. This method of solving optimisation problems is called linear programming.

▶ See video: 23DS at www.everythingmaths.co.za

Worked example 2: Optimisation using graphs

QUESTION

Consider again the example of Mr. Hunter who manufactures Mountees and Roadees:

Mr. Hunter manufactures bicycles. He produces two different models of bicycles; strong mountain bikes called Mountees and fast road bikes called Roadees. He cannot produce more than 5 Mountees on a day and he can manufacture a maximum of 3 Roadees a day. He needs 1 technician to assemble a Mountee and 2 technicians to assemble a Roadee. The company has 8 technicians in the assembly department. The profit on a Mountee is R 800 and R 2400 on a Roadee. The demand is such that he can sell all the bikes he manufactures.

Determine the number of each model of bicycle that must be manufactured in order to make a maximum profit.

SOLUTION

Step 1: Assign variables

In this situation there are two variables that we need to consider: let the number of Mountees produced be M and let the number of Roadees produced be R .

Notice that the values of M and R are limited to positive integers; Mr. Hunter cannot sell negative numbers of bikes nor can he sell a fraction of a bike.

Step 2: Organise the information

We can write these constraints as inequalities:

$$\text{number of Mountees: } 0 \leq M \leq 5$$

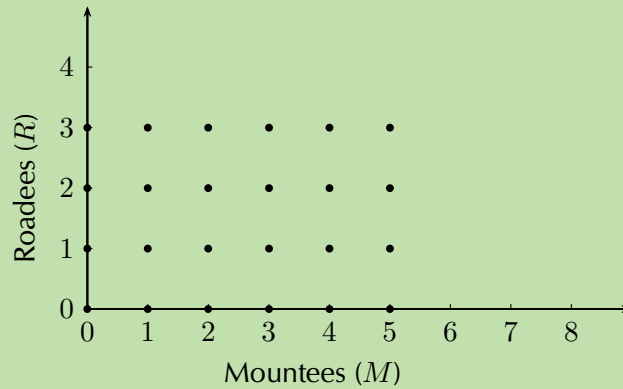
$$\text{number of Roadees: } 0 \leq R \leq 3$$

$$\text{total number of technicians: } M + 2R \leq 8$$

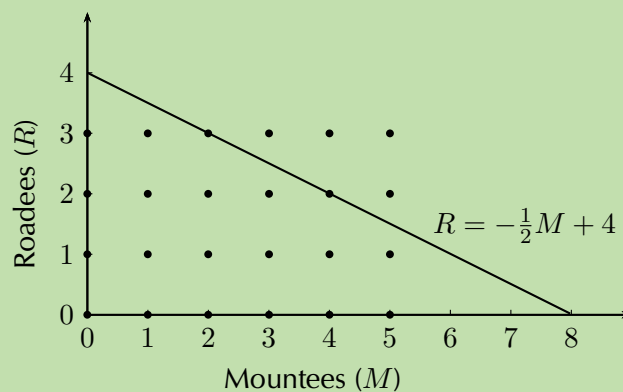
We also know that $P = 800M + 2400R$. This is called the objective function, sometimes also referred to as the search line, because the objective (goal) is to determine the maximum value of P .

Step 3: Solve using graphs

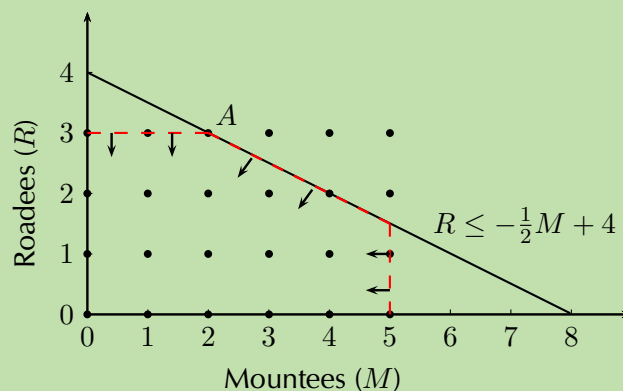
We represent the number of Mountees manufactured daily on the horizontal axis and the number of Roadees manufactured daily on the vertical axis. Since M and R are positive integers, we only use the first quadrant of the Cartesian plane. Note that the graph only includes the integer values of M between 0 and 5 and R between 0 and 3.



For the number of technicians in the assembly department $M + 2R \leq 8$. If we make R (represented on the y -axis) the subject of the inequality we get $R \leq -\frac{1}{2}M + 4$.



The arrows indicate the region in which the solution will lie, where $R \leq -\frac{1}{2}M + 4$. This area is called the feasible region.



We substitute the possible combinations into the profit equation $P = 800M + 2400R$, and find the combination that gives the maximum profit.

$$\begin{aligned}\text{At } A(2; 3) : \quad P &= 800(2) + 2400(3) \\ &= \text{R } 8800\end{aligned}$$

Step 4: Write the final answer

Therefore the maximum profit is obtained if 2 Mountees and 3 Roadees are manufactured per day.

► See video: 23DT at www.everythingmaths.co.za

Worked example 3: Optimisation using graphs

QUESTION

Solve the “furniture store opening special” problem using graphs:

As part of their opening special, a furniture store has promised to give away at least 40 prizes. They intend to give away kettles and toasters. They decide there will be at least 10 units of each prize. A kettle costs the company R 120 and a toaster costs R 100.

Determine how many of each prize will represent the cheapest option for the company. Calculate how much this combination of kettles and toasters will cost.

SOLUTION

Step 1: Assign variables

In this situation there are two variables that we need to consider: let the number of kettles be k and the number of toasters be t , with $k, t \in \mathbb{Z}$.

Step 2: Organise the information

We can write the given information as inequalities:

$$\text{number of kettles: } k \geq 10$$

$$\text{number of toasters: } t \geq 10$$

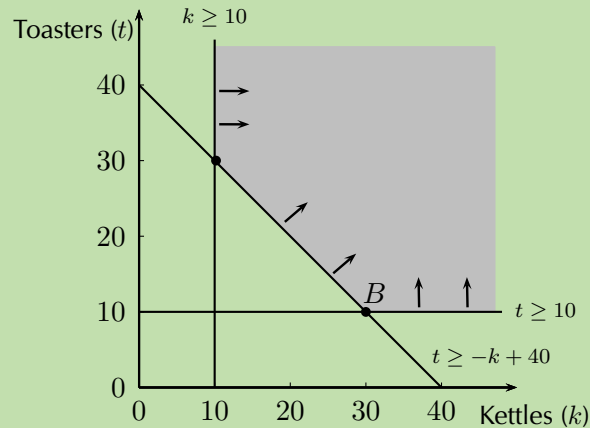
$$\text{total number of prizes: } k + t \geq 40$$

We make t the subject of the inequality:

$$t \geq -k + 40$$

Step 3: Solve using graphs

Represent the constraints on a set of axes:



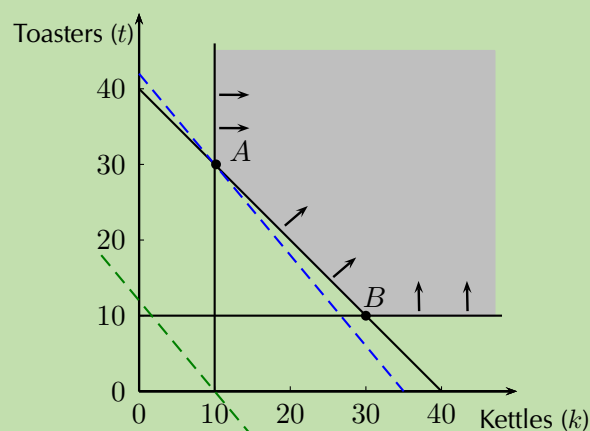
We shade the feasible region as shown in the diagram. Remember that in this situation only the points with integer coordinates inside or on the border of the feasible region are possible solutions. The combination giving the minimum cost will lie towards or on the lower border of the feasible region, which gives us many points to consider. To find the optimum value of C , we use the graph of the objective function

$$C = 120k + 100t$$

To draw the line, we make t the subject of the formula

$$t = -\frac{6}{5}k + \frac{C}{100}$$

We see that the gradient of the objective function is $-\frac{6}{5}$, but we do not know the exact value of the t -intercept ($\frac{C}{100}$). To find the minimum value of C , we need to determine the position of the objective function where it first touches the feasible region and also gives the lowest t -intercept.



We indicate the gradient of the objective function on the graph (the green search line). Keeping the gradient the same, we “slide” the objective function towards the lower border of the feasible region and find that it touches the feasible region at point

$A(10; 30)$. This optimum position of the objective function is indicated on the graph by the dotted line passing through point A .

We substitute the coordinates of A into the cost equation $C = 120k + 100t$:

$$\begin{aligned}\text{At } A(10; 30) : \quad C &= 120(10) + 100(30) \\ &= \text{R } 4200\end{aligned}$$

The minimum cost can also be determined graphically by reading off the coordinates of the t -intercept of the objective function in the optimum position:

$$\begin{aligned}t_{\text{int}} &= 42 \\ \therefore \frac{C}{100} &= 42 \\ \therefore C &= \text{R } 4200\end{aligned}$$

Step 4: Write the final answer

Therefore the minimum cost to the company is R 4200 with 10 kettles and 30 toasters.

Exercise 12 – 2: Optimisation

1. You are given a test consisting of two sections. The first section is on algebra and the second section is on geometry. You are not allowed to answer more than 10 questions from any section, but you have to answer at least 4 algebra questions. The time allowed is not more than 30 minutes. An algebra problem will take 2 minutes and a geometry problem will take 3 minutes to solve.

Let x be the number of algebra questions and y be the number of geometry questions.

- a) Formulate the equations and inequalities that satisfy the above constraints.
 - b) The algebra questions carry 5 marks each and the geometry questions carry 10 marks each. If T is the total marks, write down an expression for T .
2. A local clinic wants to produce a guide to healthy living. The clinic intends to produce the guide in two formats: a short video and a printed book. The clinic needs to decide how many of each format to produce for sale. Estimates show that no more than 10 000 copies of both items together will be sold. At least 4000 copies of the video and at least 2000 copies of the book could be sold, although sales of the book are not expected to exceed 4000 copies. Let x be the number of videos sold, and y the number of printed books sold.
 - a) Write down the constraint inequalities that can be deduced from the given information.
 - b) Represent these inequalities graphically and indicate the feasible region clearly.

- c) The clinic is seeking to maximise the income, I , earned from the sales of the two products. Each video will sell for R 50 and each book for R 30. Write down the objective function for the income.
- d) What maximum income will be generated by the two guides?
3. A certain motorcycle manufacturer produces two basic models, the Super X and the Super Y. These motorcycles are sold to dealers at a profit of R 20 000 per Super X and R 10 000 per Super Y. A Super X requires 150 hours for assembly, 50 hours for painting and finishing and 10 hours for checking and testing. The Super Y requires 60 hours for assembly, 40 hours for painting and finishing and 20 hours for checking and testing. The total number of hours available per month is: 30 000 in the assembly department, 13 000 in the painting and finishing department and 5000 in the checking and testing department.

The above information is summarised by the following table:

Department	Hours for Super X	Hours for Super Y	Hours available per month
Assembly	150	60	30 000
Painting and finishing	50	40	13 000
Checking and testing	10	20	5000

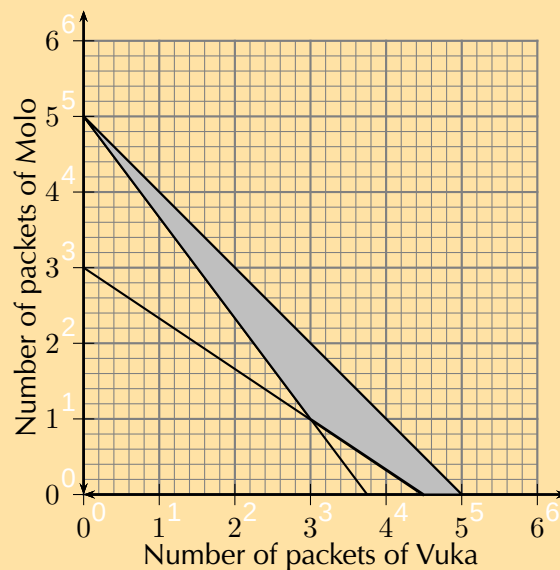
Let x be the number of Super X and y be the number of Super Y models manufactured per month.

- a) Write down the set of constraint inequalities.
- b) Use graph paper to represent the set of constraint inequalities.
- c) Shade the feasible region on the graph paper.
- d) Write down the profit generated in terms of x and y .
- e) How many motorcycles of each model must be produced in order to maximise the monthly profit?
- f) What is the maximum monthly profit?
4. A group of students plan to sell x hamburgers and y chicken burgers at a rugby match. They have meat for at most 300 hamburgers and at most 400 chicken burgers. Each burger of both types is sold in a packet. There are 500 packets available. The demand is likely to be such that the number of chicken burgers sold is at least half the number of hamburgers sold.
- a) Write the constraint inequalities and draw a graph of the feasible region.
- b) A profit of R 3 is made on each hamburger sold and R 2 on each chicken burger sold. Write the equation which represents the total profit P in terms of x and y .
- c) The objective is to maximise profit. How many of each type of burger should be sold?
5. Fashion-Cards is a small company that makes two types of cards, type X and type Y. With the available labour and material, the company can make at most 150 cards of type X and at most 120 cards of type Y per week. Altogether they cannot make more than 200 cards per week.

There is an order for at least 40 type X cards and 10 type Y cards per week. Fashion-Cards makes a profit of R 5 for each type X card sold and R 10 for each type Y card.

Let the number of type X cards manufactured per week be x and the number of type Y cards manufactured per week be y .

- One of the constraint inequalities which represents the restrictions above is $0 \leq x \leq 150$. Write the other constraint inequalities.
 - Represent the constraints graphically and shade the feasible region.
 - Write the equation that represents the profit P (the objective function), in terms of x and y .
 - Calculate the maximum weekly profit.
6. To meet the requirements of a specialised diet a meal is prepared by mixing two types of cereal, Vuka and Molo. The mixture must contain x packets of Vuka cereal and y packets of Molo cereal. The meal requires at least 15 g of protein and at least 72 g of carbohydrates. Each packet of Vuka cereal contains 4 g of protein and 16 g of carbohydrates. Each packet of Molo cereal contains 3 g of protein and 24 g of carbohydrates. There are at most 5 packets of cereal available. The feasible region is shaded on the attached graph paper.



- Write down the constraint inequalities.
- If Vuka cereal costs R 6 per packet and Molo cereal also costs R 6 per packet, use the graph to determine how many packets of each cereal must be used so that the total cost for the mixture is a minimum.
- Use the graph to determine how many packets of each cereal must be used so that the total cost for the mixture is a maximum (give all possibilities).

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1. 23DV 2. 23DW 3. 23DX 4. 23DY 5. 23DZ 6. 23F2



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1 Exponents and surds

Exercise 1 – 1: The number system

- | | | |
|---|--|--|
| 1. $\mathbb{R}; \mathbb{Q}'$ | 7. $\mathbb{R}; \mathbb{Q}$ | 13. $\mathbb{R}; \mathbb{Q}$ |
| 2. $\mathbb{R}; \mathbb{Q}$ | 8. $\mathbb{R}; \mathbb{Q}'$ | 14. $\mathbb{R}; \mathbb{Q}'$ |
| 3. $\mathbb{R}; \mathbb{Q}$ | 9. $\mathbb{R}'$ | 15. $\mathbb{R}; \mathbb{Q}$ |
| 4. $\mathbb{R}; \mathbb{Q}$ | 10. $\mathbb{R}; \mathbb{Q}'$ | 16. $\mathbb{R}; \mathbb{Q}; \mathbb{Z}$ |
| 5. $\mathbb{R}; \mathbb{Q}; \mathbb{Z}; \mathbb{N}_0$ | 11. $\mathbb{R}; \mathbb{Q}$ | |
| 6. $\mathbb{R}' \mathbb{Q}'$ | 12. $\mathbb{R}; \mathbb{Q}; \mathbb{Z}$ | |

Exercise 1 – 2: Laws of exponents

- | | | |
|----------------------|-------------------------------|-------------------------|
| 1. 4^{3a+3} | 9. $\frac{1}{m+n}$ | 16. 4 |
| 2. 72 | 10. $2p^{ts}$ | 17. $\frac{m^2 n^2}{2}$ |
| 3. $9p^{10}$ | 11. $\frac{1}{a}$ | 18. 400 |
| 4. k^{2x-2} | 12. k | 19. $\frac{1}{y^7}$ |
| 5. $5^{2z-2} + 5^z$ | 13. 2^{a+1} | 20. 8 |
| 6. 1 | 14. h^4 | 21. 2^{6a+2} |
| 7. x^{10} | 15. $\frac{a^4 b^6}{c^6 d^2}$ | 22. $2pt$ |
| 8. $\frac{b^2}{a^2}$ | | 23. $81q^{2s}y^{8a+2}$ |

Exercise 1 – 3: Rational exponents and surds

- | | |
|-----------------------------|-------------------------|
| 1. a) 7 | 2. a) $s^{\frac{1}{6}}$ |
| b) $\frac{1}{6}$ | b) $16m^4$ |
| c) $\frac{1}{\sqrt[3]{36}}$ | c) $\frac{3}{2}m^2$ |
| d) $-\frac{4}{3}$ | d) 8 |
| e) $8x^3$ | 3. $x^{\frac{31}{16}}$ |

Exercise 1 – 4: Simplification of surds

- | | | | |
|----|--------------|----|---|
| 1. | a) 4 | 2. | a) $\frac{ab}{b-a}$ |
| | b) ab^4c^2 | | b) $-(a^{\frac{1}{2}} + b^{\frac{1}{2}})$ |
| | c) 2 | | |
| | d) xy^4 | | |

Exercise 1 – 5: Rationalising the denominator

- | | |
|------------------------------|-------------------------------------|
| 1. $2\sqrt{5}$ | 6. $\frac{\sqrt{6} + \sqrt{14}}{2}$ |
| 2. $\frac{\sqrt{6}}{2}$ | 7. $\frac{3p - 4\sqrt{p}}{p}$ |
| 3. $\sqrt{6}$ | 8. $\sqrt{t} - 2$ |
| 4. $\frac{3\sqrt{5} + 3}{4}$ | 9. $\frac{1 - \sqrt{m}}{1 - m}$ |
| 5. $\frac{x\sqrt{y}}{y}$ | 10. $\sqrt{ab}$ |

Exercise 1 – 6: Solving surd equations

- | | |
|---------------------------------|-------------------------|
| 1. $x = 4$ | 6. $x = 8$ or $x = -27$ |
| 2. $p = 3$ | 7. $n = -\frac{1}{4}$ |
| 3. $y = 1$ | 8. $d = 3$ or $d = -5$ |
| 4. $t = 3$ | 9. $y = 1$ or $y = 81$ |
| 5. $z = 9$ or $z = \frac{1}{4}$ | 10. $f = 5$ |

Exercise 1 – 7: Applications of exponentials

- | | |
|--------------|-----------|
| 1. 9,7% | 3. 7 |
| 2. 4 254 691 | 4. 26 893 |

Exercise 1 – 8: End of chapter exercises

1. a) $\frac{1}{4}$
b) $4\frac{1}{4}$
2. a) x^4
b) s
c) $m^{\frac{25}{3}}$
d) $m^{\frac{8}{3}}$
e) $-m^{\frac{8}{3}}$
f) $81y^{\frac{16}{3}}$
3. a) $\frac{3b^{\frac{45}{2}}}{(a^{12}c^{\frac{5}{2}})}$
b) $3a^3b^2$
c) $a^{24}b^{12}$
d) $x^{\frac{7}{2}}$
e) $x^{\frac{4}{3}}b^{\frac{5}{3}}$
4. $\frac{1}{x^{\frac{1}{16}}}$
5. $x - 2$
6. $\frac{10\sqrt{x} + 10}{x - 1}$
7. $\frac{3\sqrt{x} + 2x\sqrt{x}}{2x}$
8. a) $6\sqrt{2}$
b) $7\sqrt{5}$
c) 2
d) $\frac{1}{4\sqrt{2}}$
e) 2
f) $\frac{16\sqrt{15}}{5}$
9. a) $6 + 4\sqrt{2}$
b) $6 + 5\sqrt{2}$
c) $4 + 2\sqrt{2} + 2\sqrt{3} + 2\sqrt{6}$
10. a) 55
b) 1
11. $15\sqrt{2}x^3$
12. a) $1 + \frac{2\sqrt{5}}{5}$
b) $\frac{2y + y\sqrt{y} - 4\sqrt{y} - 8}{y - 4}$
c) $2\sqrt{x} + 2\sqrt{10}$
13. $\frac{3}{2}$
15. 3
16. $-\sqrt{288}$
17. a) 4
b) $-\frac{1}{3}$
c) 3
d) No solution
e) $x = \frac{1}{8}$ or $x = -8$
18. b) $x = 1$

2 Equations and inequalities

Exercise 2 – 1: Solution by factorisation

1. $t = 0$ or $t = -2$
2. $y = -1$
3. $s = \pm 5$
4. $y = 3$ or $y = 2$
5. $y = 4$ or $y = -9$
6. $p = -2$
7. $y = -3$ or $y = -8$
8. $y = 6$ or $y = 7$
9. $x = -7$ or $x = -2$
10. $y = 4k$ or $y = k$
11. $y = 9$ or $y = -9$
12. $y = \pm\sqrt{5}$
13. $h = \pm 6$
14. $y = \pm\sqrt{14}$
15. $p = -2$
16. $y = \pm 6\sqrt{2}$
17. $f = \frac{5}{2}$ or $f = -3$
18. $x = \frac{1}{4}$
19. $y = \frac{1}{7}$
20. $x \in \mathbb{R}, x \neq \pm 3$
21. $y = -13$ or $y = -1$
22. $t = \frac{3}{2}$ or $t = -2$
23. $m = -6$
24. $t = 0$ or $t = 3$

Exercise 2 – 2: Solution by completing the square

1. a) $x = -5 - 3\sqrt{3}$ or $x = -5 + 3\sqrt{3}$
b) $x = -1$ or $x = -3$
c) $p = -4 \pm \sqrt{21}$
d) $x = -3 \pm \sqrt{7}$
e) No real solution
f) $t = -8 \pm 3\sqrt{6}$
g) $x = -1 \pm \sqrt{\frac{5}{3}}$
- h) $z = -4 \pm \sqrt{22}$
i) $z = \frac{11}{2}$ or $z = 0$
j) $z = 5$ or $z = -1$
2. $k = -3 \pm \sqrt{9 - a}$
3. $y = \frac{-q \pm \sqrt{q^2 - 4pr}}{2p}$

Exercise 2 – 3: Solution by the quadratic formula

1. $t = 1$ or $t = \frac{-4}{3}$
2. $x = \frac{5+\sqrt{37}}{2}$ or $t = \frac{5-\sqrt{37}}{2}$
3. No real solution
4. $p = \frac{1}{2}$ or $p = -1$
5. No real solution
6. $t = \frac{-3+\sqrt{69}}{10}$ or $t = \frac{-3-\sqrt{69}}{10}$
7. $t = 2 \pm \sqrt{2}$
8. $k = \frac{7+\sqrt{373}}{18}$ or $k = \frac{7-\sqrt{373}}{18}$
9. $f = \frac{1}{2}$ or $f = -2$
10. No real solution

Exercise 2 – 4:

1. $x = -1, x = -4, x = -2$ and $x = -3$
2. $x = 1, x = 4$ and $x = -2$
3. $x = -7, x = 4, x = -1$ and $x = -2$
4. $x = -4, x = 3, x = -3$ and $x = 2$
5. $x = \frac{8 \pm \sqrt{40}}{4}$
6. $x = -5, x = 3, x = -1 + \sqrt{10}$ and $x = -1 - \sqrt{10}$

Exercise 2 – 5: Finding the equation

1. $x^2 - x - 6 = 0$
2. $x^2 - 16 = 0$
3. $2x^2 - 5x - 3 = 0$
4. $k = 3$ and $x = \frac{3}{4}$
5. $p = 5$ and $x = -1$

Exercise 2 – 6: Mixed exercises

1. $y = \frac{1}{8}$ or $y = -\frac{8}{3}$
2. $x = \frac{3}{2}$ or $x = -\frac{7}{2}$
3. $t = \frac{2}{3}$ or $t = 2$
4. $y = 1$ or $y = -1$
5. $m = 1$ or $m = 4$
6. $y = \pm \frac{5}{7}$
7. $w = \frac{3}{2}$ or $w = 4$
8. $y = \frac{6}{5}$ or $y = \frac{1}{4}$
9. $n = \frac{8}{3}$ or $n = -\frac{9}{8}$
10. $y = -\frac{8}{3}$ or $y = \frac{3}{2}$
11. $x = -\frac{1}{2}$ or $x = 3$
12. $y = -\frac{5}{2}$ or $y = -\frac{5}{9}$
13. $y = \frac{4}{5}$ or $y = \frac{1}{5}$
14. $g = -\frac{1}{4}$ or $g = 1$
15. $y = 2$ or $y = -\frac{5}{9}$
16. $p = \frac{3}{7}$ or $p = -\frac{1}{5}$
17. $y = -\frac{2}{9}$ or $y = -1$
18. $y = \frac{9}{2}$ or $y = \frac{9}{7}$

Exercise 2 – 7: From past papers

1.
 - a) Real, unequal and rational
 - b) Real and equal
 - c) Real, unequal and irrational
 - d) Real, unequal and rational
 - e) Real, unequal and irrational
 - f) Non-real
 - g) Real, unequal and rational
 - h) Real, unequal and irrational
 - i) Non-real
 - j) Real and equal
2.
 - b) real and unequal
 - c) $k = -6 \pm 2\sqrt{6}$
4.
 - a) $k = 6$
 - b) $k = \frac{1}{3}$
5.
 - a) $k = 4$ or $k = 1$
 - b) $k = 0$ or $k = 5$
6.
 - a) all real values of a , b and p
 - b) $a = b$ and $p = 0$

Exercise 2 – 8: Solving quadratic inequalities

1.
 - a) $-3 < x < 4$
 - b) $x < -\frac{4}{3}$ or when $x > 1$
 - c) no real solutions
 - d) $-1 < t < 3$
 - e) All real values of s .
 - f) All real values of x .
 - g) $x \leq -\frac{1}{4}$ or $x \geq 0$
2.
 - i) $x < 3$ or $x > 6$ with $x \neq 3$
 - j) $-2 \leq x \leq 2$ and $x > 7$ with $x \neq 7$
 - k) $x > 0$ with $x \neq 0$
 - a) $x < -3$ or $x > 3$
 - b) $-\sqrt{5} \leq x \leq \sqrt{5}$
 - c) no solution
 - d) All real values of x

Exercise 2 – 9: Solving simultaneous equations

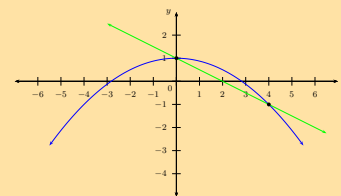
1.
 - a) $(0; 5)$ and $(2; 3)$
 - b) $x = 3 \pm \sqrt{2}$ and $y = 2 \pm \sqrt{2}$
 - c) $(-1; 0)$ and $(\frac{1}{4}; \frac{5}{8})$
 - d) $b = \frac{2 \pm \sqrt{88}}{6}$ and $a = \frac{11 \pm \sqrt{88}}{6}$
 - e) $(-3; -20)$ and $(2; 0)$
2.
 - f) $x = \frac{6 \pm \sqrt{264}}{2}$ and $y = \frac{70 \pm \sqrt{264}}{2}$
 - a) $(-3; 8)$ and $(2; 3)$
 - b) $(-4; 14)$ and $(3; 7)$
 - c) $(3; 4)$ and $(4; 3)$

Exercise 2 – 10:

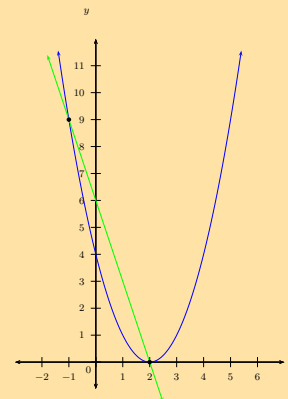
1. $b = 2 \text{ m}$, $l = 4 \text{ m}$
2. 187
3. $t = 10,5 \text{ s}$
4. $t = 5 \text{ d}$; 105 minutes; 1,4 km
5. 24 A; 70 W; 12 A

Exercise 2 – 11: End of chapter exercises

1. $x = 1,62$ or $x = -0,62$
2. $x = \pm 4$ or $x = -1$
3. $y = 0$ or $y = \pm 1$
4. $x = \pm 2$
5.
 - a) $x = 7$ or $x = 2$
 - b) $x = 2,3$ or $x = -1,3$
 - c) $x = 1,65$ or $x = -3,65$
 - d) $x = 0$ or $x = -3$
6. $x = \frac{\sqrt{16+p^2}-2}{2}$
7. $a = 3$; $b = 10$ and $c = -8$
8. $p = \pm 16$
9. $x^2 + 2x - 15$
10. Undefined: $b = -2$ Zero: $b = 2$ or $b = 3$
13. $a \geq 4$
14. $x = -\frac{3}{2}$ or $x = 1$
15.
 - a) $x < 3$ or $x \geq 7$:
 - b) $x < 1$ or $x > 5$:
 - c) $3 < x < 7$:
 - d) $x < -1$ or $x > 3$
 - e) $0,5 < x < 2,5$
 - f) $x \leq -3$ or $0 < x \leq \frac{5}{2}$
 - g) $x < \frac{2}{3}$
 - h) $-1 \leq x < 0$ or $x \geq 3$
 - i) $-4 \leq x \leq 1$
 - j) $2\frac{1}{2} \leq x < 3$
16.
 - a) $x = \pm\sqrt{3}$ and $y = \pm 2\sqrt{3}$
 - b) $a = -3$ and $b = -1$ or $a = 12$ and $b = 4$
 - c) $x = -5$ and $y = 0$ or $x = 2$ and $y = 14$
 - d) $p = \frac{5}{3}$ and $q = \frac{2}{9}$ or $p = -1$ and $q = -\frac{2}{3}$
 - e) $b = \frac{3 \pm \sqrt{5}}{2}$ and $a = \frac{7 \pm 3\sqrt{5}}{2}$
 - f) $b = \frac{-10 \pm \sqrt{140}}{4}$ and $a = \frac{-12 \pm \sqrt{140}}{2}$
 - g) $x = 3,4$ and $y = 5,4$ or $x = 3$ and $y = 5$
 - h) $b = -1,4$ and $a = 23,6$ or $b = 3$ and $a = 6$
17.
 - a)



b)



18. 35 m

20.
 - a) $y = -\frac{5}{4}$ or $y = -9$
 - b) $x = -\frac{9}{4}$ or $x = 1$
 - c) $p = -\frac{8}{7}$ or $p = -\frac{4}{3}$
 - d) $y = -\frac{1}{4}$ or $y = \frac{1}{2}$
 - e) $y = -\frac{2}{9}$ or $y = -1$
 - f) $y = \frac{7}{3}$ or $y = -\frac{1}{2}$
 - g) $y = \frac{9}{4}$ or $y = -\frac{9}{4}$
 - h) $y = \frac{8}{3}$ or $y = -6$
 - i) $y = \frac{9}{5}$ or $y = -7$
 - j) $x = \pm 4$
 - k) $y = \pm 7$
21. $k = 76$ and $\frac{4}{9}$
22. $x = 3$ or $x = -2$ and $y = \frac{1 \pm \sqrt{-7}}{2}$
23. $x = 4$ or $x = -1$
24. $y = \frac{3}{2}$, $y = \frac{1}{2}$ and $p = \frac{9}{2}$, $p = \frac{7}{2}$
25. $\frac{69}{4}$
26. 7
27. $\frac{2 \pm \sqrt{12}}{2}$
28. $t = \frac{1}{2}$, $t = 1$ or $t = \frac{3 \pm \sqrt{33}}{4}$

Exercise 3 – 1: Linear sequences

1. $-19; -35; -51$
2. a) -19
b) $T_2 = 15; T_4 = 33$
3. a) $T_n = 10 + 3n; T_{10} = 40; T_{15} = 55;$
 $T_{30} = 100$
- b) $T_n = 12 + 6n; T_{10} = 72; T_{15} = 102;$
 $T_{30} = 192$
- c) $T_n = -5 - 5n; T_{10} = -55; T_{15} = -80;$
 $T_{30} = -155$
4. $T_9 = 36$
5. a) $44; 66; 121$

Exercise 3 – 2: Quadratic sequences

1. a) 10
b) 2
c) 2
d) -2
e) 2
f) -4
g) 4
- h) -2
i) $6a$
j) 6
k) $2t$
- d) $T_2 = -3$
e) $T_4 = 63$
f) $T_1 = 2$
2. a) $T_4 = 53$
b) $T_2 = 30$
c) $T_1 = 17$
3. a) $3; 9; 17; 27$
b) $-6; -9; -14; -21$
c) $1; 8; 21; 40$
d) $0; -5; -14; -27$

Exercise 3 – 3: Quadratic sequences

1. a) 1
b) 2
c) 4
d) 8
- e) -2
2. $12; 30; 58; 96; 144$
3. $T_9 = 379$
4. $n = 4$
5. a) $T_5 = 84; T_6 = 111$
b) $T_n = 2n^2 + 5n + 9$

Exercise 3 – 4: End of chapter exercises

1. $-4; 9; 16; 25; 36$
2. a) Quadratic sequence
b) Quadratic sequence
c) Quadratic sequence
d) Quadratic sequence
e) Quadratic sequence
f) Quadratic sequence
g) Linear sequence
h) Linear sequence
i) Quadratic sequence
j) Quadratic sequence
k) Quadratic sequence
l) Linear sequence
m) Quadratic sequence
3. $x = 31$
4. $n = 11$
5. $T_{11} = 363$
6. $n = 9$
7. $T_5 = 114$
8. $n = 8$
9. a) $T_5 = 19;$
 $T_n = 4n - 1;$
 $T_{10} = 39$
b) $T_5 = -3;$
 $T_n = 22 - 5n;$
 $T_{10} = -28$
c) $T_5 = 2\frac{1}{2}; T_n = \frac{1}{2}n;$
 $T_{10} = 5$
d) $T_5 = a + 4b;$
 $T_n = a - b + bn;$
 $T_{10} = a + 9b$
e) $T_5 = -7;$
 $T_n = 3 - 2n;$
 $T_{10} = -17$
10. a) $T_n = n^2 + 3;$
 $T_{100} = 10\ 003$
b) $T_n = 6n - 4;$
 $T_{100} = 596$
c) $T_n = 2n^2 + 5;$
 $T_{100} = 20\ 005$
d) $T_n = 3n^2 + 2;$
 $T_{100} = 30\ 002$
11. a) $2; 5; 8; 11; 14$
b) Constant difference,
 $d = 3$
- c) Yes
12. a) $T_n = 4n - 19$
b) $n = 48$
13. a) Incorrect
b) Correct
14. c) Linear
15. b) Linear
d) Quadratic
e) $T_n = \frac{1}{2}n^2 + \frac{3}{2}n + 1$
f) $T_{21} = 253$
g) 31 cm
16. a) -1
b) 7
17. b) 2
c) $T_n = n^2 - n$
d) 210
e) 25
18. $4; 14; 34; 64; 104; 154$

4 Analytical geometry

Exercise 4 – 1: Revision

1. a) $2\sqrt{26}$ units
b) 7 units
c) $x + 1$ units
2. $p = 6$ or $p = 2$
3. a) $-\frac{1}{2}$
b) 3
5. 2
6. a) $(1; 2)$
- b) $(-\frac{1}{2}; -\frac{1}{2})$
7. $B(4; 2)$
8. a) $y = -4x + 3$ and
 $y = -4x + 19$
c) $AD = \sqrt{17}$ units and
 $BC = \sqrt{17}$ units
d) $y = \frac{4}{3}x - \frac{7}{3}$
e) Parallelogram (one
opposite side equal
and parallel)
9. $N(0; 3)$
10. a) $PQ = \sqrt{20}$ and
 $SR = \sqrt{20}$
b) $M(\frac{3}{2}; 1)$
d) $PS: y = -\frac{2}{5}x - \frac{1}{5}$
and $SR: y = \frac{1}{2}x - 2$
e) No
f) Parallelogram

Exercise 4 – 2: The two-point form of the straight line equation

1. $y = \frac{2}{3}x + 5$
2. $y = -3x + \frac{1}{4}$
3. $y = x + 3$
4. $y = 2x - 1$
5. $y = -5$
6. $y = \frac{3}{4}x + 3$
7. $y = -x + (s + t)$
8. $y = 5x + 2$
9. $y = \frac{q}{p}x - q$

Exercise 4 – 3: Gradient–point form of a straight line equation

1. $y = \frac{2}{3}x + 4$

4. $y = 11$

7. $y = -\frac{4}{5}x + 1$

2. $y = -x - 2$

5. $y = -2x + 7$

8. $x = 4$

3. $y = -\frac{1}{3}x$

6. $x = -\frac{3}{2}$

9. $y = 3ax + b$

Exercise 4 – 4: The gradient–intercept form of a straight line equation

1. $y = 2x + 3$

4. $y = -\frac{3}{7}x$

7. $y = -\frac{3}{2}$

2. $y = 4x - 4$

5. $y = \frac{1}{2}x - \frac{1}{5}$

8. $y = 3x + 4$

3. $y = -x - 1$

6. $y = 2x - 2$

9. $y = -5x$

Exercise 4 – 5: Angle of inclination

1. a) 1,7

g) $-0,8$

d) Horizontal line

b) -1

h) 0

e) $18,4^\circ$

c) 0

i) 3,7

f) Vertical line

d) 1,4

2. a) $36,8^\circ$

g) $71,6^\circ$

e) Undefined

b) $26,6^\circ$

h) 30°

f) 1

c) 45°

Exercise 4 – 6: Inclination of a straight line

1. a) $38,7^\circ$

f) 45°

2. $85,2^\circ$

b) 135°

g) $56,3^\circ$

c) 80°

h) $63,4^\circ$

3. 90°

d) 80°

i) $161,6^\circ$

e) $102,5^\circ$

j) Gradient undefined

4. $81,8^\circ$

Exercise 4 – 7: Parallel lines

1. a) Parallel

e) Parallel

3. $y = 3x$

b) Parallel

f) Parallel

4. $y = \frac{3}{2}x + 1$

c) Parallel

d) Not parallel

2. $y = -2x - 3$

5. $y = -\frac{7}{10}x - 1$

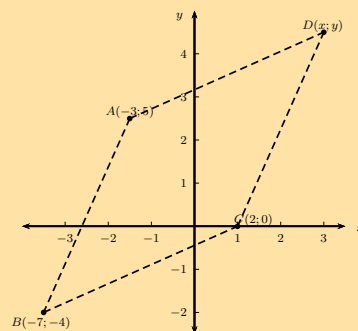
Exercise 4 – 8: Perpendicular lines

1. a) Perpendicular
b) Not perpendicular
c) Perpendicular
d) Perpendicular
- e) Perpendicular
f) Not perpendicular
g) Not perpendicular
2. $y = \frac{1}{2}x - 3$
3. $y = -5x + 3$
4. $y = -x + 2$
5. $x = -2$

Exercise 4 – 9: End of chapter exercises

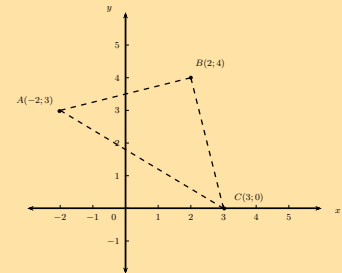
1. a) $y = \frac{1}{2}x + \frac{7}{2}$
b) $y = -x + 4$
c) $y = \frac{1}{2}x + 4$
d) $y = 2x + 4$
e) $y = 3x$
2. a) $\theta = 63,4^\circ$
b) $\theta = 18,4^\circ$
c) $\theta = 36,9^\circ$
d) $\theta = 146,3^\circ$
e) $\theta = 161,6^\circ$
3. a) $y = -2x + 7$
b) $(\frac{7}{2}; 0)$
c) $\theta = 116,6^\circ$
d) $m = \frac{1}{2}$
e) $\angle Q\hat{P}R = 90^\circ$
f) $y = -2x$
g) $(\frac{1}{2}; -\frac{3}{2})$
h) $y = -2x - \frac{1}{2}$

4. a)



- b) $D(6; 9)$
5. a) $(-1; -2)$
b) $(8; 3)$
c) $x = -1$
d) $MN = 5$ units
e) $\angle M\hat{N}P = 21,8^\circ$
f) $y = \frac{5}{2}x + \frac{11}{2}$

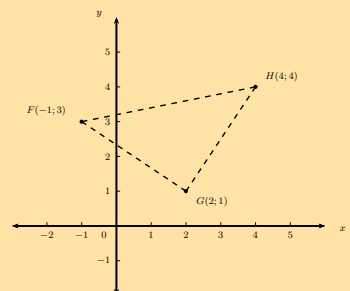
6. a)



- c) $y = \frac{1}{4}x + \frac{7}{2}$
d) $D(-1; -1)$
e) $E(\frac{5}{2}; 2)$

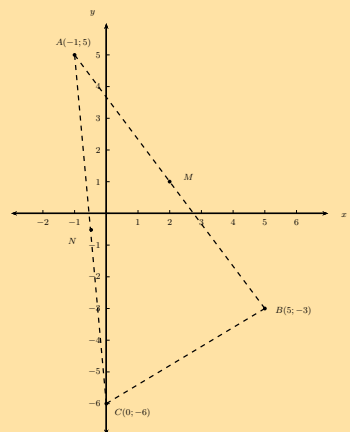
7. a) $y = \frac{3}{2}x + 2$
b) $\angle T\hat{S}V = 49,6^\circ$

8. a)



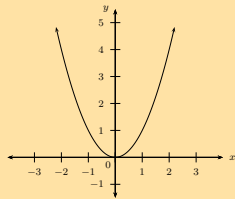
- c) $y = -5x + 11$
d) Yes
e) $y = \frac{3}{2}x + \frac{9}{2}$

9. a)

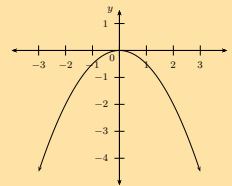


Exercise 5 – 1: Revision

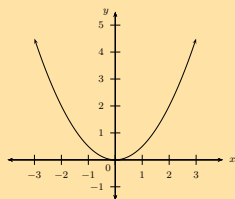
1. a)



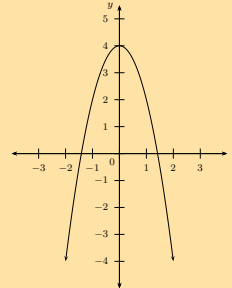
c)



b)



d)



Exercise 5 – 2: Domain and range

1. $\{x : x \in \mathbb{R}\}; \{y : y \geq -1, y \in \mathbb{R}\}$

4. $\{x : x \in \mathbb{R}\}; \{y : y \leq 0, y \in \mathbb{R}\}$

2. $\{x : x \in \mathbb{R}\}; \{y : y \leq 4, y \in \mathbb{R}\}$

3. $\{x : x \in \mathbb{R}\}; \{y : y \geq 0, y \in \mathbb{R}\}$

5. $\{x : x \in \mathbb{R}\}; \{y : y \leq 2, y \in \mathbb{R}\}$

Exercise 5 – 3: Intercepts

1. (0; 15) and (-5; 0); (-3; 0)

3. (0; -3) and (1; 0); (3; 0)

5. (0; 37) and no x -intercepts

2. (0; 16) and (4; 0)

4. (0; 35) and $(-\frac{7}{2}; 0); (-\frac{5}{2}; 0)$

6. (0; -4) and $(-0,85; 0); (-2,35; 0)$

Exercise 5 – 4: Turning points

1. (3; -1)

3. (-2; -1)

5. (1; 21)

2. (2; 1)

4. $(-\frac{1}{2}; \frac{1}{2})$

6. (-1; -6)

Exercise 5 – 5: Axis of symmetry

1. a) Axis of symmetry:
 $x = \frac{5}{4}$

$x = 2$

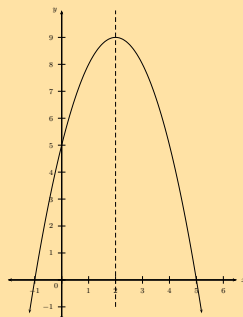
2. $y = ax^2 + q$

b) Axis of symmetry:

c) Axis of symmetry:
 $x = 2$

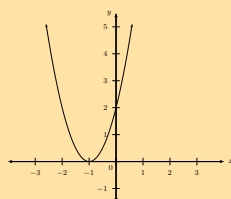
Exercise 5 – 6: Sketching parabolas

1. a)



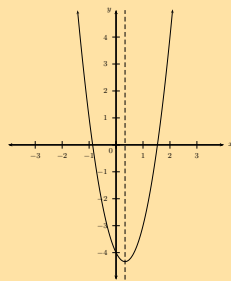
Intercepts: $(-1; 0), (5; 0), (0; 5)$
 Turning point: $(2; 9)$
 Axes of symmetry: $x = 2$
 Domain: $\{x : x \in \mathbb{R}\}$
 Range: $\{y : y \leq 9, y \in \mathbb{R}\}$

b)



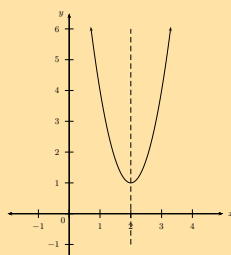
Intercepts: $(-1; 0), (0; 2)$
 Turning point: $(-1; 0)$
 Axes of symmetry: $x = -1$
 Domain: $\{x : x \in \mathbb{R}\}$
 Range: $\{y : y \geq 0, y \in \mathbb{R}\}$

c)



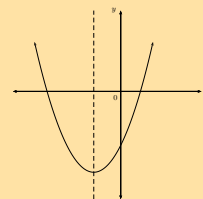
Intercepts: $(-0,87; 0), (1,54; 0), (0; -4)$
 Turning point: $(0,33; -4,33)$
 Axes of symmetry: $x = -0,33$
 Domain: $\{x : x \in \mathbb{R}\}$
 Range: $\{y : y \geq -4,33, y \in \mathbb{R}\}$

d)

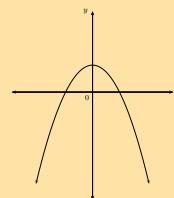


Intercepts: $(0; 13)$ Turning point: $(2; 1)$
 Axes of symmetry: $x = 2$
 Domain: $\{x : x \in \mathbb{R}\}$
 Range: $\{y : y \geq 1, y \in \mathbb{R}\}$

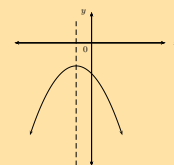
3. a)



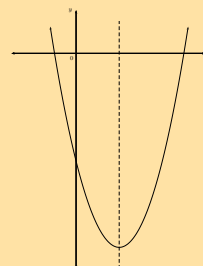
b)



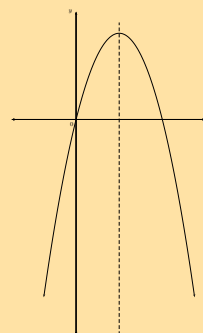
c)



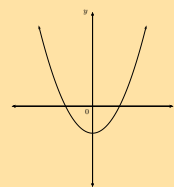
d)



e)



f)



4.

a) $y_{\text{shifted}} = 2x^2 + 16x + 32$

b) $y_{\text{shifted}} = -x^2 - 2x$

c) $y_{\text{shifted}} = 3x^2 - 16x + 22$

Exercise 5 – 7: Finding the equation

1. $y = -3(x+1)^2 + 6$ or $y = -3x^2 - 6x + 3$

3. $y = \frac{2}{3}(x+2)^2$

2. $y = \frac{1}{2}x^2 - \frac{5}{2}x$

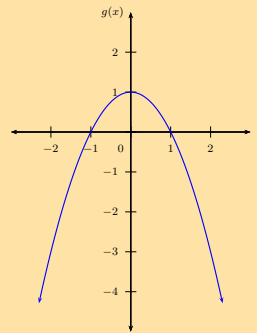
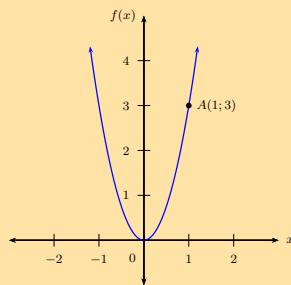
4. $y = -x^2 + 3x + 4$

Exercise 5 – 8:

1. a) 11

3. a)

2. a)



b) 6

c) $y = 6x - 3$

b) 1

c) 4

d) 0

Exercise 5 – 10: Domain and range

1. $\{x : x \in \mathbb{R}, x \neq 0\} ; \{y : y \in \mathbb{R}, y \neq 1\}$

4. $\{x : x \in \mathbb{R}, x \neq 5\} ; \{y : y \in \mathbb{R}, y \neq 3\}$

2. $\{x : x \in \mathbb{R}, x \neq 8\} ; \{y : y \in \mathbb{R}, y \neq 4\}$

3. $\{x : x \in \mathbb{R}, x \neq -1\} ; \{y : y \in \mathbb{R}, y \neq -3\}$

5. $\{x : x \in \mathbb{R}, x \neq -2\} ; \{y : y \in \mathbb{R}, y \neq 2\}$

Exercise 5 – 11: Intercepts

1. $(0; -1\frac{3}{4})$ and $(-3\frac{1}{2}; 0)$

3. $(0; 1)$ and $(\frac{1}{3}; 0)$

5. $(0; 2)$ and $(8; 0)$

2. $(\frac{5}{2}; 0)$

4. $(0; \frac{3}{2})$ and $(\frac{1}{3}; 0)$

Exercise 5 – 12: Asymptotes

1. $y = -2$ and $x = -4$

3. $y = 1$ and $x = 2$

5. $y = 0$ and $x = 2$

2. $y = 0$ and $x = 0$

4. $y = -8$ and $x = 0$

Exercise 5 – 13: Axes of symmetry

1.
 - a) For $f(x)$: $(0; 0)$; $y_1 = x$ and $y_2 = -x$
 For $g(x)$: $(0; 1)$; $y_1 = x + 1$ and $y_2 = -x + 1$
 - b) For $f(x)$: $(0; 0)$; $y_1 = x$ and $y_2 = -x$
 For $g(x)$: $(-1; 0)$; $y_1 = x + 1$ and $y_2 = -x - 1$
 - c) For $f(x)$: $(0; 0)$; $y_1 = x$ and $y_2 = -x$
 For $g(x)$: $(1; -1)$; $y_1 = x - 2$ and $y_2 = -x$
2. $k(x) = \frac{5}{x+1} + 2$

Exercise 5 – 14: Sketching graphs

1.
 - a) Asymptotes: $x = 0$; $y = 2$
 Intercepts: $(-\frac{1}{2}; 0)$
 Axes of symmetry: $y = x + 2$ and $y = -x + 2$
 Domain: $\{x : x \in \mathbb{R}, x \neq 0\}$
 Range: $\{y : y \in \mathbb{R}, y \neq 2\}$
 - b) Asymptotes: $x = -4$; $y = -2$
 Intercepts: $(-3\frac{1}{2}; 0)$ and $(0; -1\frac{3}{4})$
 Axes of symmetry: $y = x + 2$ and $y = -x - 6$
 Domain: $\{x : x \in \mathbb{R}, x \neq -4\}$
 Range: $\{y : y \in \mathbb{R}, y \neq -2\}$
 - c) Asymptotes: $x = -1$; $y = 3$
 Intercepts: $(-\frac{2}{3}; 0)$ and $(0; 2)$
 Axes of symmetry: $y = x + 4$ and $y = -x + 2$
 Domain: $\{x : x \in \mathbb{R}, x \neq -1\}$
 Range: $\{y : y \in \mathbb{R}, y \neq 3\}$
 - d) Asymptotes: $x = -2\frac{1}{2}$; $y = -2$
- Intercepts: $(0; 0)$
 Axes of symmetry: $y = x - 4\frac{1}{2}$ and $y = -x + \frac{1}{2}$
 Domain: $\{x : x \in \mathbb{R}, x \neq -4\}$
 Range: $\{y : y \in \mathbb{R}, y \neq -2\}$
- e) Asymptotes: $x = 8$; $y = 4$
 Intercepts: $(6; 0)$ and $(0; 3)$
 Axes of symmetry: $y = x - 4$ and $y = -x + 12$
 Domain: $\{x : x \in \mathbb{R}, x \neq 8\}$
 Range: $\{y : y \in \mathbb{R}, y \neq 4\}$
2. $y = \frac{1}{x+2} - 1$
3. $y = -\frac{4}{x} + 2$
4.
 - a)
 - b) Average gradient = 1
 - c) Average gradient = 12

Exercise 5 – 16: Domain and range

1. $\{x : x \in \mathbb{R}\}; \{y : y > 0, y \in \mathbb{R}\}$
2. $\{x : x \in \mathbb{R}\}; \{y : y < 1, y \in \mathbb{R}\}$
3. $\{x : x \in \mathbb{R}\}; \{y : y > -3, y \in \mathbb{R}\}$
4. $\{x : x \in \mathbb{R}\}; \{y : y > n, y \in \mathbb{R}\}$
5. $\{x : x \in \mathbb{R}\}; \{y : y > 2, y \in \mathbb{R}\}$

Exercise 5 – 17: Intercepts

1. $(0; -6)$ and $(2; 0)$
2. $(0; -17\frac{1}{3})$ and $(3; 0)$
3. $(0; -20)$ and $(-1; 0)$
4. $(0; \frac{15}{16})$ and $(-2; 0)$

Exercise 5 – 18: Asymptote

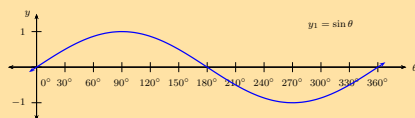
1. $y = 0$
2. $y = 1$
3. $y = -\frac{2}{3}$
4. $y = -2$
5. $y = -2$

Exercise 5 – 19: Mixed exercises

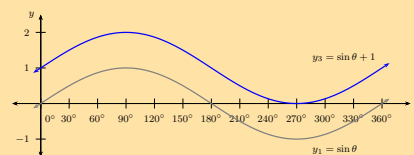
1. b) i. $y = \frac{3}{x} + 3$
 ii. $y = \frac{3}{x-3}$
 iii. $y = -\frac{3}{x}$
 iv. $y = \frac{3}{x} - \frac{1}{4}$
 v. $y = \frac{3}{x} + 4$
 vi. $y = \frac{3}{x+2} - 1$
2. a) $M(-2; 2)$
 b) $g(x) = \frac{-4}{x}$
 c) $f(x) = 2(x+1)^2$
 d) $-2 < x < 0$
 e) Range: $\{y : y \in \mathbb{R}, y \geq 0\}$
3. a) For $k(x)$:
 Intercepts: $(-2; 0), (1; 0)$ and $(0; -4)$
 Turning point: $(-\frac{1}{2}; -4\frac{1}{2})$
 Asymptote: none
 For $h(x)$:
 Intercepts: $(1, 41; 0)$
 Turning point: none
 Asymptote: $y = 0$
6. a) $f(x) = -\frac{3}{4}(x-2)^2 + 3$;
 Axes of symmetry: $x = 2$;
 Domain: $\{x : x \in \mathbb{R}\}$;
 Range: $\{y : y \in \mathbb{R}, y \leq 3\}$
- b) $g(x) = \frac{1}{4}x^2 - 2$;
 Axes of symmetry: $x = 0$;
 Domain: $\{x : x \in \mathbb{R}\}$;
 Range: $\{y : y \in \mathbb{R}, y \geq -2\}$; $h(x) = \frac{2}{x}$;
 Axes of symmetry: $y = x$
 Domain: $\{x : x \in \mathbb{R}, x < 0\}$;
 Range: $\{y : y \in \mathbb{R}, y < 0\}$;
- c) $k(x) = (\frac{1}{2})^x + \frac{1}{2}$;
 Domain: $\{x : x \in \mathbb{R}\}$;
 Range: $\{y : y \in \mathbb{R}, y > \frac{1}{2}\}$
7. b) $p = 9$
 c) Average gradient = $-2\frac{8}{9}$
 d) $y = (\frac{1}{3})^{x+2} - 2$
8. a) $f(x) = 2^x - \frac{3}{2}$ and $g(x) = -\frac{1}{4}x - \frac{1}{2}$
 b) $h(x) = -\frac{3}{x+2} + 1$
9. a) $AO = 2$ units $OB = 5$ units
 $OC = 10$ units $DE = 12,25$ units
 b) $DE = 12\frac{1}{4}$
 c) $h(x) = -2x + 10$
 d) $\{x : x \in \mathbb{R}, x < -2 \text{ and } x > 5\}$
 e) $\{x : x \in \mathbb{R}, 0 \leq x \leq 5\}$
 f) 5,25 units

Exercise 5 – 20: Revision

1.

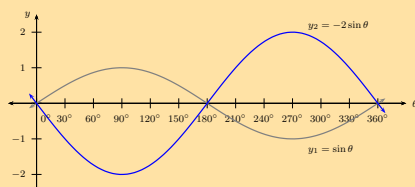


Period: $= 360^\circ$
 Amplitude: $= 1$
 Domain: $= [0^\circ; 360^\circ]$
 Range: $= [-1; 1]$
 x -intercepts: $= (0^\circ; 0); (180^\circ; 0); (360^\circ; 0)$
 y -intercepts: $= (0^\circ; 0)$
 Max. turning point: $= (90^\circ; 1)$
 Min. turning point: $= (270^\circ; -1)$



Period: 360°
 Amplitude: 1
 Domain: $[0^\circ; 360^\circ]$
 Range: $[0; 2]$
 x -intercepts: $(270^\circ; 0)$
 y -intercepts: $(0^\circ; 1)$
 Max. turning point: $(90^\circ; 2)$
 Min. turning point: $(270^\circ; 0)$

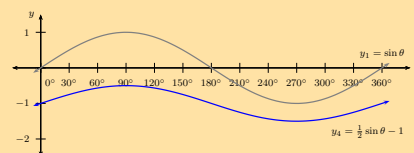
2.



Period: 360°
 Amplitude: 2
 Domain: $[0^\circ; 360^\circ]$
 Range: $[-2; 2]$
 x -intercepts: $(0^\circ; 0); (180^\circ; 0); (360^\circ; 0)$
 y -intercepts: $(0^\circ; 0)$
 Max. turning point: $(270^\circ; 2)$
 Min. turning point: $(90^\circ; -2)$

3.

4.



Period: 360°
 Amplitude: $\frac{1}{2}$
 Domain: $[0^\circ; 360^\circ]$
 Range: $[-\frac{1}{2}; -\frac{3}{2}]$
 x -intercepts: none
 y -intercepts: $(0^\circ; -\frac{1}{2})$
 Max. turning point: $(90^\circ; -\frac{1}{2})$
 Min. turning point: $(270^\circ; -\frac{3}{2})$

Exercise 5 – 21: Sine functions of the form $y = \sin k\theta$

2. a) $k = 2$

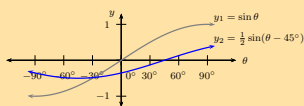
b) $k = -\frac{3}{4}$

Exercise 5 – 23: The sine function

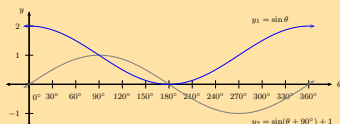
1. a)



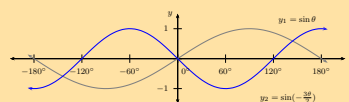
b)



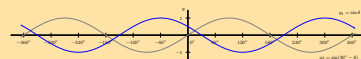
c)



d)



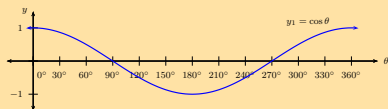
e)



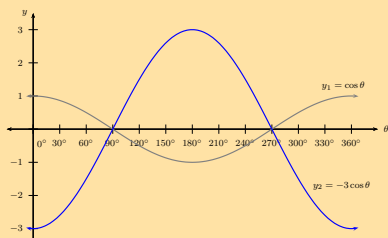
2. $a = 2$; $p = 90^\circ \therefore y = 2 \sin(\theta + 90^\circ)$ and $y = 2 \cos \theta$

Exercise 5 – 24: Revision

1.

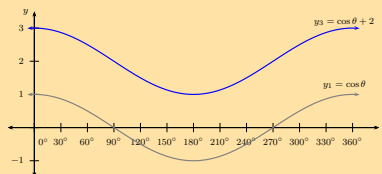


2.



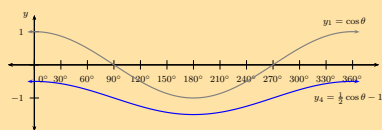
Period: 360°
 Amplitude: 1
 Domain: $[0^\circ; 360^\circ]$
 Range: $[-3; 3]$
 x-intercepts: none
 y-intercepts: $(0^\circ; 3)$
 Max. turning point: $(0^\circ; 3); (360^\circ; 3)$
 Min. turning point: $(180^\circ; -3)$

3.



Period: 360°
 Amplitude: 1
 Domain: $[0^\circ; 360^\circ]$
 Range: $[1; 3]$
 x-intercepts: none
 y-intercepts: $(0^\circ; 3)$
 Max. turning point: $(0^\circ; 3); (360^\circ; 3)$
 Min. turning point: $(180^\circ; 1)$

4.



Period: 360°
 Amplitude: $\frac{1}{2}$
 Domain: $[0^\circ; 360^\circ]$
 Range: $[-\frac{1}{2}; -\frac{3}{2}]$
 x-intercepts: none
 y-intercepts: $(0^\circ; -\frac{1}{2})$
 Max. turning point: $(0^\circ; -\frac{1}{2}); (360^\circ; -\frac{1}{2})$
 Min. turning point: $(180^\circ; -\frac{3}{2})$

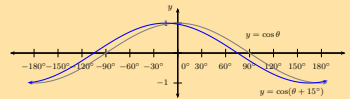
Exercise 5 – 25: Cosine functions of the form $y = \cos k\theta$

2. a) $k = \frac{3}{2}$

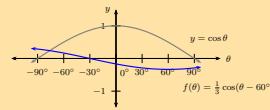
b) $k = \frac{2}{3}$

Exercise 5 – 27: The cosine function

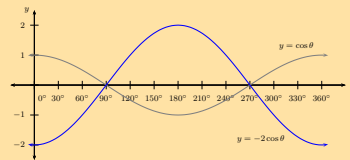
1. a)



b)



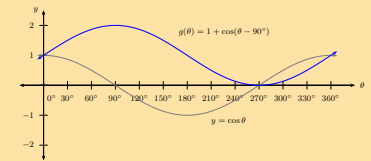
c)



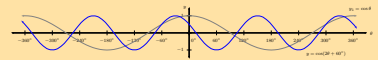
d)



e)



f)



2.

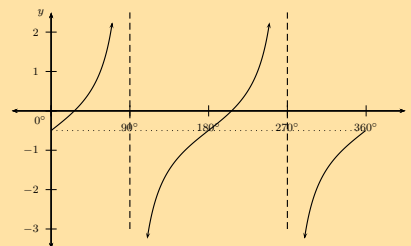
a) $a = -1$

b) $p = -180^\circ$

c) $\cos(\theta - 180^\circ) = -\cos \theta$

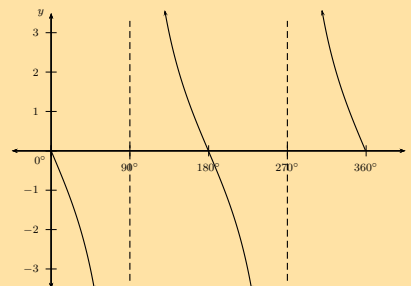
Exercise 5 – 28: Revision

1.



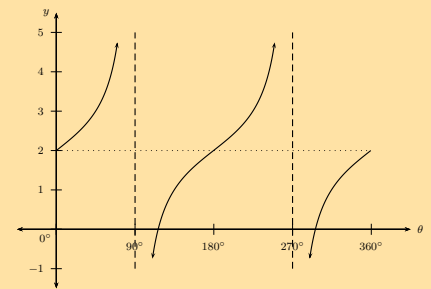
Period: 180°
 Domain: $[0^\circ; 360^\circ]$
 Range: $[-\infty; \infty]$
 x-intercepts: $(26,6^\circ; 0); (206,6^\circ; 0)$
 y-intercepts: $(0^\circ; -\frac{1}{2})$
 Asymptotes: $90^\circ; 270^\circ$

2.



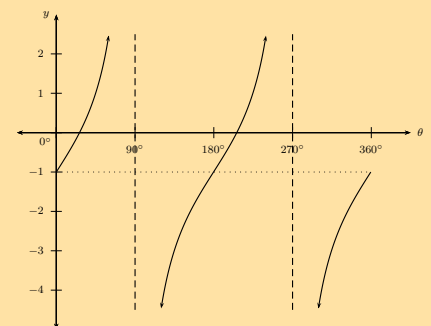
Period: 180°
 Domain: $[0^\circ; 360^\circ]$
 Range: $[-\infty; \infty]$
 x-intercepts: $(0^\circ; 0); (180^\circ; 0); (360^\circ; 0)$
 y-intercepts: $(0^\circ; 0)$
 Asymptotes: $90^\circ; 270^\circ$

3.



Period: 180°
 Domain: $[0^\circ; 360^\circ]$
 Range: $[-\infty; \infty]$
 x-intercepts: $(116,6^\circ; 0); (296,6^\circ; 0)$
 y-intercepts: $(0^\circ; 2)$
 Asymptotes: $90^\circ; 270^\circ$

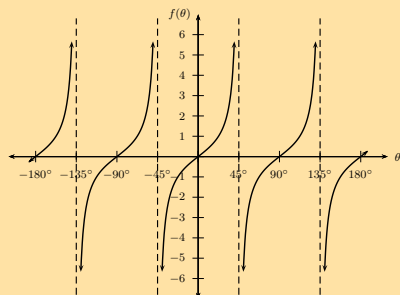
4.



Period: 180°
 Domain: $[0^\circ; 360^\circ]$
 Range: $[-\infty; \infty]$
 x-intercepts: $(26,6^\circ; 0); (206,6^\circ; 0)$
 y-intercepts: $(0^\circ; -1)$
 Asymptotes: $90^\circ; 270^\circ$

Exercise 5 – 29: Tangent functions of the form $y = \tan k\theta$

1.



Period: 90°

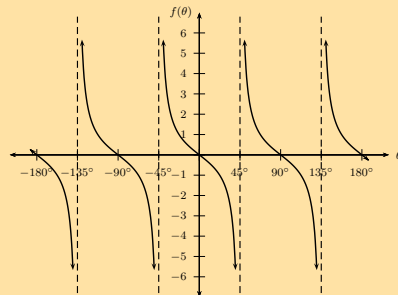
Domain: $[-180^\circ; 180^\circ]$ Range: $[-\infty; \infty]$

x -intercepts: $(-180^\circ; 0)$; $(-90^\circ; 0)$; $(0^\circ; 0)$; $(90^\circ; 0)$; $(180^\circ; 0)$

y -intercepts: $(0^\circ; 0)$

Asymptotes: -135° ; -45° ; 45° ; 135°

3.



Period: 90°

Domain: $[-180^\circ; 180^\circ]$

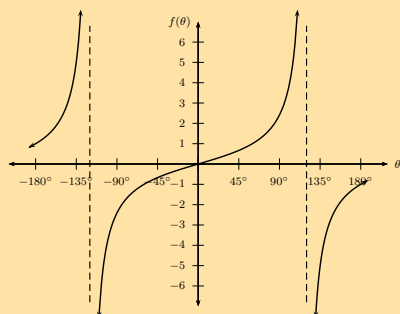
Range: $[-\infty; \infty]$

x -intercepts: $(-180^\circ; 0)$; $(-90^\circ; 0)$; $(0^\circ; 0)$; $(90^\circ; 0)$; $(180^\circ; 0)$

y -intercepts: $(0^\circ; 0)$

Asymptotes: -135° ; -45° ; 45° ; 135°

2.



Period: 240°

Domain: $[-180^\circ; 180^\circ]$

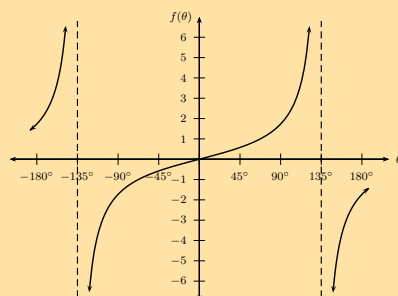
Range: $[-\infty; \infty]$

x -intercepts: $(0^\circ; 0)$

y -intercepts: $(0^\circ; 0)$

Asymptotes: -120° ; 120°

4.



Period: 270°

Domain: $[-180^\circ; 180^\circ]$

Range: $[-\infty; \infty]$

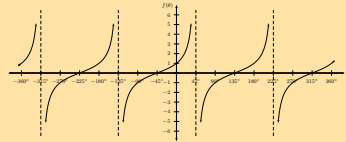
x -intercepts: $(0^\circ; 0)$

y -intercepts: $(0^\circ; 0)$

Asymptotes: -135° ; 135°

Exercise 5 – 30: Tangent functions of the form $y = \tan(\theta + p)$

1.



Period: 180°

Domain: $[-360^\circ; 360^\circ]$

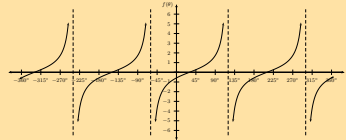
Range: $[-\infty; \infty]$

x -intercepts: $(-330^\circ; 0)$; $(-150^\circ; 0)$; $(30^\circ; 0)$; $(210^\circ; 0)$

y -intercepts: $(0^\circ; -0.58)$

Asymptotes: -315° ; -135° ; 45° ; 225°

2.



Period: 180°

Domain: $[-360^\circ; 360^\circ]$

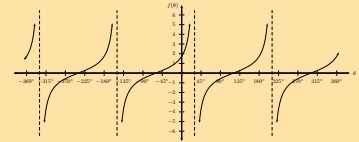
Range: $[-\infty; \infty]$

x -intercepts: $(-225^\circ; 0)$; $(-45^\circ; 0)$; $(135^\circ; 0)$; $(315^\circ; 0)$

y -intercepts: $(0^\circ; 0)$

Asymptotes: -315° ; -135° ; 45° ; 225°

3.



Period: 180°

Domain: $[-360^\circ; 360^\circ]$

Range: $[-\infty; \infty]$

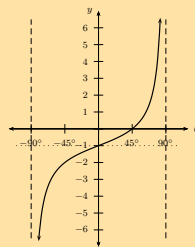
x -intercepts: $(-240^\circ; 0)$; $(-60^\circ; 0)$; $(120^\circ; 0)$; $(300^\circ; 0)$

y -intercepts: $(0^\circ; 1.73)$

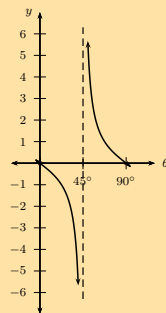
Asymptotes: -330° ; -150° ; 30° ; 210°

Exercise 5 – 31: The tangent function

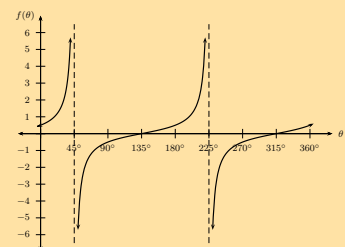
1. a)



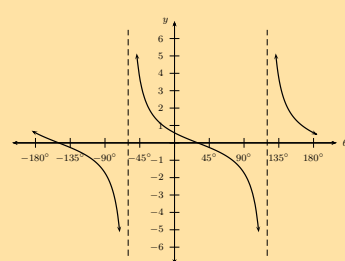
b)



c)



d)

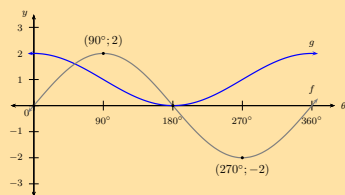


2. $a = -1$; $k = \frac{1}{2}$

Exercise 5 – 32: Mixed exercises

1. a) $f(\theta) = \frac{3}{2} \sin 2\theta$ and $g(\theta) = -\frac{3}{2} \tan \theta$
 b) $f(\theta) = -2 \sin \theta$ and $g(\theta) = 2 \cos(\theta + 360^\circ)$
 c) $y = 3 \tan \frac{\theta}{2}$
 d) $y = y = 2 \cos \theta + 2$

2. a)



- b) 360°

- c) 1

- d) At $\theta = 180^\circ$

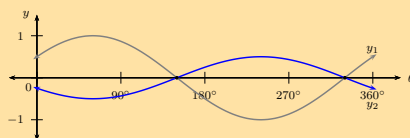
3. a) $a = 2$, $b = -1$ and $c = 240^\circ$

- b) 180°

- c) $\theta = 60^\circ; 300^\circ$

- d) $y = -\tan(\theta - 45^\circ)$

- 4.



Exercise 5 – 33: End of chapter exercises

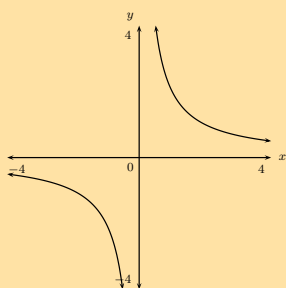
2. $a = -2$; $k = -1$

4. a) $y = x + 2^2 + 2$

- b) $y = x - 1^2 + 5$

5. $(-1; 0)$

- 6.



$$y = \frac{2}{x-3} - 1$$

7. $y = \frac{1}{(x-1)} + 2$

9. a) $a = -1$

- b) $f(-15) = 0,99997$

- c) $x = -1$

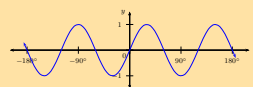
- d) $h(x) = -2^{(x-2)} + 1$

10. a) $a = 256$

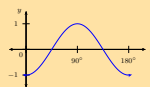
- b) $f(x) = 256 \left(\frac{3}{4}\right)^x$

- c) $f(13) = 6,08$

11. a)



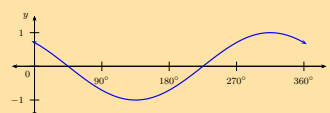
- b)



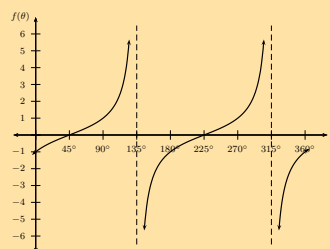
- d)



- e)



- f)



6 Trigonometry

Exercise 6 – 1: Revision

- | | | |
|--------------------|-----------------------|--------------------|
| 1. a) True | f) $17,7^\circ$ | b) 0 |
| b) True | g) $69,4^\circ$ | c) $-1\frac{1}{2}$ |
| c) False | 3. a) 17,3 cm | d) 1 |
| d) True | b) 10 cm | e) 1 |
| 2. a) $50,2^\circ$ | c) $64,8^\circ$ | 6. a) 60° |
| b) $40,5^\circ$ | 4. a) 10 cm | b) $\frac{1}{2}$ |
| c) $26,6^\circ$ | b) 5,2 cm and 19,3 cm | c) 1 |
| d) $109,8^\circ$ | c) 50 cm^2 | |
| e) No solution | 5. a) 2 | 7. No |

Exercise 6 – 2: Trigonometric identities

- | | |
|---------------------|--------------------|
| 1. a) $\cos \alpha$ | c) $\cos^2 \theta$ |
| b) $\tan^2 \theta$ | d) 0 |

Exercise 6 – 3: Reduction formulae for function values of $180^\circ \pm \theta$

- | | | |
|----------------------------|---|-------------------|
| 1. a) $\frac{\sqrt{3}}{3}$ | 2. a) $\frac{1-\cos^2 \theta}{\cos \theta}$ | 3. a) $2t$ |
| b) $\frac{1}{8}$ | b) -1 | b) $-\frac{1}{t}$ |
| c) 1 | | |

Exercise 6 – 4: Using reduction formula

- | | | |
|----------------------|----------------------------|---------------------------------|
| 1. a) $-\tan \theta$ | 3. a) $\frac{1}{\sqrt{3}}$ | e) $-\frac{4\sqrt{3}}{5}$ |
| b) 1 | b) 2 | 5. a) $-t$ |
| c) 1 | c) 2 | b) $1-t^2$ |
| 2. $-\cos \beta$ | d) $-\frac{3}{2}$ | c) $\pm \frac{t}{\sqrt{1-t^2}}$ |

Exercise 6 – 5: Co-functions

- | | | |
|---------------------|-------------------|------------------------------|
| 1. a) $\cos \theta$ | 2. a) p | c) $-\frac{p}{\sqrt{1-p^2}}$ |
| b) $\frac{3}{2}$ | b) $\sqrt{1-p^2}$ | d) p |

Exercise 6 – 6: Reduction formulae

- | | | | | | |
|----|---------------------|----|---------------------|----|-------------------------|
| 1. | a) $\sin^2 \theta$ | 2. | a) $\sin 17^\circ$ | 3. | a) $\sqrt{3}$ |
| | b) $\cos^2 \theta$ | | b) $\cos 33^\circ$ | | b) $\frac{\sqrt{3}}{2}$ |
| | c) i. 1 | | c) $\tan 68^\circ$ | | c) $\frac{1}{4}$ |
| | ii. $\tan^2 \theta$ | | d) $-\cos 33^\circ$ | | d) 1 |

Exercise 6 – 7: Solving trigonometric equations

- | | | | |
|----|--|----|--|
| 1. | a) $\alpha = 60^\circ; 300^\circ$ | 2. | a) $\theta = -323,1^\circ; -216,9^\circ; 36,9^\circ; 143,1^\circ$ |
| | b) $\alpha = 220,5^\circ; 319,5^\circ$ | | b) $\theta = -221,4^\circ; -138,6^\circ; 138,6^\circ; 221,4^\circ$ |
| | c) $\alpha = 79,2^\circ; 259,2^\circ$ | | c) $\theta = -278,5^\circ; -98,5^\circ; 81,5^\circ; 261,5^\circ$ |
| | d) $\alpha = 200,1^\circ; 339,9^\circ$ | | d) $\theta = -90^\circ; 270^\circ$ |
| | e) $\alpha = 36,9^\circ; 143,1^\circ$ | | e) $\theta = -293,6^\circ; -66,4^\circ; 66,4^\circ; 293,6^\circ$ |
| | f) $\alpha = 109,7^\circ; 289,7^\circ$ | | |

Exercise 6 – 8: General solution

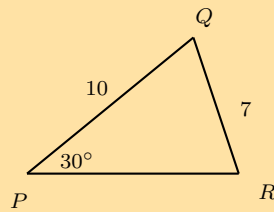
- | | | | |
|----|--|----|---|
| 1. | a) $\theta = -128,36^\circ; -101,64^\circ; 51,64^\circ$ | 2. | a) $\theta = -20^\circ + n \cdot 360^\circ$ |
| | b) $\theta = -80,45^\circ; -9,54^\circ; 99,55^\circ; 170,46^\circ$ | | b) $\alpha = 30^\circ + n \cdot 120^\circ$ |
| | c) $\theta = -53,27^\circ; 126,73^\circ$ | | c) $\beta = 10,25^\circ + n \cdot 45^\circ$ or $\beta = 55,25^\circ + n \cdot 45^\circ$ |
| | d) $\alpha = 0^\circ$ | | d) $\alpha = 70^\circ + n \cdot 360^\circ$ or $\alpha = 340^\circ + n \cdot 360^\circ$ |
| | e) $\theta = -180^\circ; 0^\circ; 180^\circ$ | | e) $\theta = 140^\circ + n \cdot 240^\circ$ or $\theta = 220^\circ + n \cdot 240^\circ$ |
| | f) $\theta = -180^\circ; 180^\circ$ | | f) $\beta = 15^\circ + n \cdot 180^\circ$ |
| | g) $\theta = 84^\circ$ | | |
| | h) $\theta = -120^\circ; 120^\circ$ | | |
| | i) $\theta = -60^\circ; -30^\circ; 120^\circ; 150^\circ$ | | |

Exercise 6 – 9: Solving trigonometric equations

- | | | | |
|----|---|----|--|
| 1. | a) $\theta = 45^\circ + k \cdot 180^\circ$ or $\theta = 135^\circ + k \cdot 180^\circ$ | 3. | a) $\theta = 120^\circ + k \cdot 360^\circ$ or $\theta = 240^\circ + k \cdot 360^\circ$ |
| | b) $\alpha = 50^\circ + k \cdot 360^\circ$ or $\alpha = 110^\circ + k \cdot 360^\circ$ | | b) $\theta = 0^\circ + k \cdot 180^\circ$ or $\theta = 146,3^\circ + k \cdot 180^\circ$ |
| | c) $\theta = 60^\circ + k \cdot 720^\circ$ or $\theta = 660^\circ + k \cdot 720^\circ$ | | c) $\alpha = 36,9^\circ + k \cdot 360^\circ$ or $\alpha = 143,1^\circ + k \cdot 360^\circ$ or $\alpha = 216,9^\circ + k \cdot 360^\circ$ or $\alpha = 323,1^\circ + k \cdot 360^\circ$ |
| | d) $\beta = 146,6^\circ + k \cdot 180^\circ$ | | d) $\beta = 15^\circ + k \cdot 120^\circ$ or $\beta = 75^\circ + k \cdot 120^\circ$ |
| | e) $\theta = 110,27^\circ + k \cdot 360^\circ$ or $\theta = 249,73^\circ + k \cdot 360^\circ$ | | e) $\alpha = 48,4^\circ + k \cdot 180^\circ$ |
| | f) $\alpha = 210^\circ + k \cdot 360^\circ$ or $\alpha = 330^\circ + k \cdot 360^\circ$ | | f) $\theta = 63,4^\circ + k \cdot 180^\circ$ or $\theta = 116,6^\circ + k \cdot 180^\circ$ |
| | g) $\beta = 23,3^\circ + k \cdot 120^\circ$ | | g) $\theta = 54,8^\circ + k \cdot 180^\circ$ or $\theta = 95,25^\circ + k \cdot 180^\circ$ |
| | h) $\theta = 122^\circ + k \cdot 180^\circ$ | | |
| | i) $\alpha = 21^\circ + k \cdot 180^\circ$ or $\alpha = 39,5^\circ + k \cdot 90^\circ$ | | |
| | j) $\beta = 22,5^\circ + k \cdot 90^\circ$ | | |
| 2. | $\theta = 0^\circ, 180^\circ, 210^\circ, 330^\circ$ or 360° | 4. | $\beta = -70,5^\circ$ or $\beta = 109,5^\circ$ |

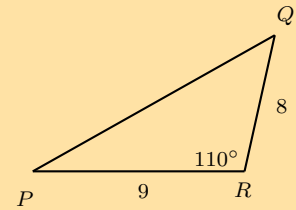
Exercise 6 – 10: The area rule

1. a)



Area $\triangle PQR = 17,5$ square units

b)



Area $\triangle PQR = 33,8$ square units

2. Area $\triangle XYZ = 645,6$ square units

3. Area = 106,5 square units

4. $\hat{C} = 72,2^\circ$ or $\hat{C} = 107,8^\circ$

Exercise 6 – 11: Sine rule

1. a) $\hat{P} = 92^\circ, q = 6,6, p = 7,4$
b) $\hat{L} = 87^\circ, l = 1,3, k = 0,89$
c) $\hat{B} = 76,8^\circ, b = 94,3, c = 91,3$
d) $\hat{Y} = 84^\circ, y = 60, z = 38,8$

2. $\hat{B} = 32^\circ, AB = 23, BC = 39$

3. $ST = 78,1$ km

4. $m = 26,2$

5. $BC = 3,2$

Exercise 6 – 12: The cosine rule

1. a) $a = 8,5, \hat{C} = 83,9^\circ, \hat{B} = 26,1^\circ$
b) $\hat{R} = 120^\circ, \hat{S} = 32,2^\circ, \hat{T} = 27,8^\circ$
c) $\hat{M} = 27,7^\circ, \hat{L} = 40,5^\circ, \hat{K} = 111,8^\circ$
d) $h = 19,1, \hat{J} = 18,2^\circ, \hat{K} = 31,8^\circ$
e) $\hat{D} = 34^\circ, \hat{E} = 44,4^\circ, \hat{F} = 101,6^\circ$

2. a) $x = 4,4$ km
b) $y = 63,5$ cm

3. a) $\hat{K} = 117,3^\circ$
b) $\hat{Q} = 78,5^\circ$

Exercise 6 – 13: Area, sine and cosine rule

1. a) 7,78 km
b) 6 km

2. $XZ = 1,73$ km, $XY = 0,87$ km

3. a) 1053 km
b) $4,42^\circ$

4. $DC = \frac{x \sin a \sin(b+c)}{\sin(a+c) \sin b}$

5. b) 438,5 km

6. 9,38 m²

7. $DC = \frac{x \sin \alpha}{\sin \beta}$

Exercise 6 – 14: End of chapter exercises

1. $\sin^2 A$

2. $1\frac{1}{4}$

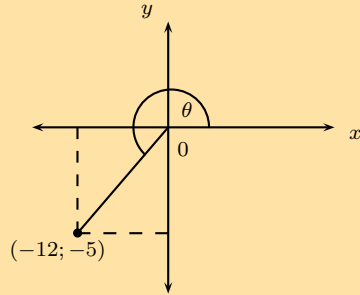
3. $\cos \alpha$

4. 3

7. a) -1

b) $\theta = 135^\circ$ or $\theta = 315^\circ$

8. a)



b) $-\frac{5}{13}$ and $\frac{12}{13}$

c) $\theta = 202,62^\circ$

9. a) $a = 1$ and $b = -\sqrt{3}$

b) $-\frac{\sqrt{3}}{2}$

10. a) $x = 50,9^\circ$ or $x = 309,1^\circ$

b) $x = 127,3^\circ$ or $x = 307,3^\circ$

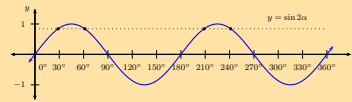
c) $x = 26,6^\circ; 153,4^\circ; 206,6^\circ$ or $333,4^\circ$

11. a) $x = 55^\circ + k \cdot 360^\circ$ or
 $x = 175^\circ + k \cdot 360^\circ$

b) $x = 180^\circ + k \cdot 360^\circ$

12. a) $x = 28,6^\circ + k \cdot 180^\circ$ or
 $x = 61,4^\circ + k \cdot 180^\circ$

b)



c) $28,6^\circ; 61,4^\circ; 208,6^\circ; 241,4^\circ$

13. a) $\hat{A}GN = \alpha - \beta$

b) $\hat{A} = 90^\circ - \alpha$

d) $H = 5$ m

14. a) $AC = 9,43$ m

b) $AD = 6,2$ m

c) Area = $49,25$ m²

d) Area = $49,23$ m²

7 Measurement

Exercise 7 – 1: Area of a polygon

1. b) 240 cm

c) $0,6$ m²

e) Wood: 233,2 cm and paper: $0,6$ m²

2. a) 25π units²

b) 20π units²

3. a) $1,2$ m²

b) Perimeter: 414,8 cm; Area 11 700 cm²

c) 108×108 cm²

Exercise 7 – 2: Calculating surface area

1. 273 cm²

2. Yes

Exercise 7 – 3: Calculating volume

1. a) 67,5 m²

b) 3,39 ℓ

2. b) 13,86 cm

c) 554,24 m³

Exercise 7 – 4: Finding surface area and volume

- | | | | |
|----|-----------------------|--------------|-------------|
| 1. | a) 120 cm^2 | c) 40 | ii. 165 mm |
| | b) 124 cm^3 | d) i. 120 mm | iii. 589 mm |

Exercise 7 – 5: The effects of k

- | | | | |
|----|----------------------------|----|--------------------|
| 1. | a) Is halved | 2. | a) $0,5W^3$ |
| | b) Approx. 50 times bigger | | b) $0,93 \times W$ |

Exercise 7 – 6: End of chapter exercises

- | | | | | | |
|----|--------------------------|-------------------------|--|--|----------------------------|
| 2. | a and d | iii. 960 cm^3 | 7. | No | |
| 3. | a) Triangular prism | b) 600 cm^2 | 8. | a) $10 \text{ cm} \times 10 \text{ cm} \times 10 \text{ cm}$ | |
| | b) Triangular pyramid | 5. $\sqrt{5}x^2$ | | b) 12,6 cm | |
| | c) Rhombic prism | 6. | a) $72\,000 \text{ cm}^3$ | 9. | a) Volume triples |
| 4. | a) i. 856 cm^2 | | b) $H = 54 \text{ cm}$ and $h = 60,2 \text{ cm}$ | | b) Surface area $\times 9$ |
| | ii. Rectangular prism | | c) $12\,732 \text{ cm}^2$ | | c) Volume $\times 27$ |

8 Euclidean geometry

Exercise 8 – 1: Perpendicular line from center bisects chord

- | | | |
|--------------------|----------------------|--------------------|
| 1. $x = \sqrt{41}$ | 3. $x = 10$ units | 5. $x = 3,6$ units |
| 2. $x = \sqrt{84}$ | 4. $TU = 2,66$ units | |

Exercise 8 – 2: Angle at the centre of circle is twice angle at circumference

- | | | |
|---------------------|--------------------|--------------------|
| 1. $b = 90^\circ$ | 3. $d = 200^\circ$ | 5. $f = 120^\circ$ |
| 2. $c = 22,5^\circ$ | 4. $e = 55^\circ$ | |

Exercise 8 – 3: Subtended angles in the same segment

1. a) $a = 21^\circ$
b) $c = 24^\circ$ $d = 78^\circ$
c) $d = 28^\circ$
2. a) $e = 85^\circ$
3. $f = 35^\circ$

Exercise 8 – 4: Cyclic quadrilaterals

1. a) $a = 93^\circ$, $b = 74^\circ$
b) $a = 114^\circ$
- c) $a = 29^\circ$

Exercise 8 – 5: Tangents to a circle

1. $d = 9,4$ cm
2. $e = 2,5$ cm
3. $f = 3$ cm

Exercise 8 – 6: Tangent-chord theorem

1. a) $a = 33^\circ$, $b = 33^\circ$
b) $c = 72^\circ$, $d = 54^\circ$
c) $f = 38^\circ$, $g = 47^\circ$
d) $l = 48^\circ$
- e) $i = 40^\circ$, $j = 101^\circ$,
 $k = 40^\circ$
- f) $m = 56^\circ$, $n = 34^\circ$,
 $o = 56^\circ$
- g) $p = 38^\circ$, $q = 52^\circ$,
 $r = 90^\circ$
2. $a = 26^\circ$, $b = 64^\circ$, $c = 128^\circ$

Exercise 8 – 7: End of chapter exercises

1. a) $\hat{A} = x$
b) $\hat{C}\hat{O}\hat{D} = 2x$
c) $\hat{D} = 90^\circ - x$
2. a) $\hat{D}_1 = 78^\circ$
b) $\hat{M}_1 = 39^\circ$
c) $\hat{F}_2 = 51^\circ$
d) $\hat{G} = 58^\circ$
e) $\hat{E}_1 = 32^\circ$
3. a) $\hat{D}_2 = x$
b) $\hat{O}\hat{A}\hat{B} = x$
c) $\hat{O}\hat{B}\hat{A} = x$
d) $\hat{A}\hat{O}\hat{B} = 180^\circ - 2x$
- e) $\hat{C} = 90^\circ - x$
4. a) $OM = 3$ cm
b) $AM = 8$ cm
c) $AB = 4\sqrt{5}$ cm
5. $x = 35^\circ$
6. a) $R\hat{Q}S$, $Q\hat{S}O$
b) $P\hat{O}S = 2x$
7. a) $O\hat{D}C = 35^\circ$
b) $C\hat{O}\hat{D} = 110^\circ$
c) $C\hat{B}D = 55^\circ$
d) $B\hat{A}D = 90^\circ$
e) $A\hat{D}B = 45^\circ$
8. $x = 4y$
10. $OQ = 17$ mm
11. a) $Q\hat{R}P$, $Q\hat{S}P$, $R\hat{S}T$
b) $S\hat{R}T = 80^\circ$, $S\hat{T}R = 30^\circ$, $P\hat{Q}S = 30^\circ$
d) $P\hat{M}Q = 110^\circ$
13. a) $90^\circ - \frac{x}{2}$
b) $\frac{x}{2}$
c) $90^\circ - \frac{x}{2}$
14. c) $90^\circ - 2x$
d) $AO = 13$ cm

9 Finance, growth and decay

Exercise 9 – 1: Revision

- | | | |
|----------------|--------------|----------|
| 1. R 13 630 | 3. a) R 2536 | 4. 9,38% |
| 2. R 10 246,59 | b) R 2468,27 | 5. 4,56% |

Exercise 9 – 2: Simple decay

- | | |
|--------------|------------|
| 1. R 112 000 | 3. 11,66% |
| 2. R 941,18 | 4. 7 years |

Exercise 9 – 3: Compound depreciation

- | | | |
|--------------------|--------------|----------------|
| 1. R 23 766,73 | 3. R 131 072 | 5. 7,62 kg |
| 2. 2229 cormorants | 4. 132 221 | 6. R 85 997,13 |

Exercise 9 – 4: Finding i

- | | |
|----------|----------|
| 1. 14,9% | 3. 12,0% |
| 2. 16,4% | 4. 9,4% |

Exercise 9 – 5: Timelines

- | | | |
|----------------|----------------|-------------------|
| 1. R 38 588,25 | 3. R 2600 | 5. R 1 149 283,50 |
| 2. R 35 308,00 | 4. R 19 950,62 | 6. R 7359,83 |

Exercise 9 – 6: Nominal and effect interest rates

- | | | |
|---------------------------|----------|---------------------------------|
| 1. a) 12,6% | b) 17,7% | 5. a) 9,42% is the better rate. |
| b) 15,5% | c) 16,8% | b) 9,38% |
| c) 22,1% | 3. 9,1% | c) 9,52% |
| 2. a) 16,8%; 17,7%; 17,5% | 4. 9,4% | |

Exercise 9 – 7: End of chapter exercises

- | | | |
|-----------------|-----------------|------------------|
| 1. a) R 246 400 | 5. a) 7,44% | b) 11,1% |
| b) R 265 599,87 | b) R 148 826,15 | c) 11,0% |
| 2. R 229,92 | c) R 135 968,69 | 9. R 322 580,65 |
| 3. a) R 8800 | 6. a) R 8042,19 | 10. a) 4,7% |
| b) 18,1% | b) 26,82% | b) 4,8% |
| 4. R 238 191,17 | 8. a) 11,3% | 11. R 212 347,69 |

10 Probability

Exercise 10 – 1: Revision

- | | | |
|----------------------|---------------------|-------------------------------------|
| 1. $\frac{y}{r+b+y}$ | b) $\frac{8}{21}$ | f) $\frac{4}{9}$ |
| 2. $\frac{5}{12}$ | 8. a) $\frac{4}{9}$ | 9. a) $\frac{24}{116} \approx 0,21$ |
| 3. $\frac{3}{8}$ | b) $\frac{5}{9}$ | b) $\frac{65}{116} \approx 0,56$ |
| 4. 3 | c) $\frac{1}{9}$ | c) 0 |
| 6. 5 | d) $\frac{8}{9}$ | d) $\frac{21}{29} \approx 0,72$ |
| 7. a) $\frac{1}{6}$ | e) $\frac{5}{9}$ | 10. 0,28 |

Exercise 10 – 2: Venn diagram revision

- | | | |
|--------|--------|---------|
| 1. 0,5 | 2. 0,7 | 3. 0,18 |
|--------|--------|---------|

Exercise 10 – 3: Dependent and independent events

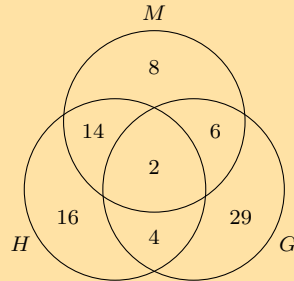
- | | | |
|------------------------------|-----------------------|---|
| 1. a) not mutually exclusive | 2. a) $\frac{17}{30}$ | d) No |
| b) dependent | b) $\frac{11}{30}$ | 3. independent and not mutually exclusive |
| | c) Yes | |

Exercise 10 – 4: Venn diagrams

1. a) $\frac{9}{40}$
b) $\frac{3}{5}$
c) $\frac{1}{10}$
d) dependent

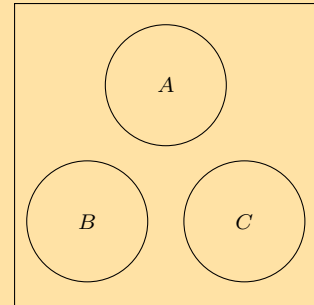
2. $P(Z \text{ and not } (X \text{ or } Y)) = \frac{4}{25}$

3. a)



- b) 6
c) 29
d) 2

4.



Exercise 10 – 5: Tree diagrams

1. $\frac{7}{36}$

2. $\frac{671}{1296}$

3. a) $\frac{1}{4}$

- b) $\frac{5}{16}$

4. a) $\frac{1}{4}$

- b) $\frac{5}{16}$

Exercise 10 – 6: Contingency tables

1. a) $\frac{5}{8}$
b) $\frac{8}{23}$
c) $\frac{12}{23}$
d) dependent

2. The events are whether a bus leaves from Location A or not and whether a bus left late or

not.

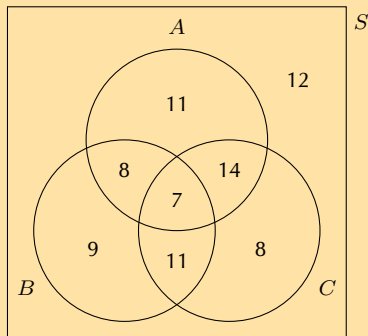
The events are dependent.

3.

	A	not A	Totals
B	14	6	20
not B	21	9	30
Totals	35	15	50

Exercise 10 – 7: End of chapter exercises

1. 0,68
2. 0,24
3. a) The experiment is the outcome of selecting a particular model and colour car from the sample space of available cars.
The sample space is {pink model A ; lime-green model A ; purple model A ; purple model B ; orange model B ; multicolour model B }.
- b) 1
- c) 0
4. 0,57
5. a) 0,38
b) 0,7
c) 0,69
d) 0,69
e) 0,3
6. a) $\frac{3}{5}$
b) $\frac{1}{2}$
c) $\frac{1}{10}$
d) 0
e) $\frac{2}{5}$
f) $\frac{9}{10}$
7. a)



- b) 12
 - c) 8
 - d) $\frac{17}{20}$
 8. a) $h = 0,3$
b) $h = 0,5$
 9. a) 0,434
b) 0,182
c) 0,47
 - 10.
-

$$P(\text{two orange balls}) = \frac{63}{136}$$

11.

	Durban	Bloemfontein	Totals
Liked living there	130	30	160
Did not like living there	140	200	340
Totals	270	230	500

The events are dependent.

12.

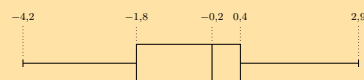
	Multivitamin A	Multivitamin B	Totals
Improvement in health	400	300	700
No improvement in health	140	120	260
Totals	540	420	960

Multivitamin A is more effective.

11 Statistics

Exercise 11 – 1: Revision

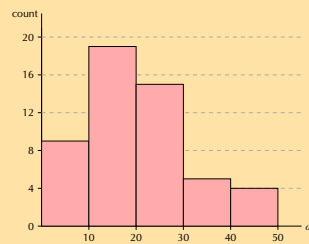
1. a) mean = $-4,3$; first quartile = $-6,2$; second quartile = $-3,4$; third quartile = $-2,9$.
b) mean = $-5,6$; first quartile = -60 ; second quartile = -6 ; third quartile = 65 .
c) mean = $18,5$; first quartile = 7 ; second quartile = $11,5$; third quartile = 33 .
2. range = $9,6$; inter-quartile range = $2,51$.
- 3.



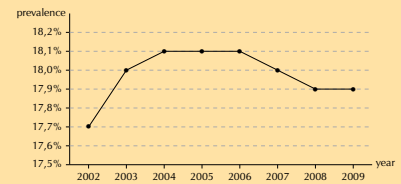
Exercise 11 – 2: Histograms

1. a) 170 million
b) increasing
c) 3 million

2.



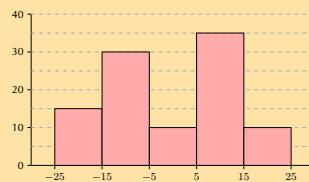
3.



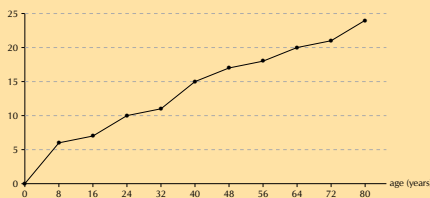
Exercise 11 – 3: Ogives

1. a) 20
b) 15
c) 60%

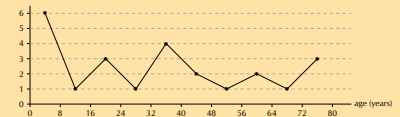
2.



3. a)



- b) 11 people
c) 19 people
d) 34
e) 25,5
f)



4. e) 49,25
f) 49,7

Exercise 11 – 4: Variance and standard deviation

1. a) Cape Town: 3,84. Durban: 3,82.
b) Cape Town: 0,121. Durban: 0,184.
c) Cape Town
2. Mean = 270,7. Variance = 27 435,2.
3. Mean = -1,95. Variance = 127,5.

4. a) 10,4
b) 0,27
c) 3
5. 13 and 20

Exercise 11 – 5: Symmetric and skewed data

1. skewed left

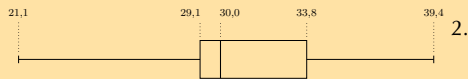
- b) skewed right

2. a) skewed right

- c) skewed left
d) symmetric

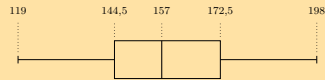
Exercise 11 – 6: Outliers

1. a)



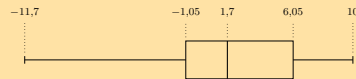
There are two outliers on the left.

b)

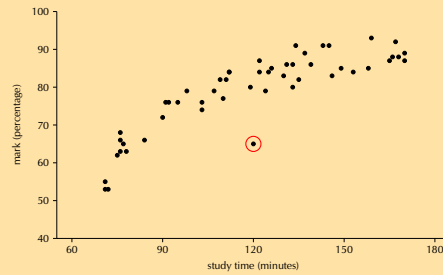


There are no outliers.

c)

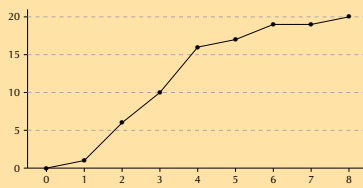
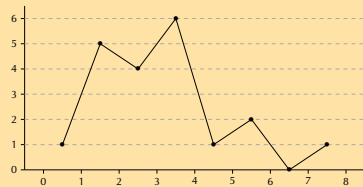
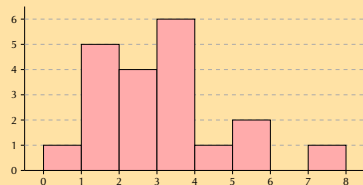


There is one outlier on the left.



Exercise 11 – 7: End of chapter exercises

1.



2.



Skewed right

3. a) Mean = $11\frac{3}{8}$. Standard deviation = 6,69.

b) Mean = $5\frac{3}{8}$. Standard deviation = 2,33.

c) There are no outliers.

4.

a) Mean = R 12 497,50. Standard deviation = R 1768,55.

b) 4

c) Mean = R 12 997,50. Standard deviation = R 1768,55.

d) Mean = R 13 747,25. Standard deviation = R 1945,41.

e) 4

f) 10%

5.

a) 14

b) approximately 88 kg.

c) 75

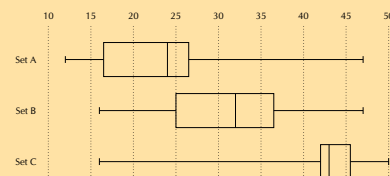
6.

a) A. Mean = 23,83. Five number summary = [12 ; 16,5 ; 24 ; 26,5 ; 47].

B. Mean = 31,17. Five number summary = [16 ; 25 ; 32 ; 36,5 ; 47].

C. Mean = 41,83. Five number summary = [16 ; 42 ; 43 ; 45,5 ; 50].

b)

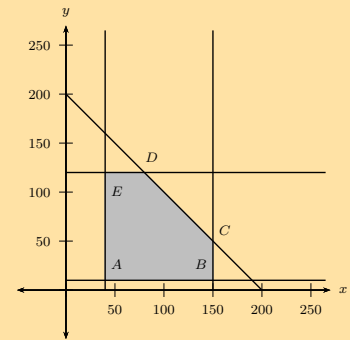


c) Set A: skewed left. Set B: slightly skewed left. Set C: skewed right.

Exercise 12 – 2: Optimisation

1. a) $x \leq 10; y \leq 10; x \geq 4; 2x + 3y \leq 30$
b) $T = 5x + 10y$
2. a) $x + y \leq 10\ 000; x \geq 4000; y \geq 2000;$
 $y \leq 4000$
c) $I = 50x + 30y$
d) R 460 000
3. a) $150x + 60y \geq 30\ 000;$
 $50x + 40y \geq 13\ 000;$
 $(10)x + (20)y \geq 5000$
d) $E = (20\ 000)x + (10\ 000)y$
e) 140 Super X and 150 Super Y
f) R 4 300 000
4. a) $x \leq 300; y \geq 0.5x; x + y \leq 500$
b) $P = (3)x + (2)y$
c) 300 hamburgers and 200 chicken burgers
5. a) $x \leq 150; y \leq 120; x + y \leq 200; x \geq 40; y \geq 10$

b)



- c) $P = (5)x + (10)y$
d) 80 of card X and 120 of card Y
6. a) $4x + 3y \geq 15; 16x + 24y \geq 72;$
 $x + y \leq 5$
b) 3 packets of Vuka and 1 packets of Molo
c) 0; 5 or 5; 0

VERSION 1 CAPS

GRADE 11 MATHEMATICS

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